EEE4114F
Digital Signal Processing

Course Notes

Department of Electrical Engineering
Faculty of Engineering and the Built Environment
University of Cape Town

M. Y. Martin
F. C. Nicolls

Preface

This book has been adapted and modified from notes originally written by Fred Nicolls, with various additions being made to supplement the existing material.

These notes are written as a reference text to support the first part of the EEE4114F course at the University of Cape Town, covering Digital Signal Processing (DSP) specifically. This includes various aspects of discrete-time signals and systems, with a particular focus on Fourier representations and the z-transform (and the analysis thereof).

This book is not meant to be exhaustive, but it aims to provide students with sufficient information on the most crucial aspects of DSP and serve as a useful supplement to the lectures and practicals.

Enjoy the course, and please contact the course lecturer if you spot any errors – mistakes *do* unfortunately happen!

— M. Y. Martin

Contents

Chapter 1

Introduction

First off, welcome to EEE4114F! Good to have you on board, and we look forward to teaching you all about digital signal processing in the coming chapters.

This first chapter aims to cover a few basic concepts related to the course – including an overview of some background theory and applications for digital signals in the real world – before delving into the rest of the material.

1.1 Digital Signals

A signal is any variable that carries information. Examples of the types of signals of interest include:

- Speech (telephony, radio, everyday communication)
- Biomedical signals (EEG brain signals)
- Sound and music
- Video and image
- Radar signals (range and bearing)

Digital signal processing (DSP) is concerned with the digital representation of signals and the use of digital processors to analyse, modify, or extract information from signals.

Many signals in DSP are derived from analogue signals which have been sampled at regular intervals and converted into digital form. The key advantages of DSP over analogue processing are:

- Guaranteed accuracy (determined by the number of bits used)
- Perfect reproducibility

- No drift in performance due to temperature or age
- Takes advantage of advances in semiconductor technology
- Greater flexibility (can be reprogrammed without modifying hardware)
- Superior performance (linear phase response possible, and filtering algorithms can be made adaptive)
- Sometimes information may already be in digital form

There are, however, still a few disadvantages:

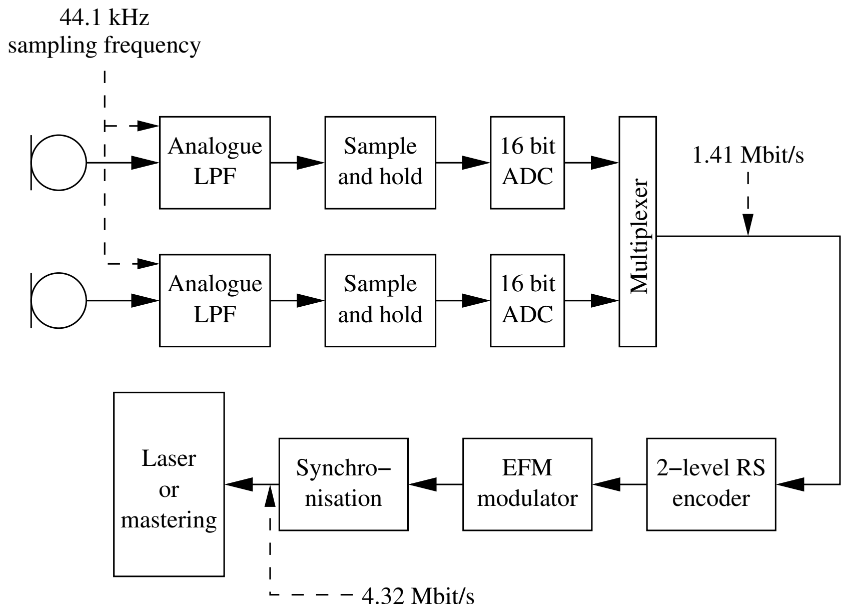
- Speed and cost (DSP design and hardware may be expensive, especially with high bandwidth signals)
- Finite wordlength problems (limited number of bits may cause degradation)

1.2 Applications of DSP

Application areas of DSP are considerable:

- Image processing (pattern recognition, robotic vision, image enhancement, facsimile, satellite weather map, animation)
- Instrumentation and control (spectrum analysis, position and rate control, noise reduction, data compression)
- Speech and audio (speech recognition, speech synthesis, text to speech, digital audio, equalisation)
- Military (secure communication, radar processing, sonar processing, missile guidance)
- Telecommunications (echo cancellation, adaptive equalisation, spread spectrum, video conferencing, data communication)
- Biomedical (patient monitoring, scanners, EEG brain mappers, ECG analysis, X-ray storage and enhancement)

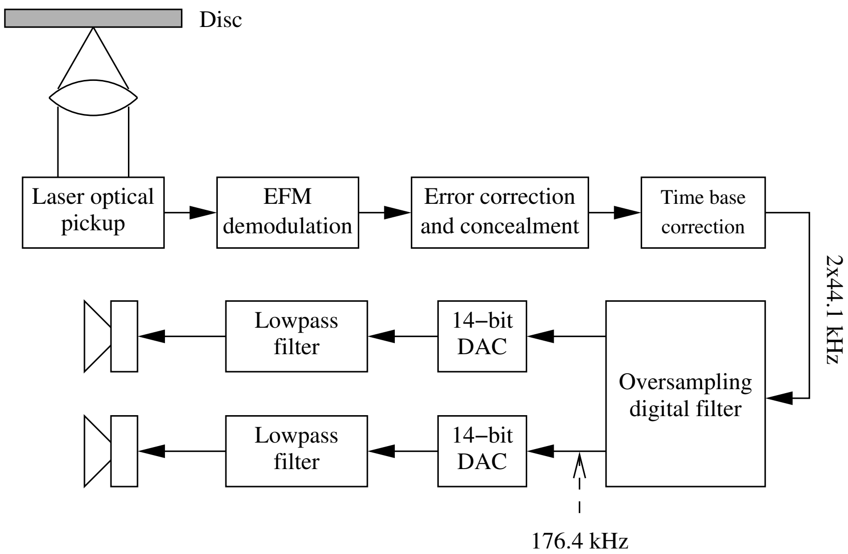
As an (outdated) example, consider audio signal reconstruction in Compact Disks (CDs), where information on a CD is recorded on a spiral track as a succession of bits (10^6 bits/mm^2). The recording or mastering process is depicted below:



This process works as follows:

- Analogue signal in each stereo channel is sampled at 44.1 kHz and digitised to 16 bits (90 dB dynamic range), resulting in 32 bits per sampling instant
- Encoded using a two-level Reed-Solomon code to enable errors to be corrected or concealed during reproduction
- An EFM (eight-to-fourteen) modulation scheme translates each byte in the stream to a 14 bit code, which is more suitable for disc storage (eliminates adjacent 1's, etc.)
- The resulting bit stream is used to control a laser beam, which records information on the disc.

The audio signal reconstruction process is then demonstrated below:

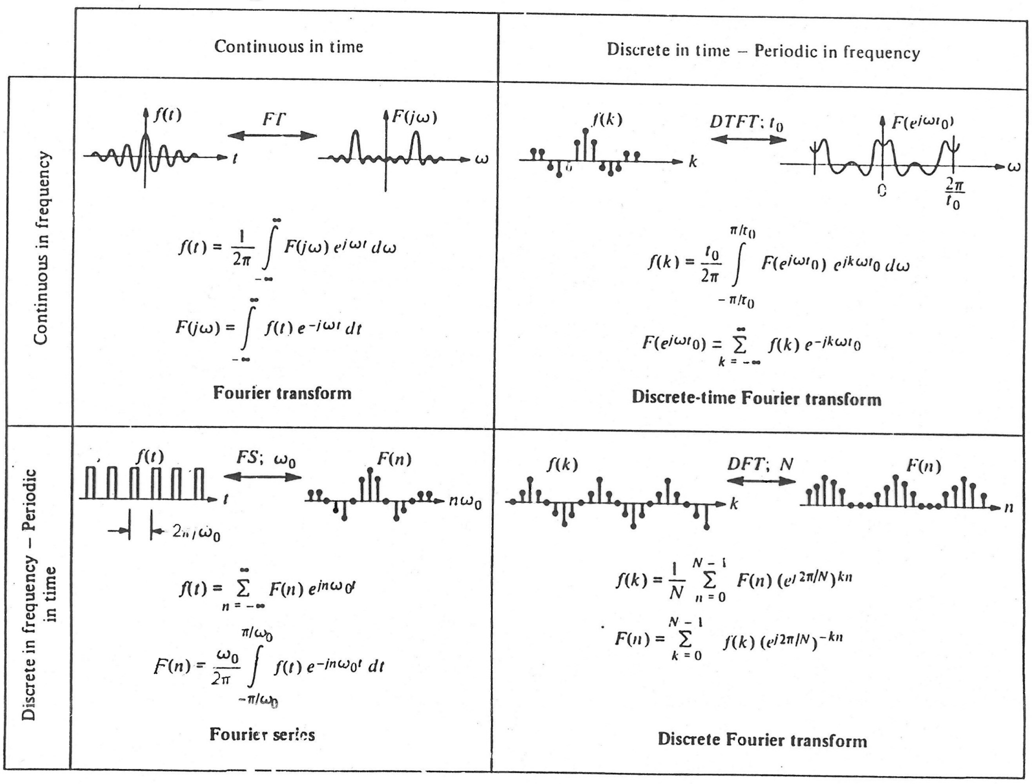


And this is achieved as follows:

- Track optically scanned at 1.2 m/s
- Signal is demodulated, errors detected and (if possible) corrected. If correction is not possible, errors are concealed by interpolation or muting
- This results in a series of 16 bit words, each representing a single audio sample. These samples could be applied directly to a DAC and analogue lowpass filtered.
- However, this would require high specification lowpass filters (20 kHz frequencies must be reduced by 50 dB), and the filter should have linear phase. To avoid this, signals are upsampled by a factor of 4. This makes the output of the DAC smoother, simplifying the analogue filtering requirements.
- The use of a digital filter also allows a linear phase response, reduces chances of intermodulation, and yields a filter that varies with clock rate.

1.3 The Course Ahead

You should remember the following figure from the *Signals and Systems II* course, where we compared four different classes of Fourier transforms.



Four classes of Fourier transform

In a previous course, we introduced the concept of the frequency domain (and Fourier analysis) by discussing the Fourier series – a "transformation" that essentially allowed us to represent a continuous-time, periodic signal as an aperiodic sequence of discrete-frequency impulses. This represented the bottom-left Fourier transform class in the figure.

We then looked at the top-left class, often referred to as simply the "Fourier transform". More accurately, and going forward in this course, we will refer to this as the Continuous-time Fourier Transform (or CTFT) since it transforms continuous-time signals into continuous-frequency representations. This was, for the most part, the extent of what you would have covered on Fourier transforms up to now – with maybe a quick look into the Discrete Fourier Transform.

In this course, we will be examining the two remaining Fourier transform representations; specifically, the rest of these notes are organised as follows:

- Chapter 2 will introduce you to the Discrete-time Fourier Transform, which allows us to represent a discrete-time signal in continuous frequency (the top-right class in the figure)

- Chapter 3 will demonstrate at the concept of a z-transform – effectively the discrete-time equivalent of the Laplace transform (which you would have covered in previous courses); to this end, Chapters 2 and 3 could be seen as a recap of the *Signals and Systems I* course, but all in discrete-time!

- Chapter 4 will feature a deep dive into the theory behind sampling, showing the extensive mathematics underpinning the relationship between continuous-time and discrete-time signals, as well as their frequency representations in the two respective frequency domains

- Chapter 5 will, finally, discuss the Discrete Fourier Transform, which allows us to represent a discrete-time signal in discrete frequency (the bottom-right class in the figure) – arguably the most useful of the Fourier transform classes.

- Chapter 6 will delve into detail on the analysis of systems, and how we can represent them in different forms to glean important information; this ties in with the concepts introduced by the z-transform in Chapter 3

After reading through this chapter, you should now have some of the necessary background to understanding why DSP is important and how it fits into the wider field of electrical engineering. And with this, we can now start getting into the more interesting stuff in the next few chapters, as discussed above – starting with a more detailed overview of digital signals, how we represent them, and how we can manipulate them using frequency-domain methods.

Chapter 2

Discrete-time Signals and Systems

(See Oppenheim and Schafer, Second Edition pages 8–93, or First Edition pages 8–79)

2.1 Introduction

This chapter serves as an overview of discrete-time signals, systems, and their properties, as well as an introduction to the Discrete-time Fourier Transform. Time to start cooking.

2.2 Discrete-time Signals

A discrete-time signal is represented as a sequence of numbers, or a set of values:

$$x = \{x[n]\}, \qquad -\infty < n < \infty.$$

Here n is an integer, and $x[n]$ is the nth sample in the sequence.

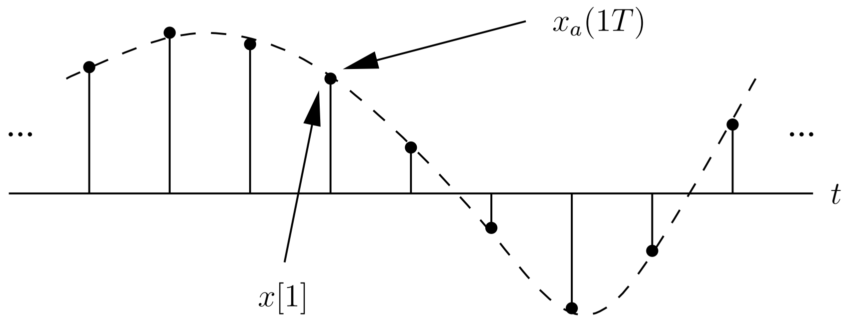
2.2.1 Sampling Theory

Discrete-time signals are often obtained by sampling continuous-time signals. In this case the nth sample of the sequence is equal to the value of the analogue signal $x_a(t)$ at time $t = nT$:

$$x[n] = x_a(nT), \qquad -\infty < n < \infty.$$

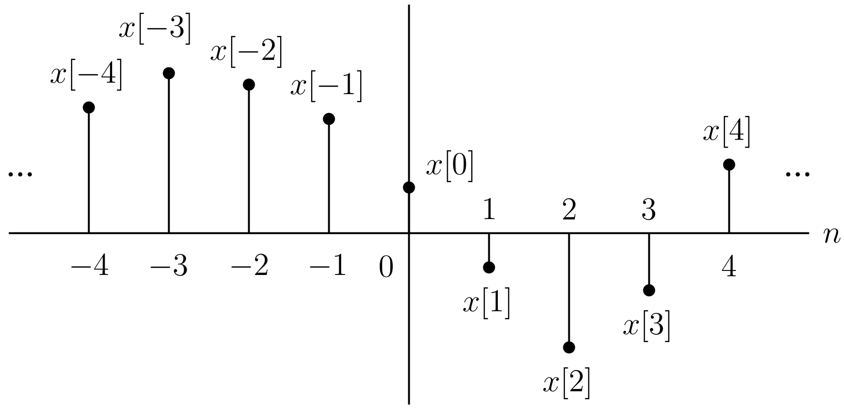
Note that this is *still* an infinitely long signal (or vector), but the discretisation results in it being countable and ordered.

The **sampling period** is then equal to T, and the sampling frequency is $f_s = 1/T$.



For this reason, although $x[n]$ is strictly the nth number in the sequence, we often refer to it as the nth **sample**. We also often refer to "the sequence $x[n]$" when we mean the entire sequence.

Discrete-time signals are often depicted graphically as follows:



These signals can be plotted using the MATLAB function stem, and an important (*crucial*) point is to note that the value $x[n]$ is **undefined** for non-integer values of n – not *zero*.

Later in the course, we will cover sampling theory in significantly more detail – so much so that you may regret ever looking into it...

2.2.2 Sequence Manipulations

Sequences can be manipulated in several ways. The **sum** and **product** of two sequences $x[n]$ and $y[n]$ are defined as the sample-by-sample sum and product respectively. Multiplication of $x[n]$ by a (which can be a real or complex constant) is defined as the pointwise multiplication of each sample value by a.

A sequence $y[n]$ is a **delayed** or **shifted** version of $x[n]$ if

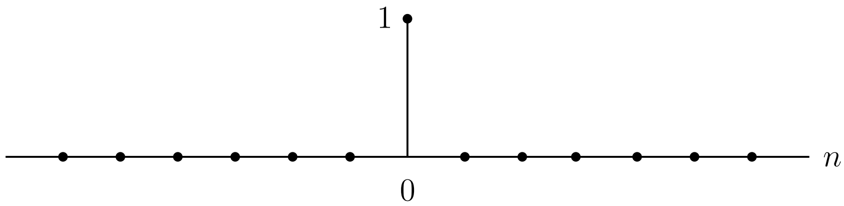
$$y[n] = x[n - n_0],$$

with n_0 an integer.

Logically, many other transformations and manipulations can be performed in the same way as for continuous-time signals; these include operations such as reflection about an axis, scaling the argument n by a constant value to compress or expand the signal in time, and more.

2.2.3 The Unit Impulse

The **unit sample sequence**

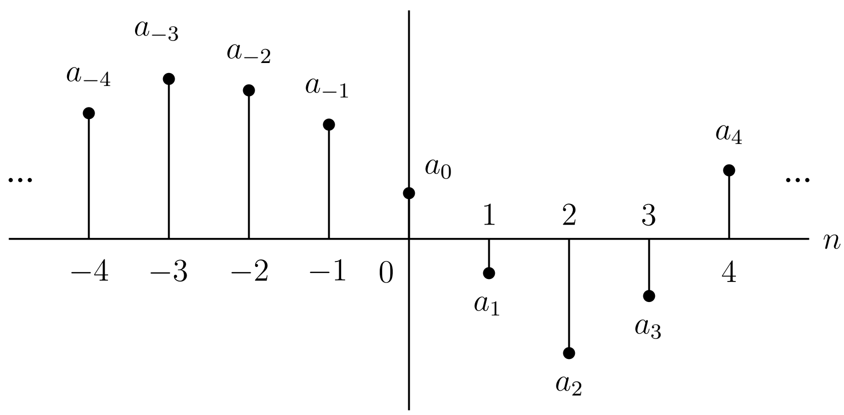


is defined as

$$\delta[n] = \begin{cases} 0 & n \neq 0 \\ 1 & n = 0. \end{cases}$$

This sequence is often referred to as a **discrete-time impulse**, or just an **impulse**. This is the most fundamental digital signal you could possibly imagine, and as you may have predicted, it plays the same role for discrete-time signals as the Dirac delta function does for continuous-time signals. However, there are no mathematical complications in its definition; in fact, it is a whole lot easier to wrap your head around since we are not dealing with infinitesimals and limits. Ah, the beauty of DSP!

An important aspect of the impulse sequence is that an arbitrary sequence can be represented as a sum of scaled, delayed impulses. For example, the sequence



can be represented as

$$x[n] = a_{-4}\delta[n+4] + a_{-3}\delta[n+3] + a_{-2}\delta[n+2] + a_{-1}\delta[n+1] + a_0\delta[n]$$
$$+a_1\delta[n-1] + a_2\delta[n-2] + a_3\delta[n-3] + a_4\delta[n-4].$$

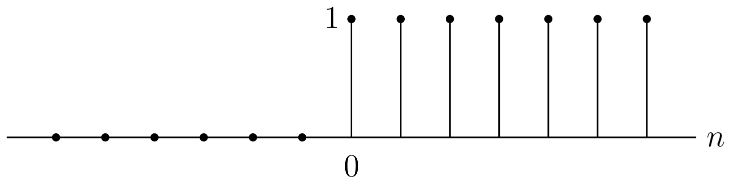
In general, this means that *any* digital signal (or sequence) can be expressed as

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k].$$

This is fundamentally what makes DSP "easier" than continuous-time signal processing, because in times of deep distress and chaos, we can always revert to simplifying any signal down to a linear sum of impulses.

2.2.4 The Unit Step

The **unit step sequence**



is defined as

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0. \end{cases}$$

As you can quickly tell, this is notably simpler than the continuous-time version since we do not need to fret about what the value of the signal is at $n = 0$ – there is no uncertainty or ambiguity here.

Additionally, the unit step is related to the impulse by

$$u[n] = \sum_{k=-\infty}^{n} \delta[k].$$

This should look vaguely familiar – it is basically the same representation we used in continuous time, where an indefinite integration of the delta function resulted in a unit step. Here, instead, an "indefinite" sum of the unit impulse gives us a unit step. Try sketching this graphically and grinding through the maths if you need to convince yourself of this!

Alternatively, if you are scared of summations that end at n, the above expression could instead be expressed as

$$u[n] = \delta[n] + \delta[n-1] + \delta[n-2] + \cdots = \sum_{k=0}^{\infty} \delta[n-k].$$

Conversely, this also means that the unit sample sequence could be expressed as the first *backward difference* of the unit step sequence as follows:

$$\delta[n] = u[n] - u[n-1]$$

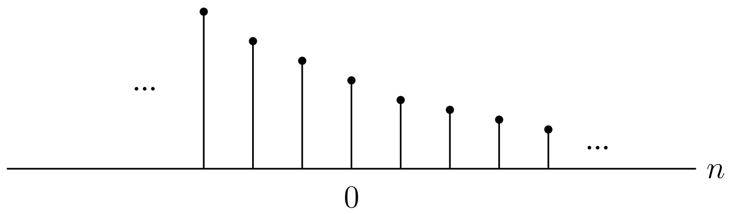
Again, sketch this yourself if you are not yet convinced.

2.2.5 Exponential Sequences

As you should expect, **exponential sequences** are quite important for analysing and representing discrete-time systems. The general form is

$$x[n] = A\alpha^n.$$

If A and α are *real* numbers then the sequence is real. If $0 < \alpha < 1$ and A is positive, then the sequence values are positive and decrease with increasing n:

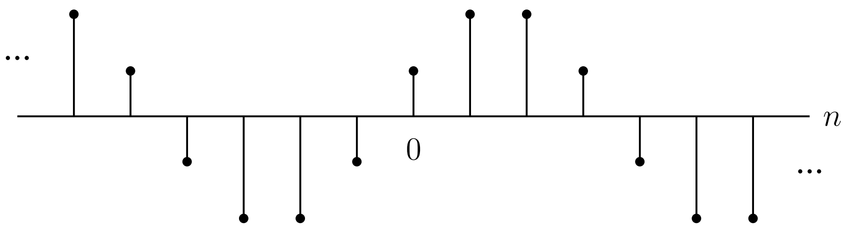


For $-1 < \alpha < 0$ the sequence alternates in sign, but decreases in magnitude. For $|\alpha| > 1$ the sequence grows in magnitude as n increases.

However, note that A and/or α could also be complex-valued. For now, think about what this would mean as we will be discussing it later!

2.2.6 Sinusoidal Sequences

A **sinusoidal sequence**



has the form

$$x[n] = A\cos[\omega_0 n + \phi] \qquad \text{for all } n,$$

with A and ϕ real constants.

The exponential sequence $A\alpha^n$ with complex $\alpha = |\alpha|e^{j\omega_0}$ and $A = |A|e^{j\phi}$ can be expressed as

$$\begin{aligned} x[n] = A\alpha^n = |A|e^{j\phi}|\alpha|^n e^{j\omega_0 n} &= |A||\alpha|^n e^{j(\omega_0 n+\phi)} \\ &= |A||\alpha|^n \cos[\omega_0 n + \phi] + j|A||\alpha|^n \sin[\omega_0 n + \phi], \end{aligned}$$

so the real and imaginary parts are exponentially weighted sinusoids.

When $|\alpha| = 1$ the sequence is called the **complex exponential sequence**:

$$x[n] = |A|e^{j(\omega_0 n+\phi)} = |A|\cos[\omega_0 n + \phi] + j|A|\sin[\omega_0 n + \phi].$$

The **frequency** of this complex sinusoid is ω_0, measured in **radians per sample** (since we are now working with *samples* in the time domain, and not *seconds*). The **phase** of the signal is ϕ.

Since the index n is always an integer, this leads to a few important differences between the properties of discrete-time and continuous-time complex exponentials:

- Consider the complex exponential with frequency $(\omega_0 + 2\pi)$:

  $$x[n] = Ae^{j(\omega_0+2\pi)n} = Ae^{j\omega_0 n}e^{j2\pi n} = Ae^{j\omega_0 n}.$$

  Thus the sequence for the complex exponential with frequency ω_0 is *exactly* the same as that for the complex exponential with frequency $(\omega_0 + 2\pi)$. More generally, complex exponential sequences with frequencies $(\omega_0 + 2\pi r)$, where r is an integer, are indistinguishable from one another. Similarly, for sinusoidal sequences

  $$x[n] = A\cos[(\omega_0 + 2\pi r)n + \phi] = A\cos[\omega_0 n + \phi].$$

- In the continuous-time case, sinusoidal and complex exponential sequences are always periodic. Discrete-time sequences are periodic (with period N) if

  $$x[n] = x[n+N] \qquad \text{for all } n.$$

  Thus the discrete-time sinusoid is only periodic if

  $$A\cos[\omega_0 n + \phi] = A\cos[\omega_0 n + \omega_0 N + \phi],$$

which requires that

$$\omega_0 N = 2\pi k \qquad \text{for } k \text{ and } N \text{ as integers.}$$

The same condition is required for the complex exponential sequence $Ce^{j\omega_0 n}$ to be periodic. If this condition is not met, the discrete sequence will *not* be periodic.

The two factors just described can be combined to reach the conclusion that there are only N distinguishable frequencies for which the corresponding sequences are periodic with period N. One such set is

$$\omega_k = \frac{2\pi k}{N}, \qquad k = 0, 1, \ldots, (N-1).$$

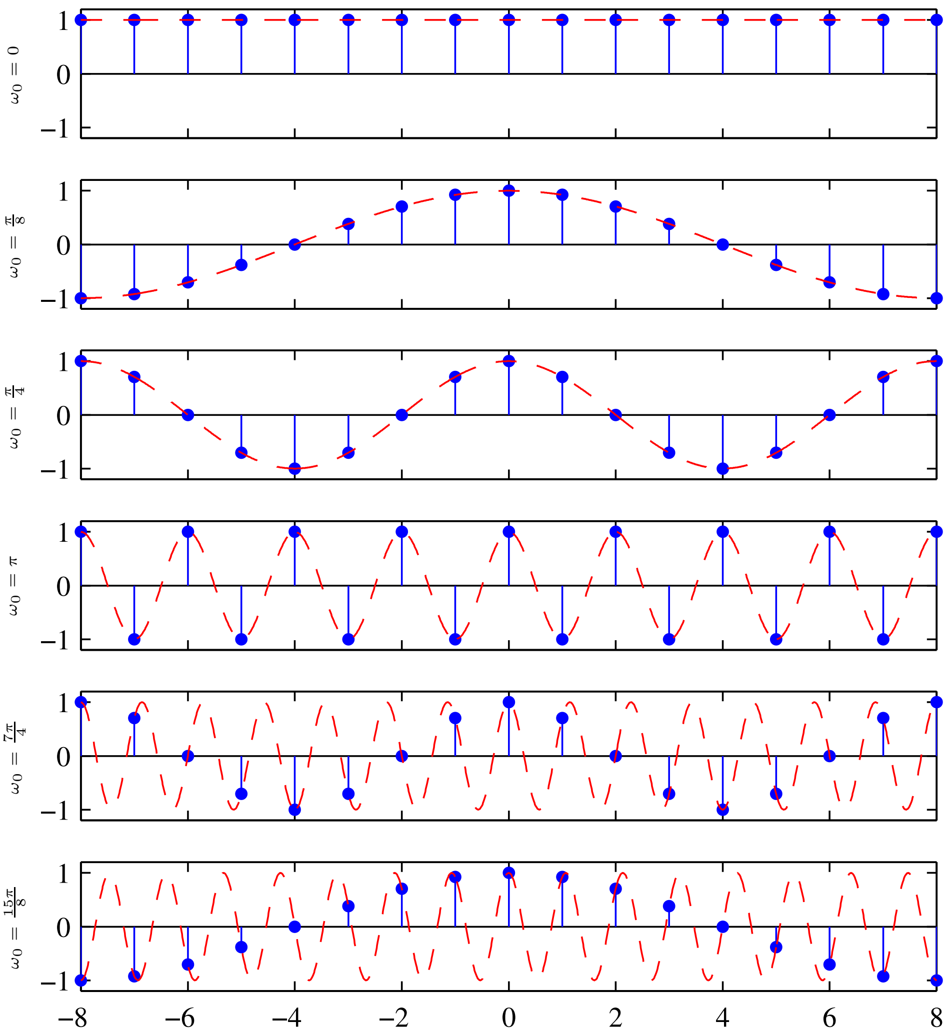
Additionally, for discrete-time sequences, the interpretation of high and low frequencies has to be modified: the discrete-time sinusoidal sequence $x[n] = A\cos[\omega_0 n + \phi]$ oscillates more rapidly as ω_0 increases from 0 to π, but the oscillations become slower as it increases further from π to 2π. This can be seen below, where the sampling rate remains constant but we vary the signal's fundamental frequency ω_0. Note that the red curves denote the sampled continuous signals, and the blue stems denote the digital/discrete sequence that we have sampled.

The first plot demonstrates a constant (DC) signal since there is no varying component ($\omega_0 = 0$), but as we increase the value of ω_0, we see that the signal samples start varying more and more quickly until we reach $\omega_0 = \pi$. At this point, we have the *fastest* possible discrete-time signal – a sequence that varies from positive to negative (or vice-versa) on every single sample.

This illustrates why $\omega_0 = \pi$ is seen as the highest possible frequency, and after this point, increasing ω_0 further actually *decreases* the speed of the signal. This can be seen in the final two plots, where ω_0 has been increased further and yet the resulting discrete sequence is varying more slowly over time (i.e., the signal has more gradual change and smoother slopes). This should make sense since, ultimately, after $\omega_0 = \pi$ we start to wrap around back towards $\omega_0 = 0$.

Thus, the sequence corresponding to $\omega_0 = 0$ is indistinguishable from that with $\omega_0 = 2\pi$, and we can see this by comparing the second and sixth plots – the underlying continuous-time signals are quite different, but the discrete-time sequences are *exactly* the same. We can also notice that the third and fifth discrete-time signals are the same; in both cases, this is due to the symmetry of the underlying cosine.

In general, any frequencies in the vicinity of $\omega_0 = 2\pi k$ for integer k are typically referred to as low frequencies, and those in the vicinity of $\omega_0 = (\pi + 2\pi k)$ are high frequencies.

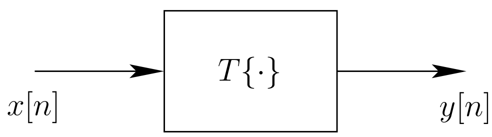


2.3 Discrete-time Systems

A discrete-time system is defined as a transformation or mapping operator that maps an input signal $x[n]$ to an output signal $y[n]$. This can be denoted as

$$y[n] = T\{x[n]\},$$

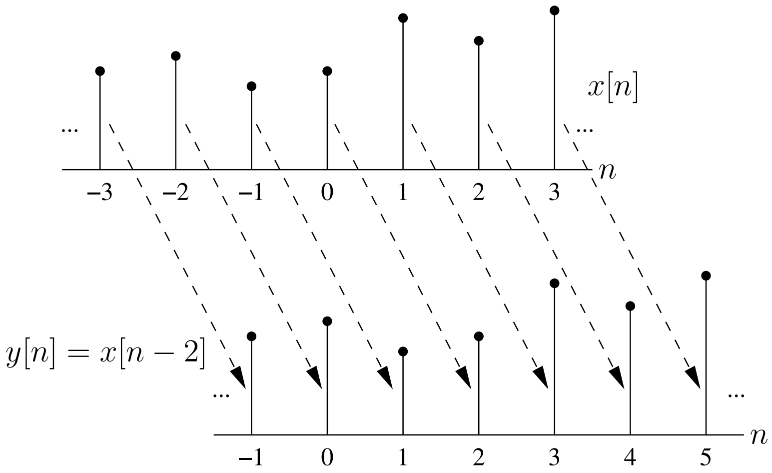
showing how the input $x[n]$ is acted upon by a recipe T to produce an output signal $y[n]$. This can be visualised as:



Example: Ideal delay

$$y[n] = x[n - n_d] :$$

This operation delays the input sequence in time by shifting each value in the signal to the right by n_d samples. For example, for the case $n_d = 2$:



Logically, we can also apply leftward shifts (bringing the signal forward in time) by setting n_d to a negative value.

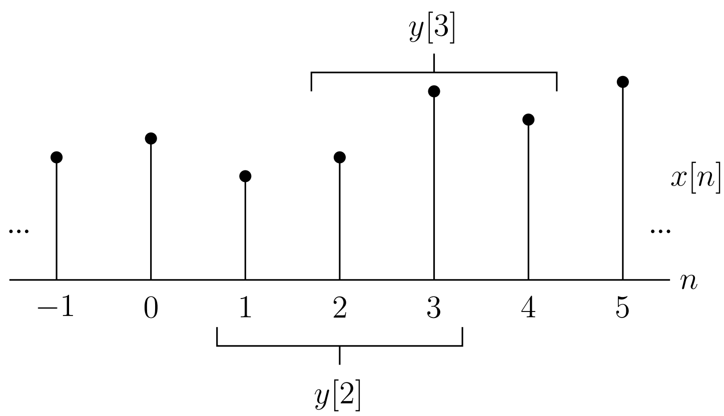
Example: Moving average

A moving-average system has an input-output relationship given by

$$y[n] = \frac{1}{M_1 + M_2 + 1} \sum_{k=-M_1}^{M_2} x[n-k],$$

enabling the system to sum multiple samples in a row and then divide by the number of samples that were summed. This is repeated for various values of k, essentially applying a moving-average operation over a specific range of the signal.

For example, for $M_1 = 1$ and $M_2 = 1$, the input sequence



yields an output with

$$\begin{aligned}
&\vdots \\
y[2] &= \frac{1}{3}(x[1] + x[2] + x[3]) \\
y[3] &= \frac{1}{3}(x[2] + x[3] + x[4]) \\
&\vdots
\end{aligned}$$

As such, we are averaging every three samples as we move through $x[n]$. However, notice that this implementation of a moving-average system produces non-causal terms; in the above example, we can only calculate $y[2]$ by including $x[3]$ in the computation – which is in the future relative to the output sample being calculated.

In general, systems can be classified (in terms of causality, stability, etc.) by placing constraints on the transformation $T\{\cdot\}$. The next few subsections will delve into more detail on these properties.

2.3.1 System Memory

A system is **memoryless** if the output $y[n]$ depends only on $x[n]$ at the same n. For example, $y[n] = (x[n])^2$ is memoryless, but the ideal delay $y[n] = x[n-2]$ is not. This system is said to have **memory** since it uses knowledge of previous input(s) to compute the present output.

2.3.2 Causality

A system is **causal** if the output at n depends only on the input *at n and earlier inputs*. For example, the backward difference system is causal:

$$y[n] = x[n] - x[n-1]$$

But the forward difference system is not

$$y[n] = x[n+1] - x[n],$$

since we need future input knowledge to compute the current output. If we can provide even a single counter-example as to why a system is non-causal, we can immediately prove that the system is not causal – but *proving* causality is a bit more involved since we need to prove it for all possible values of n.

2.3.3 Linear Systems

A system is **linear** if the principle of superposition applies. Thus if $y_1[n]$ is the response of the system to the input $x_1[n]$, and $y_2[n]$ the response to $x_2[n]$, then linearity implies the following two system properties

- **Additivity:**

$$T\{x_1[n] + x_2[n]\} = T\{x_1[n]\} + T\{x_2[n]\} = y_1[n] + y_2[n]$$

- **Scaling:**

$$T\{ax_1[n]\} = aT\{x_1[n]\} = ay_1[n].$$

These properties combine to form the general principle of superposition

$$T\{ax_1[n] + bx_2[n]\} = aT\{x_1[n]\} + bT\{x_2[n]\} = ay_1[n] + by_2[n],$$

where a linear of combination of inputs to the system produces a linear combination of outputs. In all cases, a and b are arbitrary constants, and this property generalises to many inputs; e.g., the response of a linear system to $x[n] = \sum_k a_k x_k[n]$ will be $y[n] = \sum_k a_k y_k[n]$. This is highly exploitable as it allows us to decompose complicated signals into linear combinations of primitives, each of which can be transformed by the system and then combined at the output.

2.3.4 Time-invariant Systems

A system is **time invariant** if a time shift or delay of the input sequence causes a corresponding shift in the output sequence. That is, if $y[n]$ is the response to $x[n]$, then $y[n - n_0]$ is the response to $x[n - n_0]$.

For example, the accumulator system

$$y[n] = \sum_{k=-\infty}^{n} x[k]$$

is time invariant, but the compressor system

$$y[n] = x[Mn]$$

for M a positive integer (which selects every Mth sample from a sequence) is not.

2.3.5 Stability

A system is **stable** if every bounded input sequence produces a bounded output sequence:

- **Bounded input:** $|x[n]| \leq B_x < \infty$
- **Bounded output:** $|y[n]| \leq B_y < \infty$.

For instance, if the maximum possible (absolute) value in an output signal is 1000 (for any finite-valued, or bounded, input), then this output is bounded since it does not diverge to infinity. As such, the system cannot produce an unbounded output for any bounded input, and it is thus stable.

For example, the accumulator

$$y[n] = \sum_{k=-\infty}^{n} x[k]$$

is an example of an *unbounded* system, since its response to the unit step $u[n]$ (a bounded input with a maximum value of 1) is

$$y[n] = \sum_{k=-\infty}^{n} u[k] = \begin{cases} 0 & n < 0 \\ n+1 & n \geq 0, \end{cases}$$

which has no finite upper bound – as n approaches infinity, so will the system output.

System stability can be computed directly using the following inequality

$$S = \sum_{k=-\infty}^{\infty} |h[k]| < \infty.$$

If this condition is met, and the absolute sum of the impulse response of a system converges, then the system is stable. The concept of an impulse response is further discussed in the next section, but a few examples of testing system stability are shown below:

- The **ideal delay** system $h[n] = \delta[n - n_d]$ is *stable* since $S = 1 < \infty$
- The **moving average** system

$$\begin{aligned} h[n] &= \frac{1}{M_1 + M_2 + 1} \sum_{k=-M_1}^{M_2} \delta[n-k] \\ &= \begin{cases} \frac{1}{M_1+M_2+1} & -M_1 \leq n \leq M_2 \\ 0 & \text{otherwise,} \end{cases} \end{aligned}$$

as well as the **forward difference** system $h[n] = \delta[n+1] - \delta[n]$, and the **backward difference** system $h[n] = \delta[n] - \delta[n-1]$ are each *stable* since S is the sum of a finite number of finite samples, and is therefore less than ∞

- The **accumulator** system

$$\begin{aligned} h[n] &= \sum_{k=-\infty}^{n} \delta[k] \\ &= \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases} \\ &= u[n] \end{aligned}$$

is *unstable* since $S = \sum_{n=0}^{\infty} u[n] = \infty$. It is also worth noting that the impulse response of an accumulator system is simply a unit step!

2.4 Linear Time-invariant Systems

If the linearity property is combined with the representation of a general sequence as a linear combination of delayed impulses, then it follows that a linear time-invariant (LTI) system can be completely characterised by its **impulse response**. LTI systems are generally what we call "nicely behaved systems", and are what we typically deal with in this course (as you have seen in previous courses).

2.4.1 The Convolution Sum

Suppose $h_k[n]$ is the response of a linear system to the impulse $\delta[n-k]$ at $n = k$. Since

$$y[n] = T\left\{ \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] \right\},$$

and T only operates on time-domain signals in n, we can pull $x[k]$ out of the transformation. The principle of superposition thus means that

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]T\{\delta[n-k]\} = \sum_{k=-\infty}^{\infty} x[k]h_k[n].$$

If the system is additionally time invariant, then the response to $\delta[n-k]$ is $h[n-k]$. The previous equation then becomes

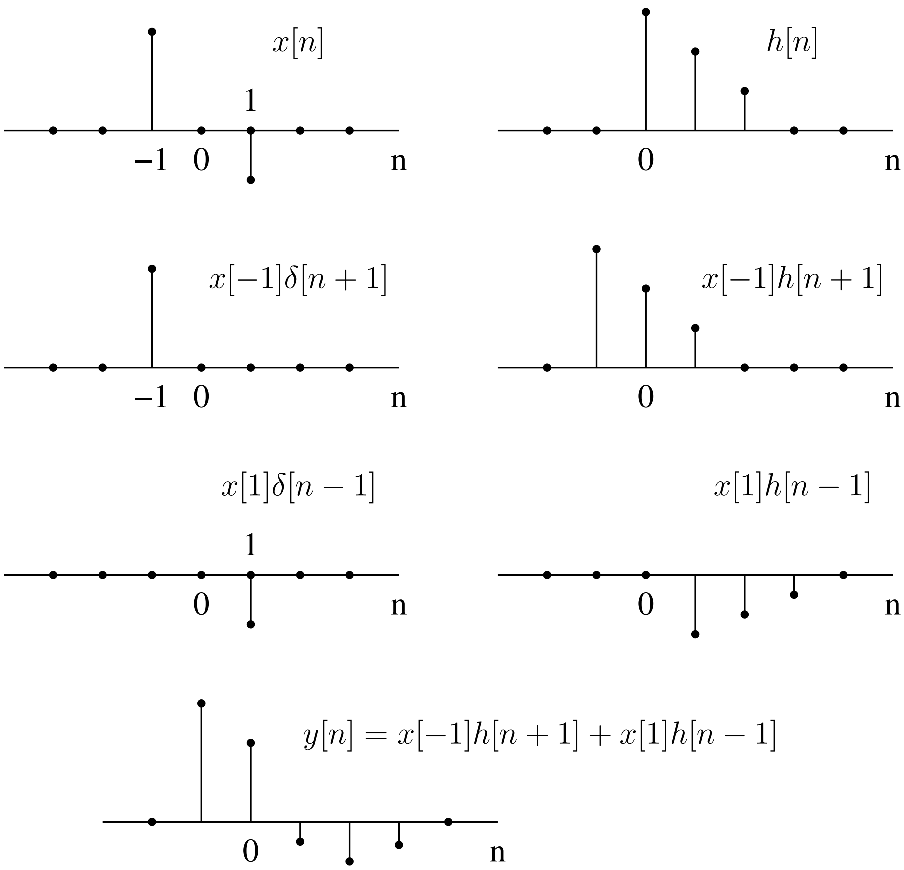
$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k].$$

This expression is called the **convolution sum** and should look familiar – yes, everyone's favourite operation makes a return in DSP too! An LTI system thus has the property that, given $h[n]$, we can find $y[n]$ for *any* input $x[n]$. Alternatively, $y[n]$ is said to be the **convolution** of $x[n]$ with $h[n]$, denoted as follows:

$$y[n] = x[n] * h[n].$$

2.4.2 Convolution Approaches

The previous derivation suggests the interpretation that the input sample at $n = k$, represented by $x[k]\delta[n-k]$, is transformed by the system into an output sequence $x[k]h[n-k]$. For each k, these sequences are superimposed to yield the overall output sequence:

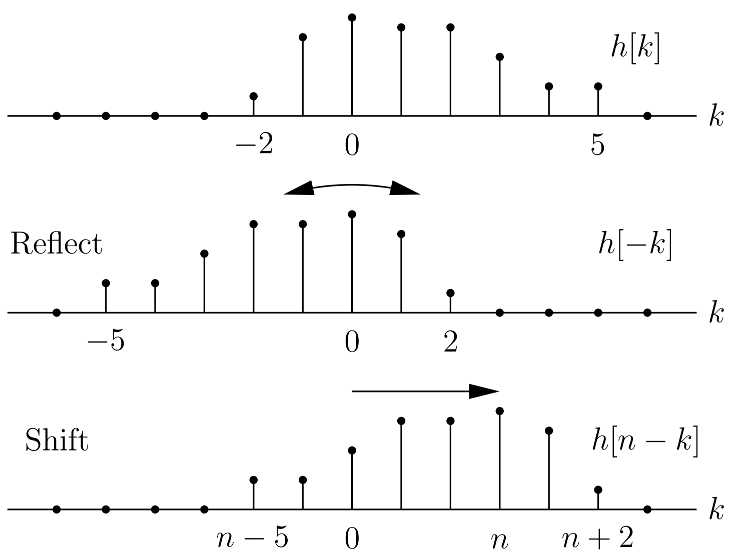


This is super easy (*barely an inconvenience!*) compared to the continuous-time approach to convolution – primarily because we avoid integrations, but also because we know that we can break down any discrete-time signal into a linear combination of scaled and shifted impulses, and each of these impulses will produce a scaled and shifted version of $h[n]$, so the output will simply be a linear combination of scaled and shifted impulse responses!

A slightly different interpretation leads to another convenient computational form: the nth value of the output, namely $y[n]$, is obtained by multiplying the input sequence (expressed as a function of k) by the sequence with values $h[n-k]$, and then summing all the values of the products $x[k]h[n-k]$. The key to this method is in understanding how to form the sequence $h[n-k]$ for all values of n of interest.

To this end, note that $h[n-k] = h[-(k-n)]$. The sequence $h[-k]$ is seen to be equivalent to the sequence $h[k]$ reflected around the origin. The sequence $h[n-k]$ is then obtained by shifting the origin of the sequence to $k = n$.

To implement discrete-time convolution, the sequences $x[k]$ and $h[n-k]$ are multiplied together for $-\infty < k < \infty$, and the products summed to obtain the value of the output sample $y[n]$. To obtain another output sample, the procedure is repeated with the origin shifted to the new sample position.



Note that this is the same as the usual "flip and shift" approach you have been practising in continuous-time for the past two years – but easier!

Example: Analytical evaluation of the convolution sum

Consider the output of a system with impulse response

$$h[n] = \begin{cases} 1 & 0 \leq n \leq N-1 \\ 0 & \text{otherwise} \end{cases}$$

to the input $x[n] = a^n u[n]$. To find the output at n, we must form the sum over all k of the product $x[k]h[n-k]$.

Since the sequences are non-overlapping for all negative n, the output must be zero:

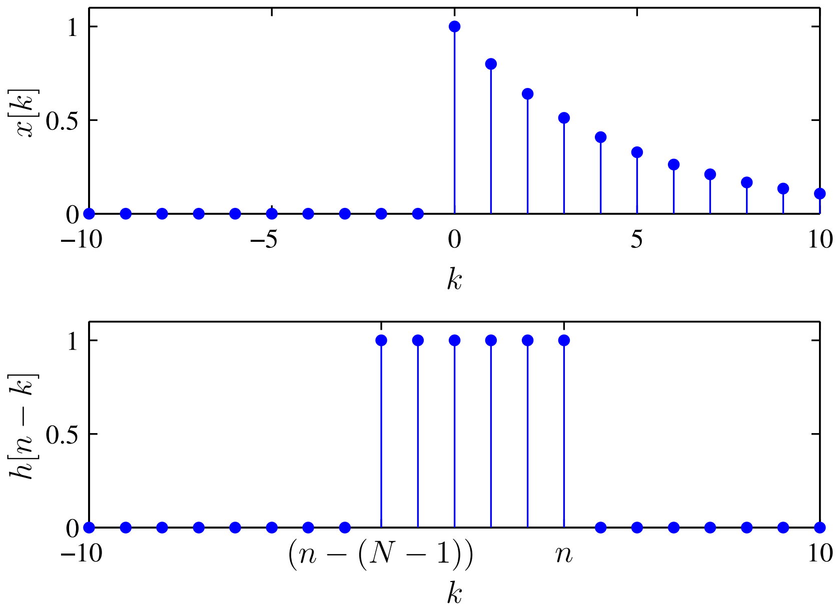
$$y[n] = 0, \qquad n < 0.$$

For $0 \leq n \leq N-1$ the product terms in the sum are $x[k]h[n-k] = a^k$, so it follows that

$$y[n] = \sum_{k=0}^{n} a^k, \qquad 0 \leq n \leq N-1.$$

Finally, for $n > N-1$, the rectangular pulse is fully "contained" within $x[k]$ and thus the product terms are $x[k]h[n-k] = a^k$ (same as before), but the lower limit on the sum is now $n-N+1$. Therefore

$$y[n] = \sum_{k=n-N+1}^{n} a^k, \qquad n > N-1.$$



Note that both of the above sums can be simplified by considering each one as a finite sum of a geometric sequence, so

$$\sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a}, \qquad a \neq 1.$$

This is a fundamental equation that we will use quite frequently, so get comfortable with it and see if you can work out the above two summations using this expression.

Note, of course, that the second summation does not start at zero, so think about how we could simplify this!

On another note, it is worth noting that the geometric sum could also be extended for infinitely long sequences, in which case we take the limit as N approaches infinity:

$$\lim_{N \to \infty} \sum_{n=0}^{N-1} a^n = \sum_{n=0}^{\infty} a^n = \frac{1}{1-a}, \qquad |a| < 1.$$

2.5 Properties of LTI Systems

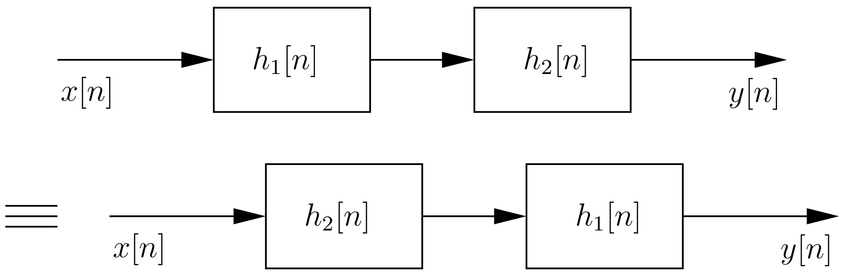
2.5.1 Basic Properties

All LTI systems are described by the convolution sum

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k].$$

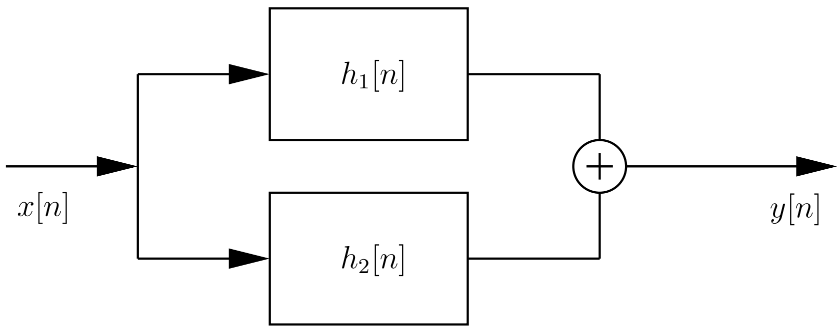
Some properties of LTI systems can therefore be found by considering the properties of the convolution operation:

- **Commutativity:** $x[n] * h[n] = h[n] * x[n]$
- **Cascade connection:**



  $y[n] = h[n] * x[n] = h_1[n] * h_2[n] * x[n] = h_2[n] * h_1[n] * x[n]$. Note that this is the same as **associativity**.

- **Parallel connection:**



$y[n] = (h_1[n] + h_2[n]) * x[n] = h_p[n] * x[n]$. Note that this is the same as **distributivity** over addition.

- Another important property relates a system's impulse response to causality. An LTI system is causal if and only if $h[n] = 0$ for $n < 0$. The ideal delay system is causal if $n_d \geq 0$; the moving average system is causal if $-M_1 \geq 0$ and $M_2 \geq 0$; the accumulator and backward difference systems are causal; the forward difference system is non-causal.

2.5.2 FIR and IIR Systems

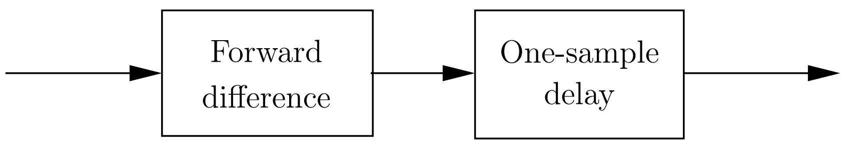
Systems with only a finite number of nonzero values in $h[n]$ are called **finite duration impulse response (FIR)** systems. FIR systems are stable if each impulse response value is finite. The ideal delay, the moving average, and the forward and backward difference described above fall into this class. **Infinite impulse response (IIR)** systems, such as the accumulator system, are more difficult to analyse. For example, the accumulator system is unstable, but the IIR system

$$h[n] = a^n u[n], \qquad |a| < 1$$

is stable since

$$S = \sum_{n=0}^{\infty} |a^n| \leq \sum_{n=0}^{\infty} |a|^n = \frac{1}{1 - |a|} < \infty$$

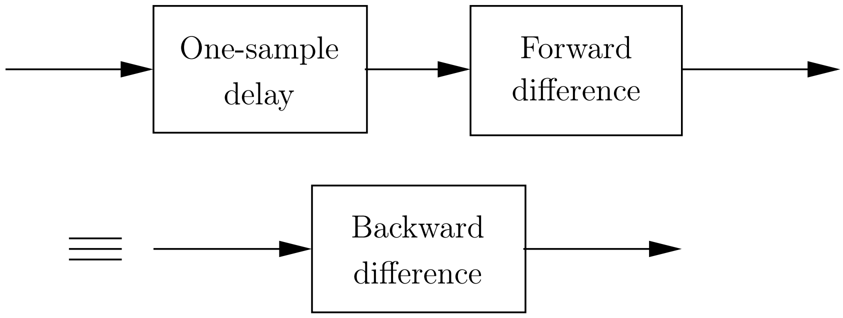
(it is the sum of an infinite geometric series).
Consider the system



which has

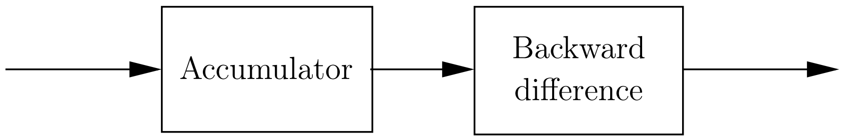
$$\begin{aligned} h[n] &= (\delta[n+1] - \delta[n]) * \delta[n-1] \\ &= \delta[n-1] * \delta[n+1] - \delta[n-1] * \delta[n] \\ &= \delta[n] - \delta[n-1]. \end{aligned}$$

This is the impulse response of a backward difference system:



Here a non-causal system has been converted to a causal one by cascading with a delay. In general, *any non-causal FIR system can be made causal by cascading with a sufficiently long delay.*

Consider the system consisting of an accumulator followed by a backward difference:



The impulse response of this system is

$$h[n] = u[n] * (\delta[n] - \delta[n-1]) = u[n] - u[n-1] = \delta[n].$$

The output is therefore equal to the input because $x[n] * \delta[n] = x[n]$. Thus the backward difference exactly compensates for (or inverts) the effect of the accumulator — the backward difference system is the **inverse system** for the accumulator, and vice versa. We define this inverse relationship for all LTI systems:

$$h[n] * h_i[n] = \delta[n],$$

but note that an inverse system does not always exist for every system. We will discuss inverse systems in more detail in later chapters!

2.6 Linear Constant Coefficient Difference Equations

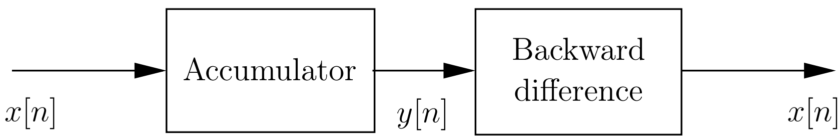
2.6.1 Form of LCCDEs

Some LTI systems can be represented in terms of linear constant coefficient difference equations (LCCDEs) – similarly to how we had linear constant coefficient differential equations back in the continuous-time case. LCCDEs always follow the form

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{m=0}^{M} b_m x[n-m].$$

Example: Difference equation representation of the accumulator

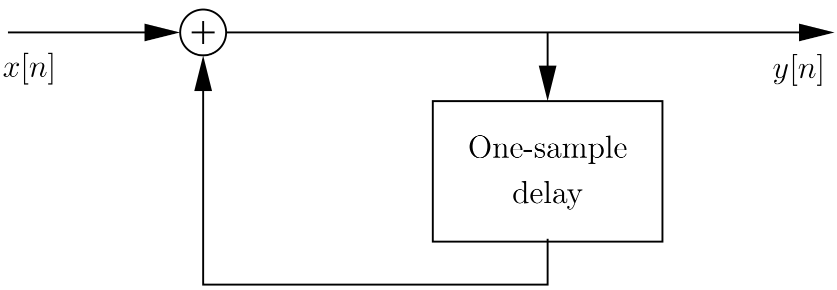
Take, for example, the accumulator:



Here $y[n] - y[n-1] = x[n]$, which can be written in the desired form with $N = 1$, $a_0 = 1$, $a_1 = -1$, $M = 0$, and $b_0 = 1$. Rewriting as

$$y[n] = y[n-1] + x[n]$$

we obtain the **recursion representation**



where at n we add the current input $x[n]$ to the previously accumulated sum $y[n-1]$.

Example: Difference equation representation of moving average

Consider now the moving average system with $M_1 = 0$:

$$h[n] = \frac{1}{M_2+1}(u[n] - u[n-M_2-1]).$$

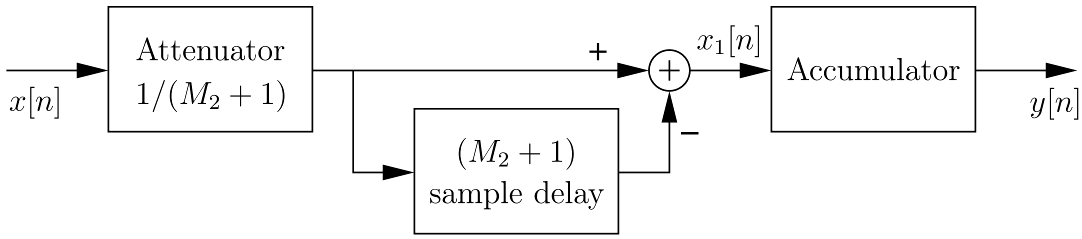
The output of the system is

$$y[n] = \frac{1}{M_2+1}\sum_{k=0}^{M_2} x[n-k],$$

which is an LCCDE with $N = 0$, $a_0 = 1$, and $M = M_2$, $b_k = 1/(M_2+1)$. Using the sifting property of $\delta[n]$,

$$h[n] = \frac{1}{M_2+1}(\delta[n] - \delta[n-M_2-1]) * u[n]$$

so



Here $x_1[n] = 1/(M_2+1)(x[n]-x[n-M_2-1])$ and for the accumulator $y[n]-y[n-1] = x_1[n]$. Therefore

$$y[n] - y[n-1] = \frac{1}{M_2+1}(x[n] - x[n - M_2 - 1]),$$

which is again a (different) LCCDE with $N = 1$, $a_0 = 1$, $a_1 = -1$, $b_0 = -b_{M_2+1} = 1/(M_2 + 1)$.

2.6.2 Initial Conditions

As for constant coefficient differential equations in the continuous case, without additional information or constraints, an LCCDE does not provide a *unique* solution for the output given an input. Specifically, suppose we have the particular output $y_p[n]$ for the input $x_p[n]$. The same equation then has the solution

$$y[n] = y_p[n] + y_h[n],$$

where $y_h[n]$ is any solution with $x[n] = 0$. That is, $y_h[n]$ is a **homogeneous solution** to the **homogeneous equation**

$$\sum_{k=0}^{N} a_k y_h[n-k] = 0,$$

i.e., when there is no input, the system produces $y_h[n]$ as an output; this can therefore be seen as the natural system output when no input is being actively provided. Through linearity, we can also think about this as

$$\begin{aligned} x[n] &\to y_p[n] \\ 0 &\to y_h[n] \\ x[n] + 0 &\to y_p[n] + y_h[n], \end{aligned}$$

and thus any particular solution can be added to the homogeneous solution to produce another valid solution.

It can thus be shown that there are N non-zero solutions to the standard form LCCDE, so a set of N auxiliary conditions are required for a unique specification of $y[n]$ for a given $x[n]$.

Note: If a system is LTI *and causal*, then the initial conditions are known as **initial rest** conditions since the system only starts responding after $n = 0$, and a unique solution can be obtained.

2.7 The Discrete-time Fourier transform

The Fourier transform considered here is, strictly speaking, what we call the **Discrete-time Fourier transform (DTFT)**, although Oppenheim and Schafer call it just the Fourier transform. Its properties are recapped here (with examples) to show nomenclature.

2.7.1 Frequency Response

Complex exponentials

$$x[n] = e^{j\omega n}, \qquad -\infty < n < \infty$$

are eigenfunctions of LTI systems:

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]e^{j\omega(n-k)} = e^{j\omega n}\left(\sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k}\right).$$

Defining

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k}$$

we have that $y[n] = H(e^{j\omega})e^{j\omega n} = H(e^{j\omega})x[n]$. Therefore, $e^{j\omega n}$ is an eigenfunction of the system, and $H(e^{j\omega})$ is the associated eigenvalue.

The quantity $H(e^{j\omega})$ is the DTFT of the impulse response, and we call this the **frequency response** of the system. As a complex quantity, this can also be written as

$$H(e^{j\omega}) = H_R(e^{j\omega}) + jH_I(e^{j\omega}) = |H(e^{j\omega})|e^{j\sphericalangle H(e^{j\omega})}.$$

Example: Frequency response of ideal delay:

Consider the input $x[n] = e^{j\omega n}$ to the ideal delay system $y[n] = x[n - n_d]$: the output is

$$y[n] = e^{j\omega(n-n_d)} = e^{-j\omega n_d} e^{j\omega n}.$$

The frequency response is therefore

$$H(e^{j\omega}) = e^{-j\omega n_d}.$$

Alternatively, for the ideal delay $h[n] = \delta[n - n_d]$,

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \delta[n - n_d] e^{-j\omega n} = e^{-j\omega n_d}$$

through the **sifting** property, as the summand will only be non-zero where the impulse is non-zero (which is at $n = n_d$).

The real and imaginary parts of the frequency response are $H_R(e^{j\omega}) = \cos(\omega n_d)$ and $H_I(e^{j\omega}) = -\sin(\omega n_d)$, or alternatively

$$\begin{aligned} |H(e^{j\omega})| &= 1 \\ \sphericalangle H(e^{j\omega}) &= -\omega n_d. \end{aligned}$$

2.7.2 Periodicity of the DTFT

The frequency response of a LTI system is essentially the same for continuous and discrete time systems. However, an important distinction is that in the discrete case it is *always* periodic in frequency with a period 2π:

$$\begin{aligned} H(e^{j(\omega+2\pi)}) &= \sum_{n=-\infty}^{\infty} h[n] e^{-j(\omega+2\pi)n} \\ &= \sum_{n=-\infty}^{\infty} h[n] e^{-j\omega n} e^{-j2\pi n} \\ &= \sum_{n=-\infty}^{\infty} h[n] e^{-j\omega n} = H(e^{j\omega}). \end{aligned}$$

This last result holds since $e^{\pm j2\pi n} = 1$ for integer n. (As an aside, this is actually important to note because the same equation does not hold up in continuous time; believe it or not, $e^{-j2\pi t}$ is actually *not* always equal to 1 due to some funkiness that arises with complex numbers when you really dig into the hardcore maths underlying this!)

This property is quite crucial as it means that a DTFT is always periodic and always has a period of 2π in frequency. The reason for this periodicity is related to the observation that the sequence

$$\left\{e^{-j\omega n}\right\}, \qquad -\infty < n < \infty$$

has exactly the same values as the sequence

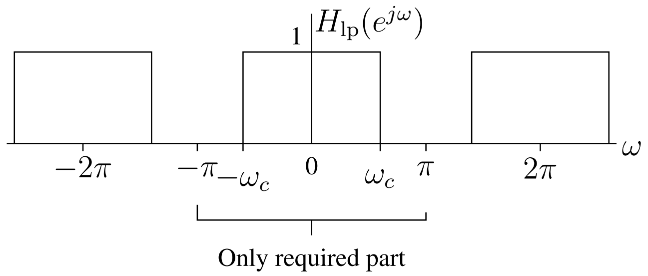
$$\left\{e^{-j(\omega+2\pi)n}\right\}, \qquad -\infty < n < \infty.$$

A system will therefore respond in exactly the same way to both sequences.

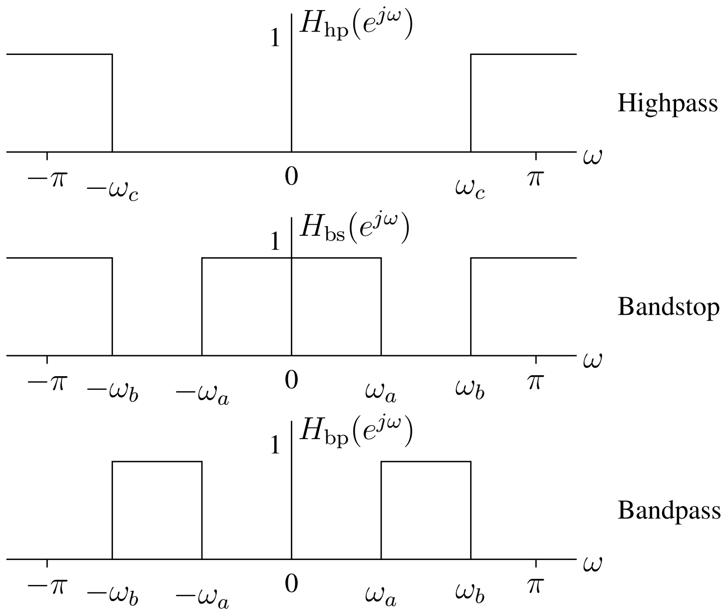
This is also an interesting observation as it represents a kind of duality with Fourier series. If you remember, a Fourier series representation essentially took a continuous-time, periodic signal and determined its discrete frequency content; here, however, we see the opposite as a discrete-time signal is being transformed into a continuous-frequency, periodic representation!

2.7.3 Basic Filters

The frequency response of an ideal lowpass filter is as follows:

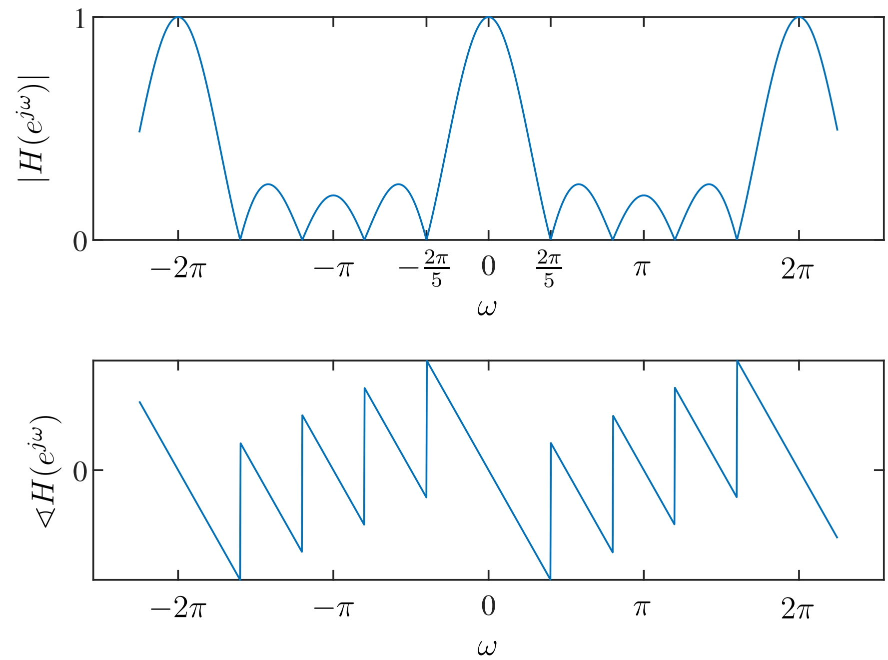


Due to the periodicity in the response, it is only necessary to consider one frequency cycle, usually chosen to be the range $-\pi$ to π. Other examples of ideal filters are:



In these cases it is implied that the frequency response repeats with period 2π outside of the plotted interval.

Example: Frequency response of a moving-average system
The frequency response of a moving average system can be plotted as follows:



This system is represented by the impulse response

$$h[n] = \begin{cases} \frac{1}{M_1+M_2+1} & -M_1 \leq n \leq M_2 \\ 0 & \text{otherwise,} \end{cases}$$

with the above plot corresponding to $M_1 = 0$ and $M_2 = 4$.

Note how both plots are periodic over 2π, and how this system attenuates high frequencies (around $\omega = \pi$) and therefore has the behaviour of a lowpass filter. This is a more realistic look at what a lowpass filter's frequency response would look like, showing quite strong sidelobes ("spectral leakage") present between the main lobes.

2.7.4 Fourier Transforms of Discrete Sequences

The DTFT of the sequence $x[n]$ is

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}.$$

This is analogous to the Fourier series analysis equation, and it is usually called the **forward transform** or **analysis** equation. The **inverse transform**, or **synthesis** formula, is given by the Fourier integral

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega,$$

but this is quite rarely used in this form.

The Fourier transform is generally a complex-valued function of ω:

$$X(e^{j\omega}) = X_R(e^{j\omega}) + jX_I(e^{j\omega}) = |X(e^{j\omega})| e^{j\sphericalangle X(e^{j\omega})}.$$

The quantities $|X(e^{j\omega})|$ and $\sphericalangle X(e^{j\omega})$ are referred to as the **magnitude** and **phase** of the Fourier transform. The Fourier transform is often referred to as the **Fourier spectrum**.

Since the frequency response of a LTI system is given by

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h[k] e^{-j\omega k},$$

it is clear that the frequency response is equivalent to the Fourier transform of the impulse response, and the impulse response is

$$h[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H(e^{j\omega}) e^{j\omega n} d\omega.$$

A sufficient condition for the existence of the DTFT of a sequence $x[n]$ is that it be absolutely summable: $\sum_{n=-\infty}^{\infty} |x[n]| < \infty$. In other words, the Fourier transform exists if the sum $\sum_{n=-\infty}^{\infty} |x[n]|$ converges. However, there are *edge cases* where a Fourier transform may exist for sequences where this is not true — such as in the case of **generalised functions**. But in this course, we will treat absolute summability as a sufficient condition for the existence of a sequence's DTFT.

2.7.5 Symmetry Properties

Any sequence $x[n]$ can be expressed as

$$x[n] = x_e[n] + x_o[n],$$

where $x_e[n]$ is **conjugate symmetric** ($x_e[n] = x_e^*[-n]$) and $x_o[n]$ is **conjugate anti-symmetric** ($x_o[n] = -x_o^*[-n]$). These two components of the sequence can be obtained as:

$$x_e[n] = \frac{1}{2}(x[n] + x^*[-n]) = x_e^*[-n]$$
$$x_o[n] = \frac{1}{2}(x[n] - x^*[-n]) = -x_o^*[-n].$$

If a real sequence is conjugate symmetric, then it is an **even** sequence (i.e., symmetric about the origin), and if conjugate antisymmetric, then it is **odd** (i.e., antisymmetric about the origin).

Similarly, the Fourier transform $X(e^{j\omega})$ can be decomposed into a sum of conjugate symmetric and antisymmetric parts:

$$X(e^{j\omega}) = X_e(e^{j\omega}) + X_o(e^{j\omega}),$$

where

$$X_e(e^{j\omega}) = \frac{1}{2}[X(e^{j\omega}) + X^*(e^{-j\omega})]$$
$$X_o(e^{j\omega}) = \frac{1}{2}[X(e^{j\omega}) - X^*(e^{-j\omega})].$$

With these definitions, and letting

$$X(e^{j\omega}) = X_R(e^{j\omega}) + jX_I(e^{j\omega}),$$

the symmetry properties of the Fourier transform can be summarised as follows:

Sequence $x[n]$	Transform $X(e^{j\omega})$
$x^*[n]$	$X^*(e^{-j\omega})$
$x^*[-n]$	$X^*(e^{j\omega})$
$\mathrm{Re}\{x[n]\}$	$X_e(e^{j\omega})$
$j\mathrm{Im}\{x[n]\}$	$X_o(e^{j\omega})$
$x_e[n]$	$X_R(e^{j\omega})$
$x_o[n]$	$jX_I(e^{j\omega})$

Most of these properties can be proved by substituting into the expression for the Fourier transform. Additionally, for real $x[n]$ the following also hold:

Real sequence $x[n]$	Transform $X(e^{j\omega})$
$x[n]$	$X(e^{j\omega}) = X^*(e^{-j\omega})$
$x[n]$	$X_R(e^{j\omega}) = X_R(e^{-j\omega})$
$x[n]$	$X_I(e^{j\omega}) = -X_I(e^{-j\omega})$
$x[n]$	$\lvert X(e^{j\omega})\rvert = \lvert X(e^{-j\omega})\rvert$
$x[n]$	$\sphericalangle X(e^{j\omega}) = -\sphericalangle X(e^{-j\omega})$
$x_e[n]$	$X_R(e^{j\omega})$
$x_o[n]$	$jX_I(e^{j\omega})$

Of these properties, the most crucial to remember are those relating to the magnitude and phase; most notably, if we take the DTFT of a real sequence $x[n]$ and plot the resulting frequency response, its magnitude will be *even* and its phase will be *odd*.

2.7.6 Transform Properties and Pairs

Let $X(e^{j\omega})$ be the DTFT of $x[n]$. The following properties then apply:

Sequences $x[n]$, $y[n]$	Transforms $X(e^{j\omega})$, $Y(e^{j\omega})$	Property
$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$	Linearity
$x[n - n_d]$	$e^{-j\omega n_d}X(e^{j\omega})$	Time shift
$e^{j\omega_0 n}x[n]$	$X(e^{j(\omega-\omega_0)})$	Frequency shift
$x[-n]$	$X(e^{-j\omega})$	Time reversal
$nx[n]$	$j\frac{dX(e^{j\omega})}{d\omega}$	Frequency diff.
$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$	Convolution
$x[n]y[n]$	$\frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\theta})Y(e^{j(\omega-\theta)})d\theta$	Modulation

Note: The Modulation property above should look roughly familiar, and it should – it is basically very similar to continuous convolution (which should make sense since it is the transform of a product of two signals), but it is a special case that we call "circular" or "periodic" convolution. We will not really discuss this concept in relation to the DTFT, but it will come up later in the course and is the start of some terrible things to come!

Some useful DTFT pairs are provided below, and keep in mind that *all* these transforms are periodic over 2π in frequency:

Sequence	Fourier transform
$\delta[n]$	1
$\delta[n-n_0]$	$e^{-j\omega n_0}$
$1 \quad (-\infty < n < \infty)$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega + 2\pi k)$
$a^n u[n] \quad (\lvert a\rvert < 1)$	$\frac{1}{1-ae^{-j\omega}}$
$u[n]$	$\frac{1}{1-e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi\delta(\omega + 2\pi k)$
$(n+1)a^n u[n] \quad (\lvert a\rvert < 1)$	$\frac{1}{(1-ae^{-j\omega})^2}$
$\frac{\sin[\omega_c n]}{\pi n}$	$X(e^{j\omega}) = \begin{cases} 1 & \lvert\omega\rvert < \omega_c \\ 0 & \omega_c < \lvert\omega\rvert \leq \pi \end{cases}$
$x[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$	$\frac{\sin[\omega(M+1)/2]}{\sin(\omega/2)} e^{-j\omega M/2}$
$e^{j\omega_0 n}$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 + 2\pi k)$

Chapter 3

The z-transform

(See Oppenheim and Schafer, Second Edition pages 94–139, or First Edition pages 149–201)

3.1 Introduction

The z-transform of a sequence $x[n]$ is

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}.$$

The z-transform can also be thought of as an operator $\mathcal{Z}\{\cdot\}$ that transforms a discrete sequence to a function:

$$\mathcal{Z}\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n]z^{-n} = X(z).$$

In both cases z is a continuous complex variable.

This can essentially be thought of as equivalent to the Laplace transform, but the input to the z-transform is a discrete signal rather than a continuous one. More on this relationship later!

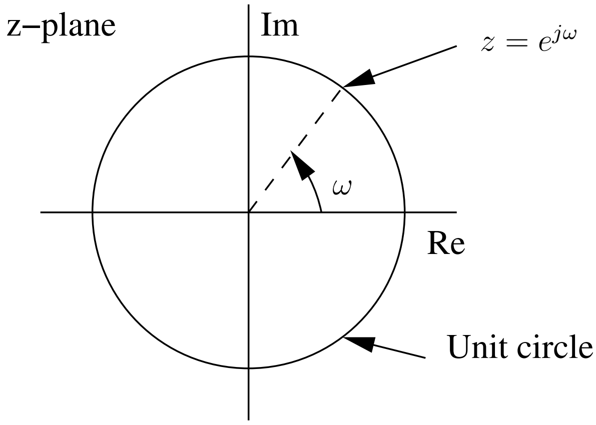
3.1.1 Relation to the DTFT

We may obtain the Fourier transform from the z-transform by making the substitution $z = e^{j\omega}$. This corresponds to restricting $|z| = 1$. Also, with $z = re^{j\omega}$,

$$X(re^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n](re^{j\omega})^{-n} = \sum_{n=-\infty}^{\infty} \left(x[n]r^{-n}\right) e^{-j\omega n}.$$

That is, the z-transform is the DTFT of the modified sequence $x[n]r^{-n}$.

For $r = 1$ this becomes the Fourier transform of $x[n]$. The DTFT therefore corresponds to the z-transform evaluated on the unit circle:



The inherent periodicity in frequency of the DTFT (i.e., being periodic over 2π) is captured naturally under this interpretation.

3.1.2 Region of Convergence

The Fourier transform does not converge for all sequences — the infinite sum may not always be finite. Similarly, the z-transform does not converge for all sequences or for all values of z. The set of values of z for which the z-transform converges is called the **Region of Convergence (ROC)**; for any other values of z, the z-transform diverges.

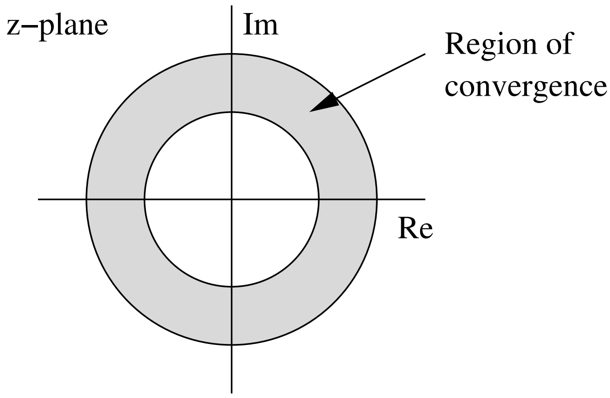
The Fourier transform of $x[n]$ exists if the sum $\sum_{n=-\infty}^{\infty} |x[n]|$ converges. However, the z-transform of $x[n]$ is just the Fourier transform of the sequence $x[n]r^{-n}$. The z-transform therefore exists (or converges) if

$$\sum_{n=-\infty}^{\infty} |x[n]r^{-n}| < \infty.$$

This leads to the condition

$$\sum_{n=-\infty}^{\infty} |x[n]||z|^{-n} < \infty$$

for the existence of the z-transform. The ROC therefore consists of a ring in the z-plane:



3.1.3 Stability and Causality

In specific cases the inner radius of this ring may include the origin, and the outer radius may extend to infinity. If the ROC includes the unit circle $|z| = 1$, then the Fourier transform will converge.

In the case of an impulse response $h[n]$, this is quite important; we know that, if a system's impulse response $h[n]$ has a DTFT $H(e^{j\omega})$, then the system must be **stable** since the impulse response is absolutely summable – we previously mentioned how this is a sufficient condition to check that the DTFT exists. We can thus extrapolate this further now and state that, if $h[n]$ has a z-transform $H(z)$ whose ROC includes the unit circle, then the DTFT of $h[n]$ converges – and thus the system must be stable!

Another interesting property that pops up from the ROC is the concept of system causality; in the previous chapter, we stated that an LTI system is causal if and only if $h[n] = 0$ for $n < 0$, and we can now relate this concept quite easily to the z-transform. Specifically, if the z-transform of $h[n]$ (the system function $H(z)$) has an ROC that extends outwards to infinity, we can immediately conclude that the system is **causal**; and if the ROC does not extend to infinity, the system is not causal. How do we know this?

Think about the definition of the z-transform of $h[n]$, namely the infinite sum

$$H(z) = \sum_{n=-\infty}^{\infty} h[n]z^{-n} = ... + h[-1]z^1 + h[0]z^0 + h[1]z^{-1} + ...$$

If the ROC includes $z \to \infty$ then the above expression cannot diverge, but we know that it will diverge due to the left-hand terms for $n < 0$ since those terms are multiplied by positive exponents of z – effectively, as we take the limit $z \to \infty$, these terms will explode to infinity. Hence, the only way to avoid this (and for our ROC to hold) is for $h[n]$ to be zero for all $n < 0$; i.e., the impulse response needs to be a right-sided sequence, and hence this corresponds to a causal system.

In summary: if the ROC of $H(z)$ includes the unit circle then the system is stable, and if the ROC extends outwards to infinity then the system is causal. Any combination of these properties can be true or false and depends purely on the ROC.

3.1.4 Poles and Zeros

Most useful z-transforms (essentially *all* useful z-transforms in engineering) can be expressed in the form

$$X(z) = K\frac{P(z)}{Q(z)},$$

where K is a multiplicative constant (often taken as 1), and $P(z)$ and $Q(z)$ are polynomials in z. Similarly to the Laplace transform, the values of z for which $P(z) = 0$ are called the **zeros** of $X(z)$, and the values with $Q(z) = 0$ are called the **poles**. The zeros and poles completely specify $X(z)$ to within a multiplicative constant (i.e., K, which only affects the gain).

Note that no poles can be contained within the ROC, and the poles will usually bound the ROC. This is fairly logical since the z-transform will *diverge* at all pole locations – this is the very definition of a pole.

Example: Right-sided exponential sequence
Consider the signal $x[n] = a^n u[n]$. This has the z-transform

$$X(z) = \sum_{n=-\infty}^{\infty} a^n u[n] z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n.$$

Convergence requires that

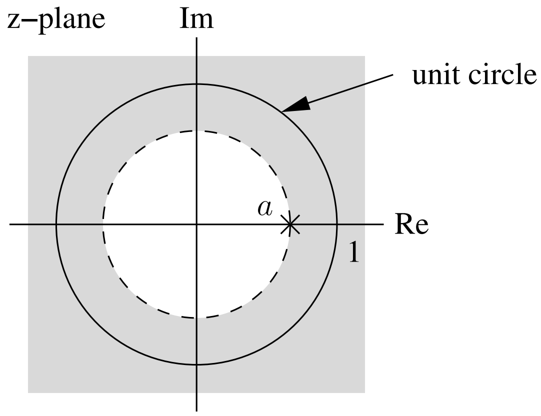
$$\sum_{n=0}^{\infty} |az^{-1}|^n < \infty,$$

which is only the case if $|az^{-1}| < 1$, or equivalently $|z| > |a|$. In the ROC, the series converges to

$$X(z) = \sum_{n=0}^{\infty} (az^{-1})^n = \frac{1}{1 - az^{-1}} = \frac{z}{z-a}, \qquad |z| > |a|,$$

since it is just a geometric series. The z-transform has a region of convergence for any finite value of a.

The Fourier transform of $x[n]$ only exists if the ROC includes the unit circle, which requires that $|a| < 1$. On the other hand, if $|a| > 1$ then the ROC does not include the unit circle, and the Fourier transform does not exist. This is consistent with the fact that for these values of a the sequence $a^n u[n]$ is exponentially growing, and the sum therefore does not converge.



We can also firmly state that, if this was the ROC of a system function $H(z)$ then the system would be causal since the ROC extends outwards to infinity.

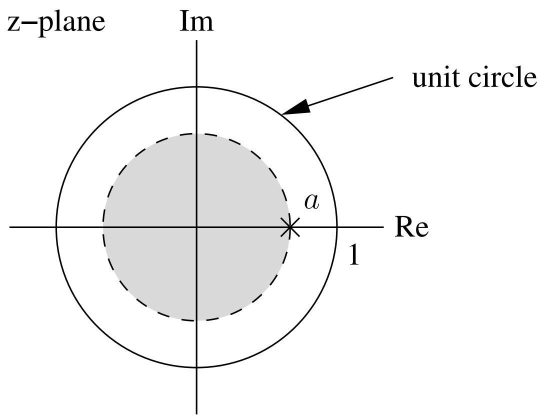
Example: Left-sided exponential sequence

Now consider the sequence $x[n] = -a^n u[-n-1]$. This sequence is left-sided because it is non-zero only for $n \leq -1$. Through some tricky algebra, the z-transform can be found to be

$$\begin{aligned}
X(z) &= \sum_{n=-\infty}^{\infty} -a^n u[-n-1] z^{-n} = -\sum_{n=-\infty}^{-1} a^n z^{-n} \\
&= -\sum_{n=1}^{\infty} a^{-n} z^n = -\sum_{n=1}^{\infty} a^{-n} z^n + (a^{-0} z^0 - a^{-0} z^0) \\
&= (a^{-0} z^0) - \sum_{n=0}^{\infty} a^{-n} z^n = 1 - \sum_{n=0}^{\infty} (a^{-1} z)^n.
\end{aligned}$$

For $|a^{-1} z| < 1$, or $|z| < |a|$, the series converges to

$$X(z) = 1 - \frac{1}{1 - a^{-1} z} = \frac{1}{1 - a z^{-1}} = \frac{z}{z - a}, \qquad |z| < |a|.$$



We can also firmly state that, if this was the ROC of a system function $H(z)$ then the system would *not* be causal since the ROC does not extend outwards to infinity. And similarly to before, the DTFT of $x[n]$ may or may not exist depending on the value of a.

Also note that the expression for the z-transform, and the pole-zero plot, are exactly the same as for the right-handed exponential sequence — *only the region of convergence is different*. Specifying the ROC is therefore critical when dealing with the z-transform – an expression for the z-transform, on its own, is an incomplete solution without the ROC.

Example: Sum of two exponentials

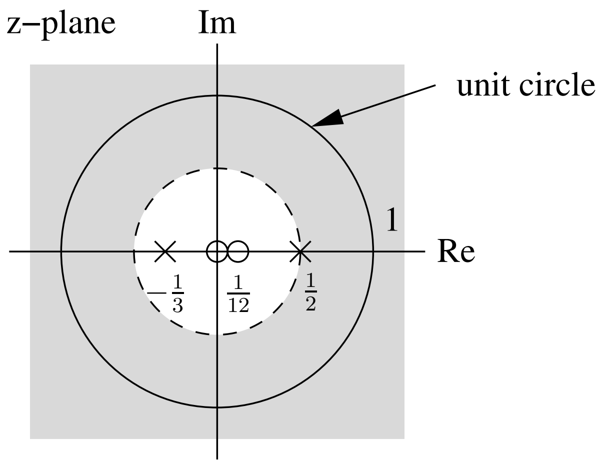
The signal $x[n] = \left(\frac{1}{2}\right)^n u[n] + \left(-\frac{1}{3}\right)^n u[n]$ is the sum of two real exponentials. The z-transform is

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} \left\{ \left(\frac{1}{2}\right)^n u[n] + \left(-\frac{1}{3}\right)^n u[n] \right\} z^{-n} \\ &= \sum_{n=-\infty}^{\infty} \left(\frac{1}{2}\right)^n u[n] z^{-n} + \sum_{n=-\infty}^{\infty} \left(-\frac{1}{3}\right)^n u[n] z^{-n} \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{2} z^{-1}\right)^n + \sum_{n=0}^{\infty} \left(-\frac{1}{3} z^{-1}\right)^n . \end{aligned}$$

From the example for the right-handed exponential sequence, the first term in this sum converges for $|z| > 1/2$, and the second for $|z| > 1/3$. The combined transform $X(z)$ therefore converges in the intersection of these regions, namely when $|z| > 1/2$. In this case

$$X(z) = \frac{1}{1 - \frac{1}{2} z^{-1}} + \frac{1}{1 + \frac{1}{3} z^{-1}} = \frac{2z(z - \frac{1}{12})}{(z - \frac{1}{2})(z + \frac{1}{3})}.$$

The pole-zero plot and region of convergence of the signal is



Example: Finite-length sequence
The signal

$$x[n] = \begin{cases} a^n & 0 \leq n \leq N-1 \\ 0 & \text{otherwise} \end{cases}$$

has z-transform

$$\begin{aligned} X(z) &= \sum_{n=0}^{N-1} a^n z^{-n} = \sum_{n=0}^{N-1} (az^{-1})^n \\ &= \frac{1-(az^{-1})^N}{1-az^{-1}} = \frac{1}{z^{N-1}} \frac{z^N - a^N}{z-a}. \end{aligned}$$

Since there are only a finite number of non-zero terms, the sum always converges when az^{-1} is finite. There are no restrictions on a ($|a| < \infty$), and the ROC is the entire z-plane – with the exception of the origin $z = 0$ (where the terms in the sum are infinite). The N roots of the numerator polynomial are at

$$z_k = ae^{j(2\pi k/N)}, \qquad k = 0, 1, \ldots, N-1,$$

since these values satisfy the equation $z^N = a^N$. The zero at $k = 0$ cancels the pole at $z = a$, so there are no poles except at the origin, and the zeros are at

$$z_k = ae^{j(2\pi k/N)}, \qquad k = 1, \ldots, N-1.$$

3.2 Properties of the ROC

The properties of the ROC depend on the nature of the signal. Assuming that the signal has a finite amplitude and that the z-transform is a rational function:

- The ROC is a ring or disk in the z-plane, centred on the origin ($0 \leq r_R < |z| < r_L \leq \infty$).

- The Fourier transform of $x[n]$ converges absolutely if and only if the ROC of the z-transform includes the unit circle. In the case of an impulse response $h[n]$, this also corresponds to the system being stable.

- The ROC cannot contain any poles.

- If $x[n]$ is finite duration (i.e., zero except on finite interval $-\infty < N_1 \leq n \leq N_2 < \infty$), then the ROC is the entire z-plane except perhaps at $z = 0$ or $z = \infty$.

- If $x[n]$ is a right-sided sequence then the ROC extends outward from the outermost finite pole to infinity. In the case of an impulse response $h[n]$, this also corresponds to the system being causal.

- If $x[n]$ is left-sided then the ROC extends inward from the innermost non-zero pole to $z = 0$. In the case of an impulse response $h[n]$, this also corresponds to the system being anticausal.

- A two-sided sequence (neither left nor right-sided) generally has an ROC consisting of a ring in the z-plane, bounded on the interior and exterior by a pole (and not containing any poles). In the case of an impulse response $h[n]$, this also corresponds to the system being non-causal. However, some two-sided sequences do not have a z-transform at all since it would not converge for any values of z.

- The ROC is a connected region.

3.3 The Inverse z-transform

Formally, the inverse z-transform can be performed by evaluating a Cauchy integral. However, for discrete LTI systems, simpler methods are often sufficient.

3.3.1 Inspection Method

If one is familiar with (or has a table of) common z-transform pairs, the inverse can be found by inspection. For example, one can invert the z-transform

$$X(z) = \left(\frac{1}{1 - \frac{1}{2}z^{-1}}\right), \qquad |z| > \frac{1}{2},$$

using the z-transform pair

$$a^n u[n] \quad \overset{z}{\longleftrightarrow} \quad \frac{1}{1 - az^{-1}}, \qquad \text{for } |z| > |a|.$$

Note that we specifically need to look at the ROC here, as a z-transform could easily have multiple different inverses – each with their own ROC.

By inspection, we recognise that

$$x[n] = \left(\frac{1}{2}\right)^n u[n].$$

Also, if $X(z)$ is a sum of terms then one may be able to do a term-by-term inversion by inspection, yielding $x[n]$ as a linear combination of terms.

3.3.2 Partial Fraction Expansion

For any rational function, we can obtain a partial fraction expansion and identify each term's z-transform. Assume that $X(z)$ is expressed as a ratio of polynomials in z^{-1}:

$$X(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}.$$

It is always possible to factor $X(z)$ as

$$X(z) = \frac{b_0}{a_0} \frac{\prod_{k=1}^{M}(1 - c_k z^{-1})}{\prod_{k=1}^{N}(1 - d_k z^{-1})},$$

where the c_k's and d_k's are the nonzero zeros and poles of $X(z)$.

Below is a good summary of the different forms of typical rational functions and their associated partial fraction expansions:

S.No.	Form of the rational function	Form of the partial fraction
1.	$\frac{px+q}{(x-a)(x-b)}, a \neq b$	$\frac{A}{x-a} + \frac{B}{x-b}$
2.	$\frac{px+q}{(x-a)^2}$	$\frac{A}{x-a} + \frac{B}{(x-a)^2}$
3.	$\frac{px^2+qx+r}{(x-a)(x-b)(x-c)}$	$\frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$
4.	$\frac{px^2+qx+r}{(x-a)^2(x-b)}$	$\frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-b}$
5.	$\frac{px^2+qx+r}{(x-a)(x^2+bx+c)}$	$\frac{A}{x-a} + \frac{Bx+C}{x^2+bx+c}$
	• where $x^2 + bx + c$ cannot be factorised further	

For this course, you are expected to know how to do partial fraction expansions, so be sure to remember your rules!

Example: Inverse by partial fractions

Consider the sequence $x[n]$ with z-transform

$$X(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}} = \frac{(1 + z^{-1})^2}{(1 - \frac{1}{2}z^{-1})(1 - z^{-1})}, \qquad |z| > 1.$$

Since $M = N = 2$ this can be expressed as

$$X(z) = B_0 + \frac{A_1}{1 - \frac{1}{2}z^{-1}} + \frac{A_2}{1 - z^{-1}}.$$

The value B_0 can be found by long division:

$$\begin{array}{r|l} & \quad\;\; 2 \\ \cline{2-2} \frac{1}{2}z^{-2} - \frac{3}{2}z^{-1} + 1 & z^{-2} + 2z^{-1} + 1 \\ & z^{-2} - 3z^{-1} + 2 \\ \cline{2-2} & \qquad 5z^{-1} - 1 \end{array}$$

so

$$X(z) = 2 + \frac{-1 + 5z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - z^{-1})}.$$

The coefficients A_1 and A_2 can be found using the following approach – or whichever method you prefer from your MAM days!

$$A_k = (1 - d_k z^{-1})X(z)\Big|_{z=d_k},$$

so

$$A_1 = \left.\frac{1 + 2z^{-1} + z^{-2}}{1 - z^{-1}}\right|_{z^{-1}=2} = \frac{1 + 4 + 4}{1 - 2} = -9$$

and

$$A_2 = \left.\frac{1 + 2z^{-1} + z^{-2}}{1 - \frac{1}{2}z^{-1}}\right|_{z^{-1}=1} = \frac{1 + 2 + 1}{1/2} = 8.$$

Therefore

$$X(z) = 2 - \frac{9}{1 - \frac{1}{2}z^{-1}} + \frac{8}{1 - z^{-1}}.$$

Using the fact that the ROC is $|z| > 1$ (as given), the terms can be inverted one at a time by inspection to yield

$$x[n] = 2\delta[n] - 9(1/2)^n u[n] + 8u[n].$$

3.3.3 Power Series Expansion

If the z-transform is given as a power series in the form

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} x[n]z^{-n} \\ &= \ldots + x[-2]z^2 + x[-1]z^1 + x[0] + x[1]z^{-1} + x[2]z^{-2} + \ldots, \end{aligned}$$

then any value in the sequence can be found by identifying the coefficient of the appropriate power of z^{-1}.

Example: Finite-length sequence
The z-transform

$$X(z) = z^2(1 - \frac{1}{2}z^{-1})(1 + z^{-1})(1 - z^{-1})$$

can be multiplied out to give

$$X(z) = z^2 - \frac{1}{2}z - 1 + \frac{1}{2}z^{-1}.$$

By inspection, the corresponding sequence can therefore be found from the definition of the z-transform and mapping each coefficient to $x[n]$ as follows:

$$x[n] = \begin{cases} 1 & n = -2 \\ -\frac{1}{2} & n = -1 \\ -1 & n = 0 \\ \frac{1}{2} & n = 1 \\ 0 & \text{otherwise} \end{cases}$$

or equivalently

$$x[n] = 1\delta[n+2] - \frac{1}{2}\delta[n+1] - 1\delta[n] + \frac{1}{2}\delta[n-1].$$

Example: Power series expansion by long division

Consider the transform

$$X(z) = \frac{1}{1 - az^{-1}}, \qquad |z| > |a|.$$

where the numerator is the dividend and the denominator is the divisor. Since the ROC is the exterior of a circle, the sequence is right-sided. We therefore perform long division (you may need to brush up on your rules!) to get a power series in powers of z^{-1} as follows:

$$\begin{array}{r l}
 & 1 + az^{-1} + a^2z^{-2} + \cdots \\ \cline{2-2}
1 - az^{-1} \big) & 1 \\
 & \underline{1 - az^{-1}} \\
 & \quad az^{-1} \\
 & \quad \underline{az^{-1} - a^2z^{-2}} \\
 & \qquad\qquad a^2z^{-2} + \cdots
\end{array}$$

The result of this long division is thus the quotient

$$\frac{1}{1 - az^{-1}} = 1 + az^{-1} + a^2z^{-2} + \cdots$$

Therefore $x[n] = a^n u[n]$.

Example: Power series expansion for left-sided sequence
Consider instead the z-transform

$$X(z) = \frac{1}{1 - az^{-1}}, \qquad |z| < |a|.$$

Because of the ROC, the sequence is now a left-sided one. Thus we perform long division and swap the order of the divisor such that we obtain a series in powers of z (since we know there are no right-sided terms):

$$\begin{array}{r l}
 & -a^{-1}z - a^{-2}z^2 - \cdots \\ \cline{2-2}
-a + z \big) & z \\
 & \underline{z - a^{-1}z^2} \\
 & \quad a^{-1}z^2
\end{array}$$

Thus $x[n] = -a^n u[-n-1]$.

Note: We could apply the same strategy to the previous example, where we work in positive powers of z in the dividend and divisor, but the crucial change is that we would then divide z by $z - a$ (instead of $-a + z$) so that we yield negative powers of z in the quotient. It is a small change, but quite important since the z-transform itself is the same in both examples – the only difference is the ROC, which allows us to know whether the sequence (i.e., the inverse z-transform) will be left or right sided.

3.4 Properties of the z-transform

In this section, if $X(z)$ denotes the z-transform of a sequence $x[n]$ and the ROC of $X(z)$ is indicated by R_x, then this relationship is indicated as

$$x[n] \quad \overset{\mathcal{Z}}{\longleftrightarrow} \quad X(z), \qquad \text{ROC} = R_x.$$

Furthermore, with regard to nomenclature, we have two sequences such that

$$\begin{aligned} x_1[n] \quad &\overset{\mathcal{Z}}{\longleftrightarrow} \quad X_1(z), \qquad \text{ROC} = R_{x_1} \\ x_2[n] \quad &\overset{\mathcal{Z}}{\longleftrightarrow} \quad X_2(z), \qquad \text{ROC} = R_{x_2}. \end{aligned}$$

3.4.1 Linearity

The linearity property is as follows:

$$ax_1[n] + bx_2[n] \quad \overset{\mathcal{Z}}{\longleftrightarrow} \quad aX_1(z) + bX_2(z), \qquad \text{ROC contains} R_{x_1} \cap R_{x_2}.$$

Notice that we use the word "contains" in the ROC; it is possible that the ROC may be exactly equal to $R_{x_1} \cap R_{x_2}$, but there are some cases where the ROC is larger than the intersection, such as when pole-zero cancellation occurs.

3.4.2 Time Shifting

The time-shifting property is as follows:

$$x[n - n_0] \quad \overset{\mathcal{Z}}{\longleftrightarrow} \quad z^{-n_0}X(z), \qquad \text{ROC} = R_x.$$

The ROC is thus unchanged, but note that it could be changed by the possible addition (or deletion) of $z = 0$ or $z = \infty$. This is one of those weird cases where we try not to think too hard on how added poles may affect the ROC, but just note that a time shift could potentially cause the new ROC to exclude $z = 0$ or $z = \infty$ – even if it was previously included.

The proof for this property is easily shown:

$$Y(z) = \sum_{n=-\infty}^{\infty} x[n-n_0]z^{-n} = \sum_{m=-\infty}^{\infty} x[m]z^{-(m+n_0)}$$
$$= z^{-n_0} \sum_{m=-\infty}^{\infty} x[m]z^{-m} = z^{-n_0}X(z).$$

Example: Shifted exponential sequence
Consider the z-transform

$$X(z) = \frac{1}{z-\frac{1}{4}}, \qquad |z| > \frac{1}{4}.$$

From the ROC, this is a right-sided sequence. Rewriting,

$$X(z) = \frac{z^{-1}}{1-\frac{1}{4}z^{-1}} = z^{-1}\left(\frac{1}{1-\frac{1}{4}z^{-1}}\right), \qquad |z| > \frac{1}{4}.$$

The term in brackets corresponds to an exponential sequence $(1/4)^n u[n]$. The factor z^{-1} shifts this sequence one sample to the right. The inverse z-transform is therefore

$$x[n] = (1/4)^{n-1}u[n-1].$$

Note that this result could also have been easily obtained using a partial fraction expansion; interestingly though, this could result in a different expression for $x[n]$ – even if the resulting sequence is exactly the same!

3.4.3 Multiplication With an Exponential Sequence

The exponential multiplication property is

$$z_0^n x[n] \quad \overset{\mathcal{Z}}{\longleftrightarrow} \quad X(z/z_0), \qquad \text{ROC} = |z_0|R_x,$$

where the notation $|z_0|R_x$ indicates that the ROC is scaled by $|z_0|$ (that is, inner and outer radii of the ROC scale by $|z_0|$). All pole-zero locations are similarly scaled by a factor z_0: if $X(z)$ had a pole at $z = z_1$, then $X(z/z_0)$ will have a pole at $z = z_0 z_1$.

- If z_0 is positive and real, this operation can be interpreted as a shrinking or expanding of the z-plane — poles and zeros change along radial lines in the z-plane.

- If z_0 is complex with unit magnitude ($z_0 = e^{j\omega_0}$) then the scaling operation corresponds to a rotation in the z-plane by an angle ω_0. That is, the poles and zeros rotate along circles centered on the origin. This can be interpreted as a shift in the frequency domain, associated with modulation in the time domain by $e^{j\omega_0 n}$. If the Fourier transform exists, this becomes

$$e^{j\omega_0 n}x[n] \quad \overset{\mathcal{F}}{\longleftrightarrow} \quad X(e^{j(\omega-\omega_0)}).$$

Example: Exponential multiplication
The z-transform pair

$$u[n] \quad \overset{\mathcal{Z}}{\longleftrightarrow} \quad \frac{1}{1-z^{-1}}, \qquad |z| > 1$$

can be used to determine the z-transform of $x[n] = r^n \cos(\omega_0 n)u[n]$. Since $\cos(\omega_0 n) = 1/2e^{j\omega_0 n} + 1/2e^{-j\omega_0 n}$, the signal can be rewritten as

$$x[n] = \frac{1}{2}(re^{j\omega_0})^n u[n] + \frac{1}{2}(re^{-j\omega_0})^n u[n].$$

From the exponential multiplication property,

$$\begin{aligned}
\frac{1}{2}(re^{j\omega_0})^n u[n] \quad &\overset{\mathcal{Z}}{\longleftrightarrow} \quad \frac{1/2}{1-re^{j\omega_0}z^{-1}}, \qquad |z| > r \\
\frac{1}{2}(re^{-j\omega_0})^n u[n] \quad &\overset{\mathcal{Z}}{\longleftrightarrow} \quad \frac{1/2}{1-re^{-j\omega_0}z^{-1}}, \qquad |z| > r,
\end{aligned}$$

so

$$\begin{aligned}
X(z) &= \frac{1/2}{1-re^{j\omega_0}z^{-1}} + \frac{1/2}{1-re^{-j\omega_0}z^{-1}}, \qquad |z| > r \\
&= \frac{1-r\cos(\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}, \qquad |z| > r.
\end{aligned}$$

Interestingly, while we cannot use partial fractions to invert this transform, an inverse *does* exist as shown by this example. A useful transform pair also pops up out of this equation, as we can obtain the following form by setting $\omega_0 = \pi/2$ and $r = 1$:

$$\cos\left(\frac{\pi}{2}n\right)u[n] \qquad \overset{\mathcal{Z}}{\longleftrightarrow} \qquad \frac{1}{1+z^{-2}}, \qquad |z| > 1$$

This is quite an interesting transform as it is (pretty much) the only time that we see this kind of polynomial form in the denominator of a z-transform. But note that this discrete-time sequence will have no DTFT since the unit circle is not included in the ROC!

3.4.4 Differentiation

The differentiation (in z) property states that

$$nx[n] \quad \overset{\mathcal{Z}}{\longleftrightarrow} \quad -z\frac{dX(z)}{dz}, \qquad \text{ROC} = R_x.$$

This can be proven through a slightly weird derivation; since

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n},$$

we have

$$-z\frac{dX(z)}{dz} = -z\sum_{n=-\infty}^{\infty}(-n)x[n]z^{-n-1} = \sum_{n=-\infty}^{\infty} nx[n]z^{-n} = \mathcal{Z}\{nx[n]\}.$$

Example: Second-order pole
The z-transform of the sequence

$$x[n] = na^n u[n]$$

can be found using

$$a^n u[n] \quad \overset{\mathcal{Z}}{\longleftrightarrow} \quad \frac{1}{1-az^{-1}}, \qquad |z| > a,$$

to be

$$X(z) = -z\frac{d}{dz}\left(\frac{1}{1-az^{-1}}\right) = \frac{az^{-1}}{(1-az^{-1})^2}, \qquad |z| > a.$$

This can be computed using the chain rule or quotient rule.

3.4.5 Conjugation

This property is

$$x^*[n] \quad \overset{\mathcal{Z}}{\longleftrightarrow} \quad X^*(z^*), \qquad \text{ROC} = R_x.$$

3.4.6 Time Reversal

This property says that

$$x[-n] \overset{\mathcal{Z}}{\longleftrightarrow} X(1/z), \qquad \text{ROC} = \frac{1}{R_x}.$$

Taken a step further, we can also include the conjugation property:

$$x^*[-n] \overset{\mathcal{Z}}{\longleftrightarrow} X^*(1/z^*), \qquad \text{ROC} = \frac{1}{R_x}.$$

The notation $1/R_x$ means that the ROC is inverted, so if R_x is the set of values such that $r_R < |z| < r_L$, then the new ROC is the set of values of z such that $1/r_L < |z| < 1/r_R$.

For instance, an ROC of $2 < |z| < 3$ would become $1/3 < |z| < 1/2$ after applying time reversal, so we change both the values and the directions of the ROC boundaries.

Example: Time-reversed exponential sequence

The signal $x[n] = a^{-n}u[-n]$ is a time-reversed version of $a^n u[n]$. The z-transform is therefore

$$X(z) = \frac{1}{1 - az} = \frac{-a^{-1}z^{-1}}{1 - a^{-1}z^{-1}}, \qquad |z| < |a^{-1}|.$$

3.4.7 Convolution

As you should be able to foresee, this property states that

$$x_1[n] * x_2[n] \overset{\mathcal{Z}}{\longleftrightarrow} X_1(z)X_2(z), \qquad \text{ROC contains } R_{x_1} \cap R_{x_2}.$$

Once again, notice the word "contains" being used in the ROC; an ROC can become somewhat tricky to determine when there is any pole-zero cancellation involved.

Example: Evaluating a convolution using the z-transform

The z-transforms of the signals $x_1[n] = a^n u[n]$ and $x_2[n] = u[n]$ are

$$X_1(z) = \sum_{n=0}^{\infty} a^n z^{-n} = \frac{1}{1 - az^{-1}}, \qquad |z| > |a|$$

and

$$X_2(z) = \sum_{n=0}^{\infty} z^{-n} = \frac{1}{1 - z^{-1}}, \qquad |z| > 1.$$

For $|a| < 1$, the z-transform of the convolution $y[n] = x_1[n] * x_2[n]$ is

$$Y(z) = \frac{1}{(1 - az^{-1})(1 - z^{-1})} = \frac{z^2}{(z - a)(z - 1)}, \qquad |z| > 1.$$

Using a partial fraction expansion, we can find

$$Y(z) = \frac{1}{1 - a}\left(\frac{1}{1 - z^{-1}} - \frac{a}{1 - az^{-1}}\right), \qquad |z| > 1,$$

so

$$y[n] = \frac{1}{1 - a}(u[n] - a^{n+1}u[n]).$$

The same result would be derived if $|a| > 1$, but the ROC would be $|z| > |a|$ instead.

Interestingly, if we think of $x_1[n]$ as an impulse response, then technically the above output is the **step response** of the system since $x_2[n]$ is a unit step.

3.4.8 Initial Value Theorem

If $x[n]$ is zero for $n < 0$ (i.e., it is a right-sided sequence), then

$$x[0] = \lim_{z \to \infty} X(z).$$

3.4.9 Final Value Theorem

Once again, if $x[n]$ is zero for $n < 0$, then

$$\lim_{n \to \infty} x[n] = \lim_{z \to 1}[(1 - z^{-1})X(z)].$$

3.4.10 Relationship With the Laplace Transform

Continuous-time systems and signals are usually described by the Laplace transform. Letting $z = e^{sT}$, where T is the sampling period and s is the complex Laplace variable

$$s = \sigma + j\omega,$$

we have

$$z = e^{(\sigma+j\omega)T} = e^{\sigma T} e^{j\omega T}.$$

Therefore

$$|z| = e^{\sigma T} \quad \text{and} \quad \sphericalangle z = \omega T = 2\pi f/f_s = 2\pi\omega/\omega_s,$$

where ω_s is the sampling frequency. As ω varies from $-\infty$ to ∞, the s-plane is mapped to the z-plane:

- The $j\omega$ axis in the s-plane is mapped to the unit circle in the z-plane; in the s-plane, the $j\omega$ axis corresponds to the Continuous-time Fourier Transform, whereas in the z-plane, the unit circle corresponds to the DTFT
- The left-hand s-plane is mapped to the inside of the unit circle; i.e., this is where the "good poles" are
- The right-hand s-plane maps to the outside of the unit circle; i.e., this is where the "bad poles" are

3.5 Transforms Table

Some common z-transform pairs are:

Sequence	Transform	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{1}{1-z^{-1}}$	$\|z\| > 1$
$-u[-n-1]$	$\frac{1}{1-z^{-1}}$	$\|z\| < 1$
$\delta[n-m]$	z^{-m}	All z except 0 or ∞
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$\|z\| > \|a\|$
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$\|z\| < \|a\|$
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\|z\| > \|a\|$
$-na^n u[-n-1]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\|z\| < \|a\|$
$\begin{cases} a^n & 0 \leq n \leq N-1, \\ 0 & \text{otherwise} \end{cases}$	$\frac{1-a^N z^{-N}}{1-az^{-1}}$	$\|z\| > 0$
$\cos(\omega_0 n)u[n]$	$\frac{1-\cos(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$	$\|z\| > 1$
$r^n \cos(\omega_0 n)u[n]$	$\frac{1-r\cos(\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}$	$\|z\| > r$

Chapter 4

Sampling of Continuous-time Signals

(See Oppenheim and Schafer, Second Edition pages 140–239, or First Edition pages 80–148)

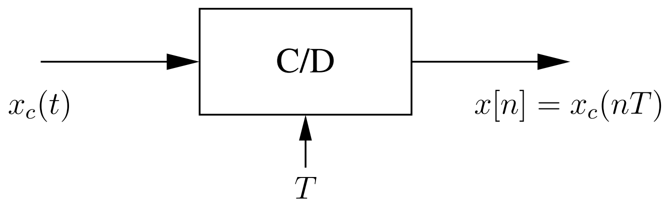
4.1 Introduction

Welcome to the most confusing chapter in the course! While it is true that you will likely come away from this chapter knowing even less about sampling than before, one could also argue that this is the most important chapter in the course; ultimately, it is absolutely fundamental to all of signal processing.

Let us start with the basics: a discrete-time signal $x[n]$ often arises from the periodic sampling of a continuous-time signal $x_c(t)$:

$$x[n] = x_c(nT), \qquad -\infty < n < \infty.$$

This system is called an ideal continuous-to-discrete-time (C/D) converter or sampler,



and is described by the following:

- Sampling period: T seconds.
- Sampling frequency: $f_s = 1/T$ samples per second, or $\Omega_s = 2\pi/T$ radians per second. Note how we use Ω [radians per second] to denote frequency in relation to continuous time (which is what you have been considering in previous courses),

and we use ω [radians per sample] to denote frequency in relation to discrete time (as we have been doing in previous chapters of these notes). This disambiguation is important and crucial to understanding this chapter!

In practice, sampling is usually approximately implemented using analog-to-digital (A/D) converter, which is (of course) non-ideal.

The sampling process is also not generally invertible: one cannot always reconstruct $x_c(t)$ unambiguously from $x[n]$. However, ambiguity can be removed by *restricting the input signals to the sampler* through bandlimiting.

4.2 Frequency-domain Representation of Sampling

4.2.1 Sampled Signals

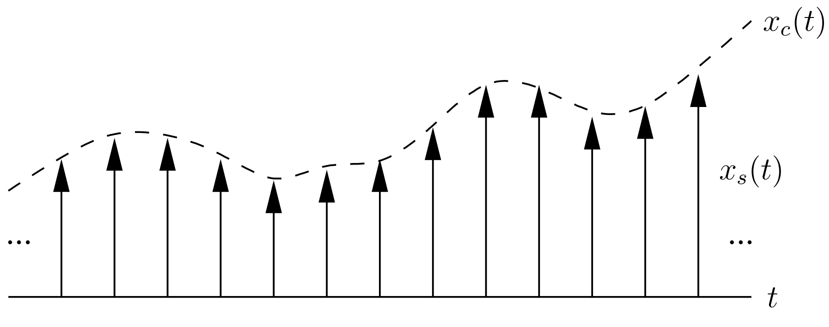
Consider converting a continuous-time signal $x_c(t)$ to a sampled version $x_s(t)$ by modulating it with the periodic impulse train

$$s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT),$$

which has the CTFT given by

$$S(j\Omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\Omega - k\Omega_s).$$

Notice that we use the argument $j\Omega$ for purely notational reasons to denote a CTFT. Sampling $x_c(t)$ thus results in the following output:



And through the sifting property of the impulse function, we can write this as

$$\begin{aligned} x_s(t) &= x_c(t)s(t) = x_c(t) \sum_{n=-\infty}^{\infty} \delta(t - nT) \\ &= \sum_{n=-\infty}^{\infty} x_c(nT)\delta(t - nT). \end{aligned}$$

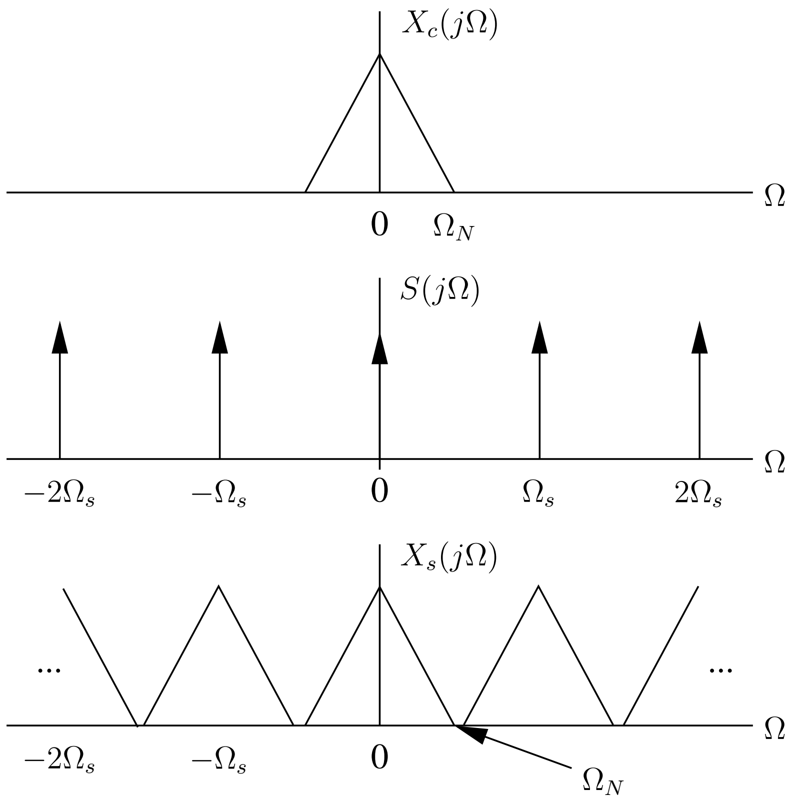
This is an interesting representation as it allows us to "change" to either $x_c(t)$ or $x[n]$, making it the most important reference signal for us to keep track of. This is also an interesting signal because it is inherently *continuous* (a function of t), but at the same time, it has a *discrete* interpretation since it is only non-zero at the specific points shown by the delta functions.

4.2.2 Frequency Domain

What about the frequency domain? Well, the CTFT of $x_s(t) = x_c(t)s(t)$ is $X_s(j\Omega)$ – which can be computed as the continuous-frequency convolution of Fourier transforms $X_c(j\Omega)$ and $S(j\Omega)$, so

$$X_s(j\Omega) = \frac{1}{2\pi} X_c(j\Omega) * S(j\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s)).$$

Therefore, the Fourier transform of $x_s(t)$ consists of copies of $X_c(j\Omega)$, shifted by integer multiples of the sampling frequency Ω_s, and then superimposed:



4.2.3 Aliasing

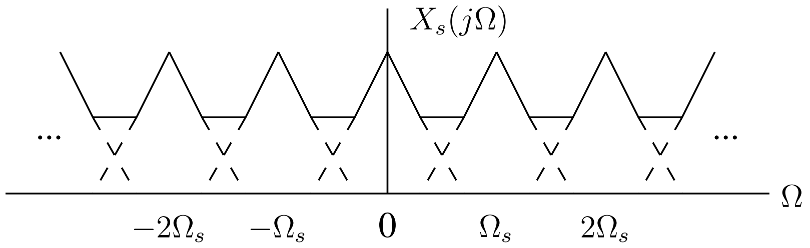
If $x_c(t)$ is bandlimited, with highest non-zero frequency at Ω_N, then the replicas do not overlap when

$$\Omega_s > 2\Omega_N.$$

The frequency Ω_N is referred to as the **Nyquist frequency**, and the frequency $2\Omega_N$ that must be exceeded in the sampling is the **Nyquist rate**.

But how do we get to this expression? One simple way of visualising it is to think about where the replicas start and end. For instance, if we look at the first two replicas (at $\Omega = 0$ and $\Omega = \Omega_s$), then the halfway point between them is at $\Omega = \Omega_s/2$, and if we want to avoid any overlap, then we need the central replica to end before this midway point; and since the end of the central replica is given by $\Omega = \Omega_N$, we therefore require that $\Omega_N < \Omega_s/2$ – or equivalently, $\Omega_s > 2\Omega_N$.

From this, we could recover the original $x_c(t)$ from $x_s(t)$ by using an ideal lowpass filter to pass the central replica and attenuate the other replicas. Otherwise, $X_c(j\Omega)$ cannot be recovered using lowpass filtering and **aliasing** results, where the replicas have some degree of overlap between them (and the spectra sum together in these regions of overlap):



4.2.4 Relating the DTFT and CTFT

The objective now is to express the DTFT $X(e^{j\omega})$ of $x[n]$ in terms of $X_c(j\Omega)$ and $X_s(j\Omega)$. Directly taking the Fourier transform of our (very valuable) reference signal

$$x_s(t) = \sum_{n=-\infty}^{\infty} x_c(nT)\delta(t - nT)$$

will, through linearity, yield the CTFT given by

$$X_s(j\Omega) = \sum_{n=-\infty}^{\infty} \mathcal{F}\left\{x_c(nT)\delta(t - nT)\right\} = \sum_{n=-\infty}^{\infty} x_c(nT)\mathcal{F}\left\{\delta(t - nT)\right\} = \sum_{n=-\infty}^{\infty} x_c(nT)e^{-j\Omega Tn}.$$

Now, since $x[n] = x_c(nT)$ and

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n},$$

it follows that

$$X_s(j\Omega) = X(e^{j\omega})\big|_{\omega=\Omega T} = X(e^{j\Omega T}).$$

Consequently, this expression should also be equivalent to the expression for $X_s(j\Omega)$ that we derived using convolution, so we can write

$$X(e^{j\Omega T}) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s)),$$

and thus, because we know that $\Omega_s = 2\pi/T$, we can write

$$X(e^{j\omega}) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega}{T} - \frac{2\pi k}{T}\right)\right).$$

Well done if you were able to follow all of that! The conclusion is thus that $X(e^{j\omega})$ is just a frequency-scaled version of $X_s(j\Omega)$, with the scaling specified by $\omega = \Omega T$ (changing from [rad/sec] to [rad/sample]).

Alternatively, the effect of sampling may be thought of as a *normalisation* of the frequency axis, so that the frequency $\Omega = \Omega_s = 2\pi/T$ of $X_s(j\Omega)$ is normalised to $\omega = 2\pi$ for $X(e^{j\omega})$ (since $\omega = \Omega T$). Yes, this *is* a mouthful!

4.3 Reconstructing Bandlimited Signals From Samples

If samples of a bandlimited continuous-time signal are taken frequently enough (think about the Nyquist criterion!), then they are sufficient to represent the signal exactly and the continuous-time signal could be recovered from the samples. This task is ideally performed by a discrete-to-continuous-time (D/C) converter. The form and behaviour of such a converter is discussed in this section.

Given sequence of samples $x[n]$, we can form the modulated impulse train

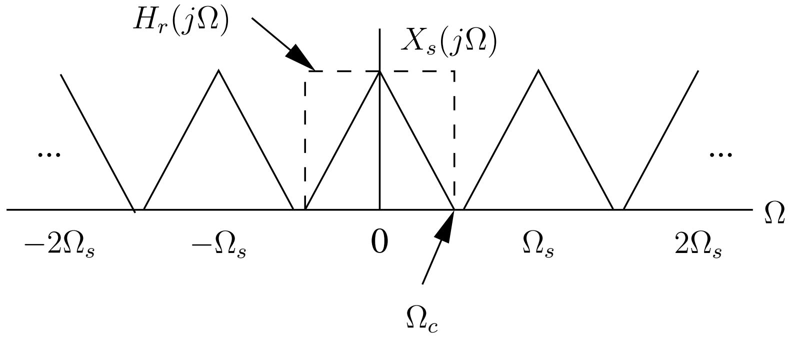
$$x_s(t) = \sum_{n=-\infty}^{\infty} x[n]\delta(t - nT).$$

The nth sample corresponds to the impulse at time $t = nT$.

If appropriate sampling conditions are met, namely the signal is bandlimited and the Fourier transform replicas do not overlap, then $x(t)$ can be reconstructed as $x_r(t)$ (from $x_s(t)$) by ideal continuous-time lowpass filtering. This can be seen as a convolution of $x_s(t)$ with an ideal LPF with the impulse response $h_r(t)$:

$$x_r(t) = \sum_{n=-\infty}^{\infty} x[n] h_r(t - nT).$$

This can be visualised as follows, where $h_r(t)$ has a cutoff frequency of Ω_c:



Note that, due to the normalisation discussed earlier, the frequency $\Omega = \Omega_s = 2\pi/T$ for $X_s(j\Omega)$ corresponds to $\omega = 2\pi$ for $X(e^{j\omega})$; as such, $\Omega = \Omega_s/2 = \pi/T$ [rad/sec] corresponds to $\omega = \pi$ [rad/sample]. This is thus a convenient choice for the filter cutoff frequency Ω_c, corresponding to the ideal reconstruction filter

$$H_r(j\Omega) = \begin{cases} T & |\Omega| \leq \pi/T \\ 0 & |\Omega| > \pi/T. \end{cases}$$

Note that we use an amplitude scaling of T to normalise $X_s(j\Omega)$ to a value of 1 after filtering.

The reconstructed signal is therefore

$$\begin{aligned} X_r(j\Omega) = H_r(j\Omega)X_s(j\Omega) &= H_r(j\Omega)X(e^{j\Omega T}) \\ &= \begin{cases} TX(e^{j\Omega T}) & |\Omega| \leq \pi/T \\ 0 & |\Omega| > \pi/T. \end{cases} \end{aligned}$$

In the time domain, the ideal reconstruction filter has impulse response

$$h_r(t) = \frac{\sin(\pi t/T)}{(\pi t/T)},$$

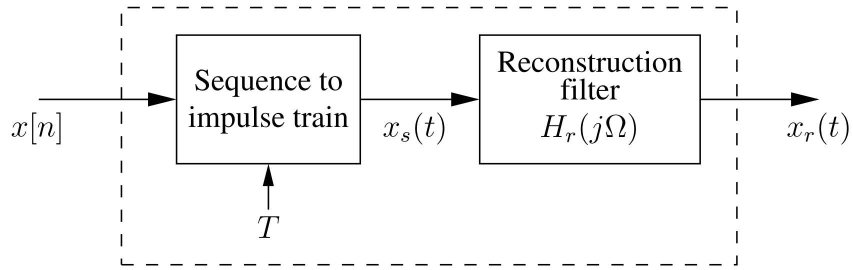
and the reconstructed signal is given by

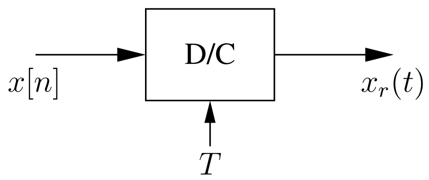
$$x_r(t) = \sum_{n=-\infty}^{\infty} x[n] \frac{\sin[\pi(t - nT)/T]}{\pi(t - nT)/T}.$$

This should look (tragically) familiar as you would have learned about it in a previous course!

From the previous frequency-domain argument, if $x[n] = x_c(nT)$ with $X_c(j\Omega) = 0$ for $|\Omega| \geq \pi/T$, then $x_r(t) = x_c(t)$. Note that the filter $h_r(t)$ is not realisable since it has infinite duration – more on this later.

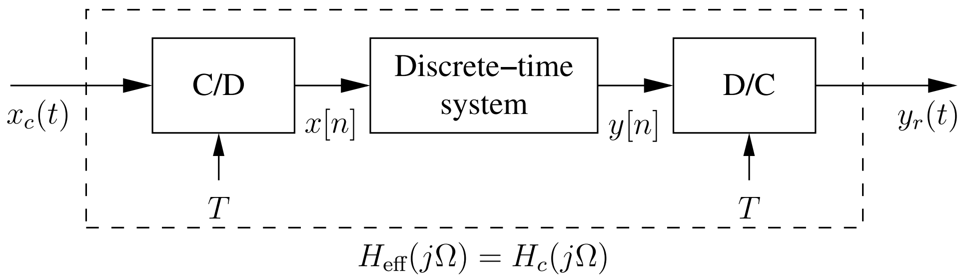
An ideal discrete-to-continuous (D/C) reconstruction system therefore has the form





4.4 Discrete-time Processing of Continuous Signals

Discrete-time systems are often used to process continuous-time signals, such as for filtering, since they are flexible and reprogrammable. This can be accomplished using:



Usually there is also an **anti-aliasing filter** applied to the input signal right at the beginning – an analog low-pass filter placed before the C/D converter to remove signal components higher than half the sampling rate (the Nyquist frequency). Furthermore, it is assumed that the C/D and D/C converters have the same sampling rate.

The C/D converter produces the discrete-time signal

$$x[n] = x_c(nT),$$

with Fourier transform

$$X(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega}{T} - \frac{2\pi k}{T}\right)\right).$$

The D/C converter creates a continuous-time output of the form

$$y_r(t) = \sum_{n=-\infty}^{\infty} y[n] \frac{\sin[\pi(t - nT)/T]}{\pi(t - nT)/T}.$$

The continuous-time Fourier transform of $y_r(t)$, namely $Y_r(j\Omega)$, and the discrete-time Fourier transform of $y[n]$, namely $Y(e^{j\Omega T})$, are related by

$$\begin{aligned} Y_r(j\Omega) &= H_r(j\Omega)Y(e^{j\Omega T}) \\ &= \begin{cases} TY(e^{j\Omega T}) & |\Omega| < \pi/T \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

If the discrete-time system is LTI, then

$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega}),$$

where $H(e^{j\omega})$ is the frequency response of the system. Therefore

$$\begin{aligned} Y_r(j\Omega) &= H_r(j\Omega)H(e^{j\Omega T})X(e^{j\Omega T}) \\ &= H_r(j\Omega)H(e^{j\Omega T})\frac{1}{T}\sum_{k=-\infty}^{\infty} X_c\left(j\left(\Omega - \frac{2\pi k}{T}\right)\right). \end{aligned}$$

If $X_c(j\Omega) = 0$ for $|\Omega| \geq \pi/T$, then the ideal LPF $H_r(j\Omega)$ selects only the term for $k = 0$ in the sum, and scales the result:

$$Y_r(j\Omega) = \begin{cases} H(e^{j\Omega T})X_c(j\Omega) & |\Omega| < \pi/T \\ 0 & |\Omega| \geq \pi/T. \end{cases}$$

Thus if $X_c(j\Omega)$ is bandlimited and sampled above the Nyquist rate, then the output is related to the input by

$$Y_r(j\Omega) = H_{\text{eff}}(j\Omega)X_c(j\Omega),$$

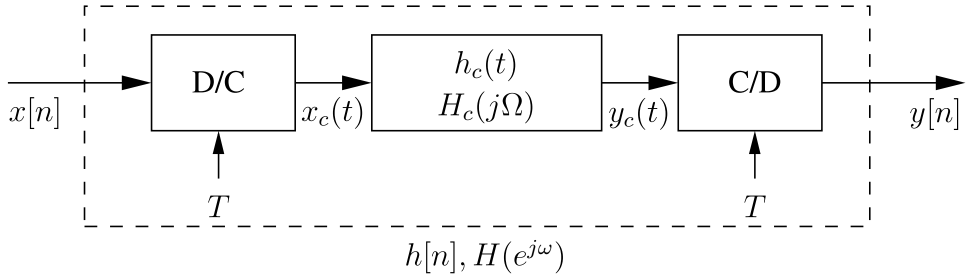
where

$$H_{\text{eff}}(j\Omega) = \begin{cases} H(e^{j\Omega T}) & |\Omega| < \pi/T \\ 0 & |\Omega| \geq \pi/T \end{cases}$$

is the effective frequency response of the system.

4.5 Continuous-time Processing of Discrete Signals

It is conceptually useful (not always *practically* useful) to consider continuous-time processing of discrete-time signals. A system to perform this task is:



Since the D/C converter includes an ideal LPF, $X_c(j\Omega)$ (and therefore also $Y_c(j\Omega)$) will be zero for $|\Omega| \geq \pi/T$. Thus the C/D converter samples $y_c(t)$ without aliasing, so

$$x_c(t) = \sum_{n=-\infty}^{\infty} x[n] \frac{\sin[\pi(t-nT)/T]}{\pi(t-nT)/T}$$

and

$$y_c(t) = \sum_{n=-\infty}^{\infty} y[n] \frac{\sin[\pi(t-nT)/T]}{\pi(t-nT)/T},$$

where the latter equation is essentially applied backwards (since we want to find $y[n]$ *from* $y_c(t)$), and we have $x[n] = x_c(nT)$ and $y[n] = y_c(nT)$. In the frequency domain,

$$\begin{aligned} X_c(j\Omega) &= TX(e^{j\Omega T}), && |\Omega| < \pi/T, \\ Y_c(j\Omega) &= H_c(j\Omega)X_c(j\Omega), && |\Omega| < \pi/T, \\ Y(e^{j\omega}) &= \frac{1}{T} Y_c\left(j\frac{\omega}{T}\right), && |\omega| < \pi. \end{aligned}$$

The overall system therefore behaves like a discrete-time system (such that $Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega})$) with frequency response

$$H(e^{j\omega}) = H_c(j\Omega) = H_c\left(j\frac{\omega}{T}\right) \qquad |\omega| < \pi.$$

Equivalently, the overall frequency response of the system will be equal to a given $H(e^{j\omega})$ if the frequency response of the continuous-time system is

$$H_c(j\Omega) = H(e^{j\Omega T}), \qquad |\Omega| < \pi/T.$$

Since $X_c(j\Omega) = 0$ for $|\Omega| \geq \pi/T$, $H_c(j\Omega)$ may be chosen arbitrarily above π/T.

4.6 Changing Sampling Rate Using Discrete-time Processing

Given the sequence

$$x[n] = x_c(nT)$$

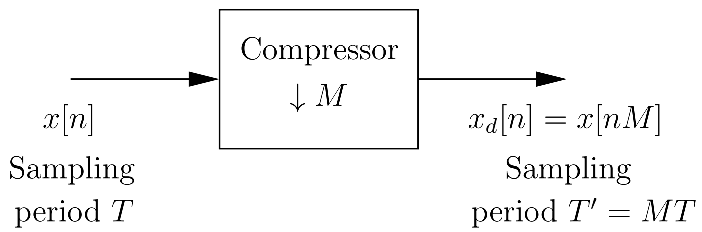
obtained by sampling (with period T) the signal $x_c(t)$, we often want to change the sampling rate (to period T'):

$$x'[n] = x_c(nT').$$

One approach is to reconstruct $x_c(t)$ from $x[n]$, and then resample with new period T'. However, we want to do this using only discrete-time operations since these are more predictable and reproducible.

4.6.1 Decreasing Sampling Rate By Integer Factor

The sampling rate **compressor** can be visualised as follows, where M is an integer:



This implements the function

$$x_d[n] = x[nM] = x_c(nMT),$$

where $x_d[n]$ is exactly the sequence that would be obtained by sampling $x_c(t)$ with period $T' = MT$. This process is also known as **downsampling** or **decimation** by an integer factor M — not just because it sounds cool, but also because it's a dangerous operation since are we actively discarding samples of our signal. This effectively means that we increase the sampling period and thus decrease our sampling rate.

If $X_c(j\Omega) = 0$ for $|\Omega| \geq \Omega_N$, then $x_d[n]$ is an exact (unaliased) representation of $x_c(t)$ if $\pi/(MT) \geq \Omega_N$.

While it is more important to visualise the inner workings of this process rather than focus on the underlying mathematics, we can still denote what happens to our original signal algebraically. In the frequency domain we have

$$X(e^{j\omega}) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega}{T} - \frac{2\pi k}{T}\right)\right)$$

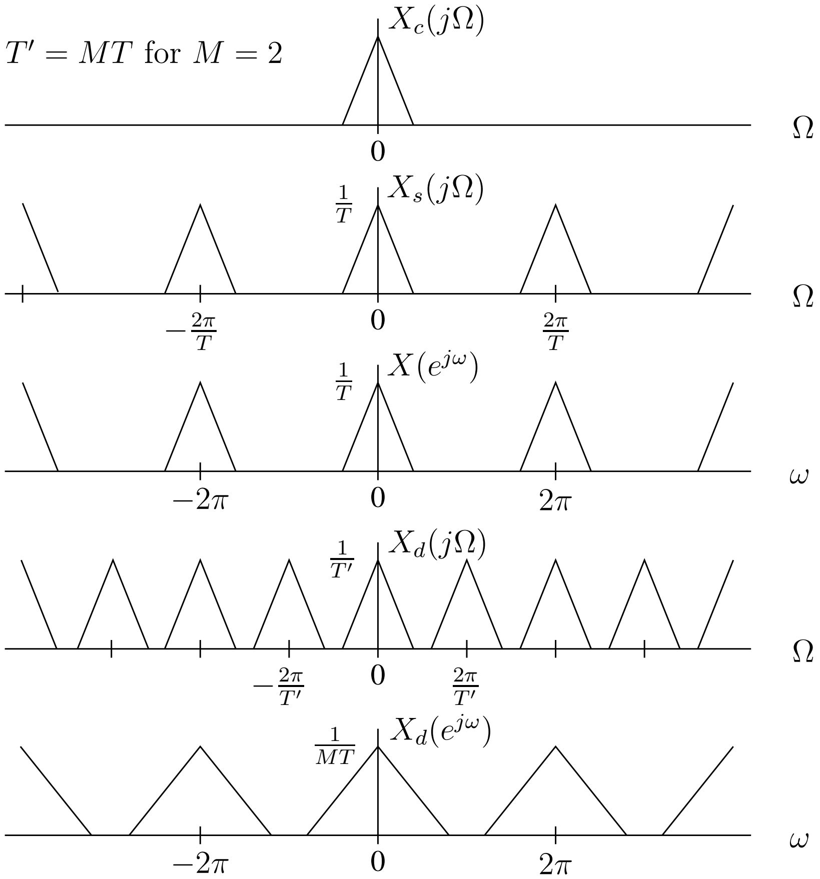
and

$$X_d(e^{j\omega}) = \frac{1}{T'}\sum_{r=-\infty}^{\infty} X_c\left(j\left(\frac{\omega}{T'} - \frac{2\pi r}{T'}\right)\right).$$

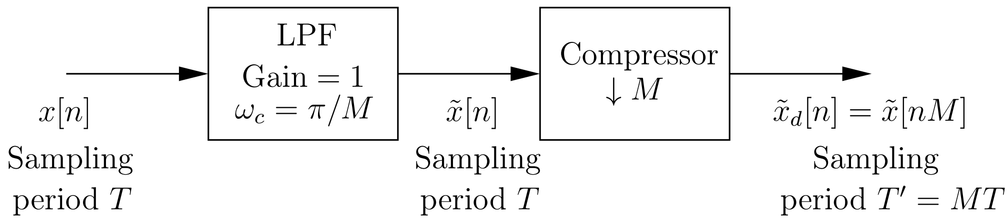
Since $T' = MT$, and noting that with $r = i + kM$ we can write the summation over r as a summation over $-\infty < k < \infty$ and $0 \leq i \leq M-1$, we obtain

$$\begin{aligned} X_d(e^{j\omega}) &= \frac{1}{M}\sum_{i=0}^{M-1}\left[\frac{1}{T}\sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega}{MT} - \frac{2\pi k}{T} - \frac{2\pi i}{MT}\right)\right)\right]. \\ &= \frac{1}{M}\sum_{i=0}^{M-1} X(e^{j(\omega/M - 2\pi i/M)}). \end{aligned}$$

The above expression is quite horrible, but the main takeaway is that decimation ends up producing expanded replicas in the frequency domain, as well as compression in the time domain. In the frequency domain, we can visualise this as below:



Note that $\omega = \Omega T'$. As a result, looking at plots 3 and 5 above, we can easily see that the DTFT has become expanded (horizontally) as well as scaled in amplitude; from this, we can see how applying a compressor to a signal could result in aliasing since the widened replicas could easily overlap. This could be avoided (at the cost of some information) by prefiltering with a lowpass filter, and then compressing the sampling rate:



Technically, looking at the previous (horrible) equation, what is really going on is that we have an expansion on the DTFT frequency axis (since we scale ω by $1/M$), causing all replicas at multiples of 2π to move to multiples of 4π; the rest of the summation is then actually adding *additional* replicas in between these expanded replicas, and these are added at multiples of 2π.

Downsampling is useful for cases where we have oversampled and want to reduce the size of our sequence, or improve processing speeds, or to simply match another signal's (lower) sampling rate – among other things. But just bear in mind that decimation can often result in a loss of signal information and should be used very carefully; oftentimes, we instead use downsampling after first *upsampling* a signal. Speaking of which...

4.6.2 Increasing Sampling Rate By Integer Factor

With underlying continuous-time signal $x_c(t)$, we want to obtain samples

$$x_i[n] = x_c(nT')$$

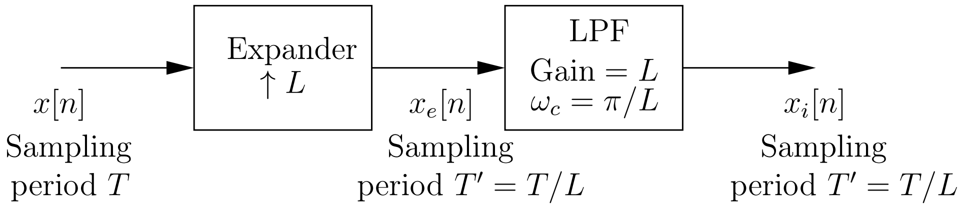
from

$$x[n] = x_c(nT),$$

where $T' = T/L$ for L an integer, leading to a lower sampling period and hence a higher sampling rate. Therefore

$$x_i[n] = x[n/L] = x_c(nT/L), \qquad n = 0, \pm L, \pm 2L, \ldots.$$

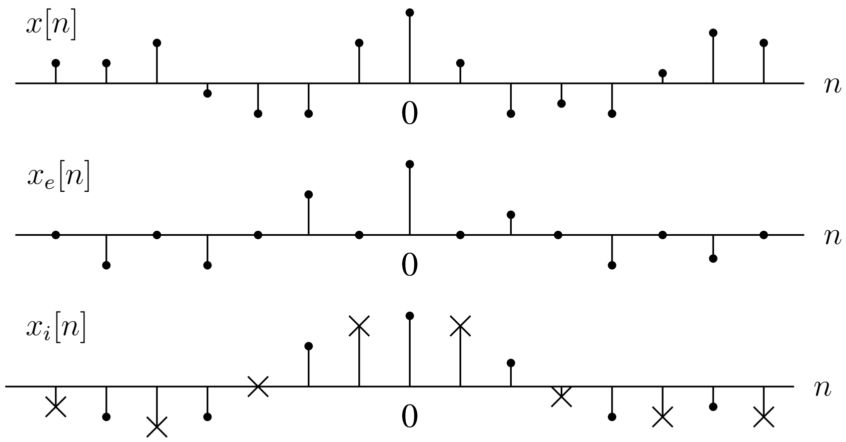
This is referred to as **upsampling** (or interpolating) by a factor L, and is performed by **expanding** the sampling rate, and then lowpass filtering:



For k an integer, the expanded signal is thus

$$x_e[n] = \begin{cases} x[n/L], & n = 0, \pm L, \pm 2L, \ldots = kL \\ 0, & \text{otherwise,} \end{cases}$$
$$= \sum_{k=-\infty}^{\infty} x[k]\delta[n - kL]$$

An example of upsampling in the discrete-time domain is shown below for $L = 2$:



As shown, the expanded signal $x_e[n]$ is simply the original sequence after being pulled apart (horizontally) away from the origin, and the intermediate values between samples are replaced by zeroes. The lowpass filter is then responsible for generating the interpolated values in place of these zeroes, resulting in $x_i[n]$ – which can be seen as the time-domain convolution of $x_e[n]$ with the filter's impulse response given by

$$h_i[n] = \frac{\sin(\pi n/L)}{\pi n/L}.$$

We can thus obtain an interpolation formula for $x_i[n]$ in terms of $x[n]$ as

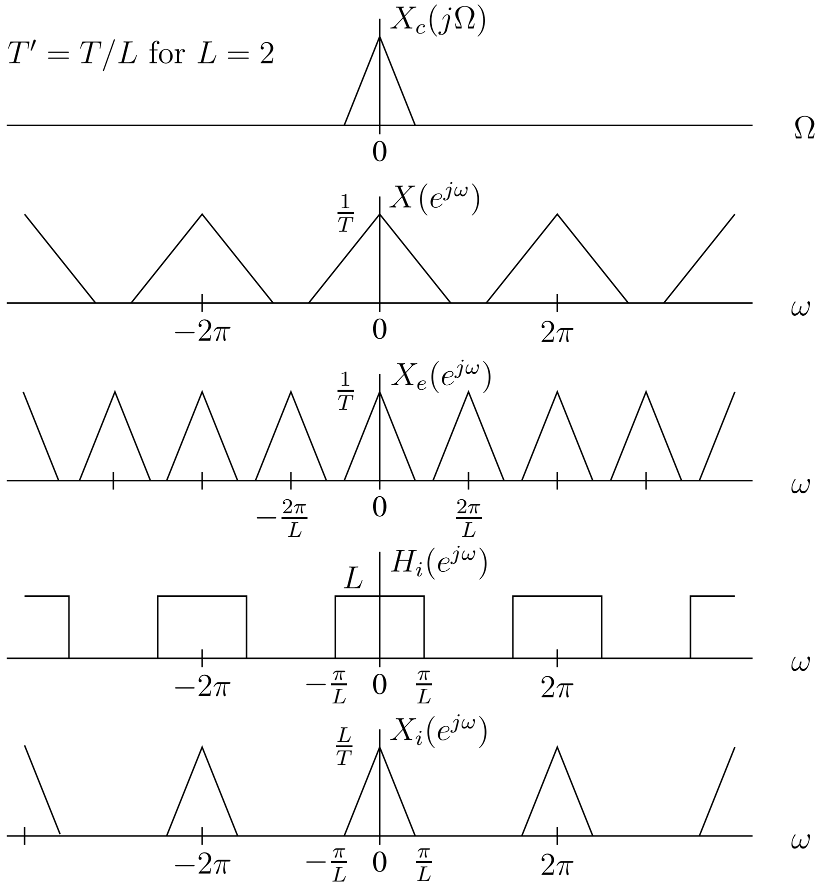
$$x_i[n] = \sum_{k=-\infty}^{\infty} x[k] \frac{\sin[\pi(n - kL)/L]}{\pi(n - kL)/L}.$$

We can also compute the Fourier transform of the expanded signal as

$$\begin{aligned}
X_e(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x_e[n]e^{-j\omega n} \\
&= \sum_{n=-\infty}^{\infty} \left(\sum_{k=-\infty}^{\infty} x[k]\delta[n-kL] \right) e^{-j\omega n} \\
&= \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x[k]\delta[n-kL]e^{-j\omega kL} \\
&= \sum_{k=-\infty}^{\infty} x[k]e^{-j\omega kL} = X(e^{j\omega L}).
\end{aligned}$$

We can make this simplification by noticing that the sifting property applies, and since the impulses only have a value of 1 when $n = kL$, the sum over n essentially falls away. For example, when $n = L$, the summand will be zero for all values of k except when $k = 1$, and thus the only term that gets summed is $x[1]e^{-j\omega L}$. As this applies for any value of n, the double sum has the same effect as the single sum in the final line of the equation above. If necessary, write out a few terms and prove it to yourself!

As stated above, the final upsampled signal is then obtained by lowpass filtering the expanded signal. For the case $L = 2$, we can visualise all of this as follows:

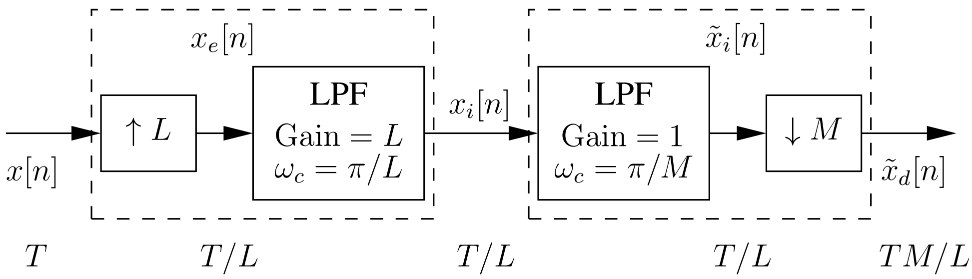


Here we can see how the DTFT of the original signal (in the second plot) gets compressed in the frequency domain into the form $X_e(e^{j\omega})$, which is then lowpass filtered to produce the interpolated version $X_i(e^{j\omega})$.

Upsampling therefore corresponds to a compression in the frequency domain and an expansion in the time domain – a much safer operation than downsampling. In signal and image processing, upsampling can be used to increase the resolution of an image/video, match the sampling rate of another signal, improve filtering, and more.

4.6.3 Changing the Sampling Rate By a Non-integer Factor

By cascading upsampling (by factor L) and downsampling (by factor M), the sampling rate can be changed by a non-integer factor as shown below:



The order of operations here is quite important, as we first upsample the signal and *then* downsample to avoid losing information. If we had done this the other way round, the downsampling process would likely have removed parts of our original signal, and the upsampler would not be able to reconstruct these destroyed samples – it would only interpolate the samples that were left.

As an example, say we wanted to resample an audio signal from 44.1 kHz to 48 kHz; naturally, this involves upsampling the original signal, but since we need to upsample by a non-integer factor (48000/44100), we can make use of a multirate system comprised of upsampling and downsampling processes. While there are other ways of doing this, the simplest approach is to represent the resampling ratio as a ratio of the smallest possible integers.

For instance, if we plug 48000/44100 into a calculator, it would return the fraction 160/147, and this is exactly what we want – the smallest possible integers that give us the required ratio of sampling rates. We can then equate this ratio to L/M, setting $L = 160$ and $M = 147$ in the above resampling pipeline to give us the exact sampling rate that we desired at the output (i.e., 48 kHz).

Another important consideration is the lowpass filtering process; the block diagram above shows two filters right after each other, which we could actually join into a *single* lowpass filter to better optimise our pipeline. We could thus reduce this down to a single filter with a gain of L (or even M, if we want to counteract the scaling of $1/M$ from the compressor), but what would the cut-off frequency need to be?

Generally, we just take the minimum of the two cut-offs shown above, such that $\omega_c = \min(\pi/L, \pi/M)$; this makes sense since we want to be sure to satisfy the requirements of both filters, so we keep the overall passband as small as possible by minimising the cut-off frequency.

The above ideas form the basis of **multirate signal processing**, where highly efficient structures are developed for implementing complicated signal processing operations. The **Discrete Wavelet Transform** can be developed in this framework, but this is not covered in this course. Definitely worth a Google though!

Chapter 5

The Discrete Fourier Transform

5.1 Introduction

The discrete-time Fourier transform (DTFT) of a sequence is a continuous function of ω, and repeats with period 2π. In practice we usually want to obtain the Fourier components using digital computation, and can only evaluate them for a discrete set of frequencies. The discrete Fourier transform (DFT) provides a means for achieving this.

The DFT is itself a sequence, and it corresponds roughly to samples, equally spaced in frequency, of the Fourier transform of the signal. The discrete Fourier transform of a length N signal $x[n]$, $n = 0, 1, \ldots, N-1$ is given by

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j(2\pi/N)kn}.$$

This is the analysis equation. The corresponding synthesis equation is

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j(2\pi/N)kn}.$$

When dealing with the DFT, it is common to define the complex quantity

$$W_N = e^{-j(2\pi/N)}.$$

With this notation the DFT analysis-synthesis pair becomes

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}$$
$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}.$$

An important property of the DFT is that it is cyclic, with period N, both in the discrete-time and discrete-frequency domains. For example, for any integer r,

$$\begin{aligned} X[k+rN] &= \sum_{n=0}^{N-1} x[n] W_N^{(k+rN)n} = \sum_{n=0}^{N-1} x[n] W_N^{kn} (W_N^N)^{rn} \\ &= \sum_{n=0}^{N-1} x[n] W_N^{kn} = X[k], \end{aligned}$$

since $W_N^N = e^{-j(2\pi/N)N} = e^{-j2\pi} = 1$. Similarly, it is easy to show that $x[n+rN] = x[n]$, implying periodicity of the synthesis equation. This is important — even though the DFT only depends on samples in the interval 0 to $N-1$, it is implicitly assumed that the signals repeat with period N in both the time and frequency domains.

To this end, it is sometimes useful to define the periodic extension of the signal $x[n]$ to be

$$\tilde{x}[n] = x[n \bmod N] = x[((n))_N].$$

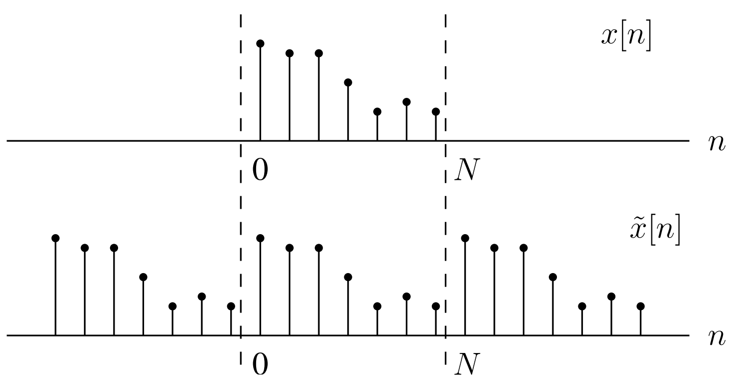
Here $n \bmod N$ and $((n))_N$ are taken to mean n modulo N, which has the value of the remainder after n is divided by N. Another way to view this is that we always check whether n lies in the range 0 to $N-1$, and if not, we add or subtract multiples of N until it is. A few examples of this "wraparound" effect are shown below:

$$\begin{aligned} 10 \bmod 7 &= (10-7) \bmod 7 = 3 \bmod 7 = 3 \\ 14 \bmod 7 &= (14-7) \bmod 7 = 7 \bmod 7 = (7-7) \bmod 7 = 0 \bmod 7 = 0 \\ -2 \bmod 7 &= (-2+7) \bmod 7 = 5 \bmod 7 = 5 \end{aligned}$$

Alternatively, if n is written in the form $n = kN + l$ for $0 \le l < N$, then

$$n \bmod N = ((n))_N = l.$$

This can be represented graphically for the case $N = 7$ as follows:



where we can see how $x[n]$ (from $n = 0$ to $n = 6$) has been periodically extended to $\tilde{x}[n]$, such that it is now a periodic signal with period $N = 7$.

Similarly, the periodic extension of $X[k]$ is defined to be

$$\tilde{X}[k] = X[k \text{ mod } N] = X[((k))_N].$$

It is sometimes better to reason in terms of these periodic extensions when dealing with the DFT. Specifically, if $X[k]$ is the DFT of $x[n]$, then the inverse DFT of $X[k]$ is $\tilde{x}[n]$. The signals $x[n]$ and $\tilde{x}[n]$ are identical over the interval 0 to $N-1$, but may differ outside of this range. Similar statements can be made regarding the transform $X[k]$.

5.2 Properties of the DFT

Many of the properties of the DFT are analogous to those of the DTFT, with the notable exception that all shifts involved must be considered to be *circular*, or modulo N, since we always want to bring/keep the signal into the range we care about (from 0 to $N-1$) – both in the time (n) and frequency (k) domains.

Defining the DFT pairs $x[n] \overset{\mathcal{D}}{\longleftrightarrow} X[k]$, $x_1[n] \overset{\mathcal{D}}{\longleftrightarrow} X_1[k]$, and $x_2[n] \overset{\mathcal{D}}{\longleftrightarrow} X[k]$, the following are properties of the DFT:

- **Symmetry** (for real $x[n]$)**:**

$$\begin{aligned} X[k] &= X^*[((-k))_N] \\ \text{Re}\{X[k]\} &= \text{Re}\{X[((-k))_N]\} \\ \text{Im}\{X[k]\} &= -\text{Im}\{X[((-k))_N]\} \\ |X[k]| &= |X[((-k))_N]| \\ \sphericalangle X[k] &= -\sphericalangle X[((-k))_N] \end{aligned}$$

- **Linearity:** $ax_1[n] + bx_2[n] \overset{\mathcal{D}}{\longleftrightarrow} aX_1[k] + bX_2[k]$.
- **Circular time shift:** $x[((n-m))_N] \overset{\mathcal{D}}{\longleftrightarrow} W_N^{km} X[k]$.
- **Circular convolution (for LTI systems):**

$$\sum_{m=0}^{N-1} x_1[m] x_2[((n-m))_N] \overset{\mathcal{D}}{\longleftrightarrow} X_1[k] X_2[k].$$

Circular convolution between two N-point signals is denoted by $x_1[n] \, \text{Ⓝ} \, x_2[n]$.

- **Modulation (circular convolution in frequency):**

$$x_1[n]x_2[n] \quad \overset{\mathcal{D}}{\longleftrightarrow} \quad \frac{1}{N}\sum_{l=0}^{N-1} X_1[l]X_2[((k-l))_N].$$

Some of these properties, such as linearity, are easy to prove. The properties involving time shifts can be quite confusing notationally, but are otherwise quite simple. For example, consider the 4-point DFT

$$X[k] = \sum_{n=0}^{3} x[n]W_4^{kn}$$

of the length 4 signal $x[n]$. This can be written as

$$X[k] = x[0]W_4^{0k} + x[1]W_4^{1k} + x[2]W_4^{2k} + x[3]W_4^{3k}$$

The product $W_4^{1k}X[k]$ can therefore be written as

$$\begin{aligned} W_4^{1k}X[k] &= x[0]W_4^{1k} + x[1]W_4^{2k} + x[2]W_4^{3k} + x[3]W_4^{4k} \\ &= x[3]W_4^{0k} + x[0]W_4^{1k} + x[1]W_4^{2k} + x[2]W_4^{3k} \end{aligned}$$

since $W_4^{4k} = W_4^{0k}$. This can be seen to be the DFT of the sequence $x[3], x[0], x[1], x[2]$, which is precisely the sequence $x[n]$ *circularly shifted* to the right by one sample. This proves the time-shift property for a shift of length 1. In general, multiplying the DFT of a sequence by W_N^{km} results in an N-point *circular* shift of the sequence by m samples. The convolution properties can be similarly demonstrated.

It is useful to note that the circularly shifted signal $x[((n-m))_N]$ is the same as the linearly shifted signal $\tilde{x}[n-m]$, where $\tilde{x}[n]$ is the N-point periodic extension of $x[n]$. This is good to know as it is sometimes easier (and less confusing) to work with the periodic extension directly and perform linear operations on that signal, rather than working with the n-point signal and having to explicitly worry about wraparounds.

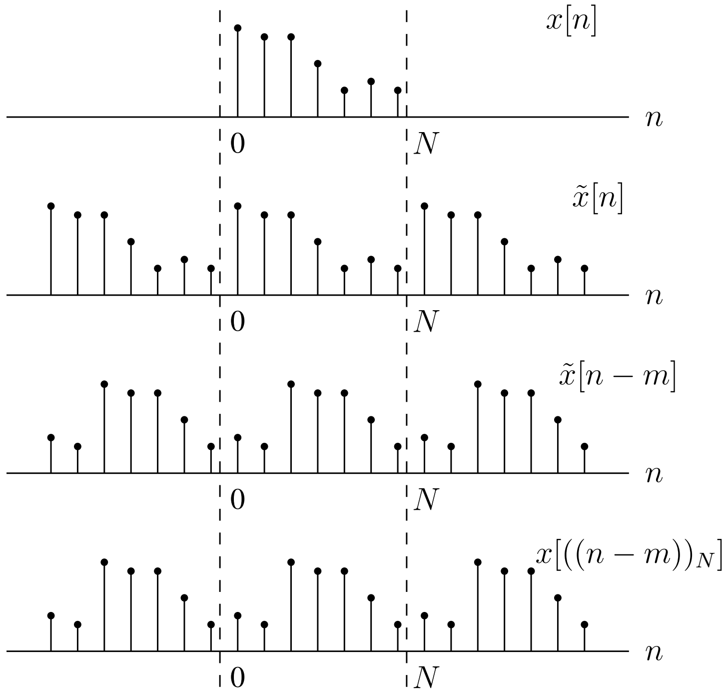
On the interval 0 to $N-1$, the circular convolution

$$x_3[n] = x_1[n] \, \textcircled{N} \, x_2[n] = \sum_{m=0}^{N-1} x_1[m]x_2[((n-m))_N]$$

can therefore be calculated using the linear convolution product below, and thus circular convolution is really just periodic convolution:

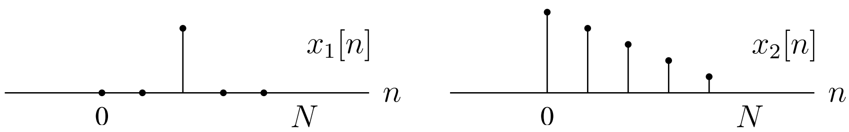
$$x_3[n] = \sum_{m=0}^{N-1} x_1[m]\tilde{x}_2[n-m].$$

The circular time shift concept is illustrated below:

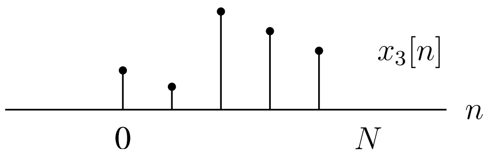


Example: Circular convolution with a delayed impulse sequence

Given the sequences below with $N = 5$



the circular convolution $x_3[n] = x_1[n] \, \text{(N)} \, x_2[n]$ is the signal $\tilde{x}_2[n]$ delayed by two samples, evaluated over the range 0 to $N-1$:



Example: Circular convolution of two rectangular pulses

Let

$$x_1[n] = x_2[n] = \begin{cases} 1 & 0 \le n \le L-1 \\ 0 & \text{otherwise.} \end{cases}$$

If $N = L$, then the N-point DFTs are

$$X_1[k] = X_2[k] = \sum_{n=0}^{N-1} W_N^{kn} = \begin{cases} N & k = 0 \\ 0 & \text{otherwise.} \end{cases}$$

This is because the summand becomes 1 when $k = 0$ and we sum it N times, and for any other value of k, we get a sum of complex exponentials that will add together to produce 0 – if you need to prove this to yourself then draw out a single case (such as for $k = 1$) and you should notice that the complex exponentials will be equally spaced around a unit circle in the complex plane, so the terms all cancel out to zero when summed.

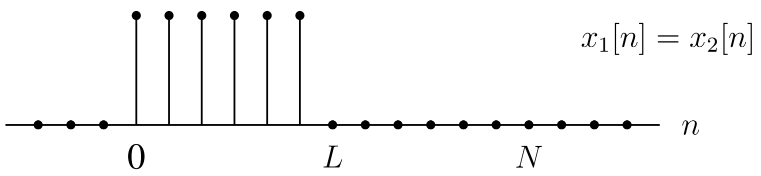
In the frequency domain, the product of these two DFTs is thus

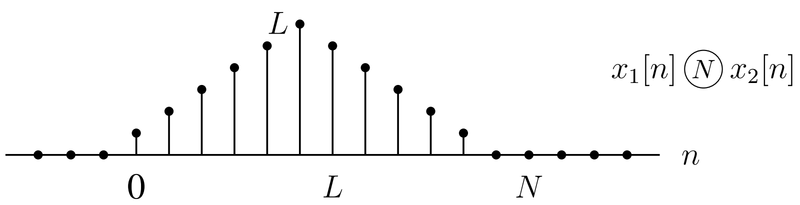
$$X_3[k] = X_1[k]X_2[k] = \begin{cases} N^2 & k = 0 \\ 0 & \text{otherwise,} \end{cases}$$

The N-point circular convolution of $x_1[n]$ and $x_2[n]$ can thus be found by taking the inverse DFT of this expression

$$\begin{aligned} x_3[n] &= \frac{1}{N}\sum_{k=0}^{N-1} X_3[k]W_N^{-kn} \\ x_3[n] &= \frac{1}{N}\left(N^2 W_N^0\right) = N \\ \therefore x_3[n] &= x_1[n] \, N \, x_2[n] = N, \qquad 0 \leq n \leq N-1. \end{aligned}$$

Suppose now that $x_1[n]$ and $x_2[n]$ are considered to be length $2L$ sequences by padding each signal with zeros. The $N = 2L$-point circular convolution is then seen to be the same as the linear convolution of the finite-duration sequences $x_1[n]$ and $x_2[n]$:





where $N = 2L$ and thus the signal spans from 0 to $2L - 1$. The next section dives into this idea of "zero padding" further.

5.3 Linear Convolution Using the DFT

Using the DFT we can compute the circular convolution as follows

- Compute the N-point DFTs $X_1[k]$ and $X_2[k]$ of the two sequences $x_1[n]$ and $x_2[n]$.
- Compute the product $X_3[k] = X_1[k]X_2[k]$ for $0 \leq k \leq N-1$.
- Compute the sequence $x_3[n] = x_1[n] \circledN x_2[n]$ as the inverse DFT of $X_3[k]$.

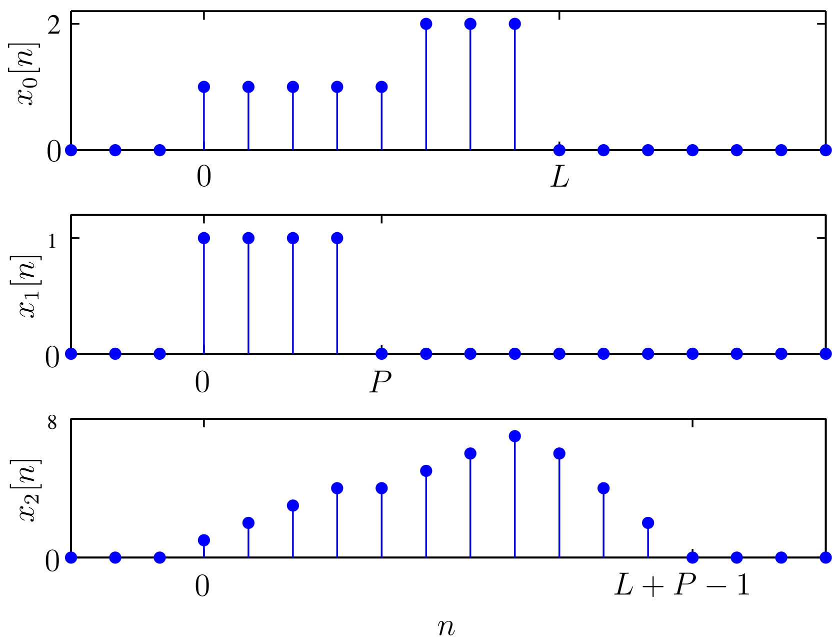
This is computationally useful due to efficient algorithms for calculating the DFT. The question that now arises is this: how do we get the *linear* convolution (required in speech, radar, sonar, image processing) from this procedure?

5.3.1 Linear Convolution of Two Finite-length Sequences

Consider a sequence $x_1[n]$ with length L points, and $x_2[n]$ with length P points. The linear convolution of the sequences,

$$x_3[n] = \sum_{m=-\infty}^{\infty} x_1[m]x_2[n-m],$$

is non-zero over a maximum length of $L+P-1$ points. The below visualisations may assist in demonstrating this:



If you want a mathematical explanation for this, think back to the definition of convolution, where we flip and shift one of our signals and then multiply it with the other signal on the m-axis and sum all the products together. Imagine if we took $x_2[n]$ above and flipped it over the vertical axis – the signal would start at $m = -P+1$ and then end at $m = 0$. So when we shift this to the right by n, the final bit of "overlap" between our signals would be when the first $x_2[n]$ impulse (at $m = -P + 1 + n$) is muliplied with the final $x_1[n]$ impulse at $m = L - 1$; this corresponds to a time shift of $n = (P - 1) + (L - 1) = L + P - 2$, and thus this is the final non-zero value that the convolution will produce.

As a result, we can see that the convolution output is non-zero only over the range $n = 0$ to $n = L + P - 2$, and thus the non-zero part of the sequence has a maximum length of $L + P - 1$.

For comparison, the N-point circular convolution of $x_1[n]$ and $x_2[n]$ is

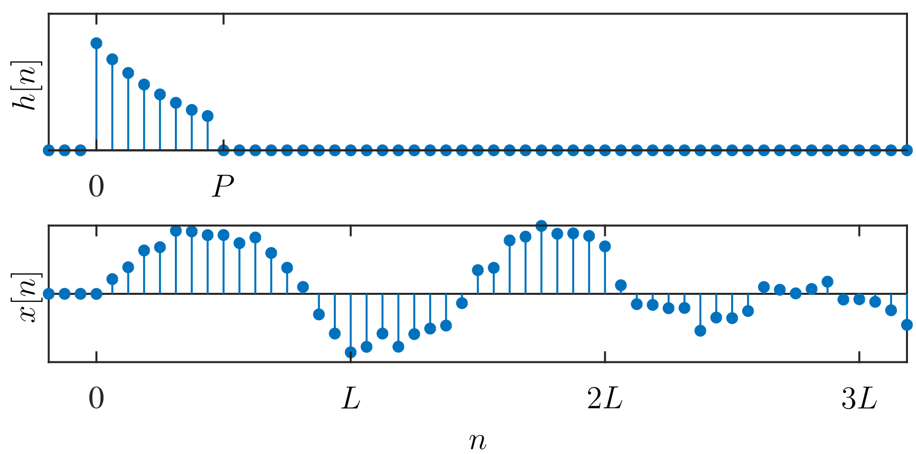
$$x_1[n] \, (N) \, x_2[n] = \sum_{m=0}^{N-1} x_1[m] x_2[((n-m))_N] = \sum_{m=0}^{N-1} x_1[m] \tilde{x}_2[n-m] :$$

It is easy to see from this that the circular convolution product will be equal to the linear convolution product on the interval 0 to $N - 1$ as long as we choose $N \geq L + P - 1$. The process of augmenting a sequence with zeros to make it of a required length is called **zero padding**; this is useful for finite signals, where we can zero pad both signals to the size of the sum of their original lengths. In doing so, the circular convolution produces the same result as linear convolution!

5.3.2 Convolution By Sectioning

Suppose that for computational efficiency we want to implement a FIR system using DFTs. It cannot in general be assumed that the input signal has a finite duration, such as in the case of working with livestreamed audio/video signals, so the methods described up to now cannot be applied directly.

The solution is to use what we call **block convolution**, where the signal to be filtered is segmented into multiple sections each of length L. For example, imagine that we had the following system impulse response $h[n]$ and input signal $x[n]$:

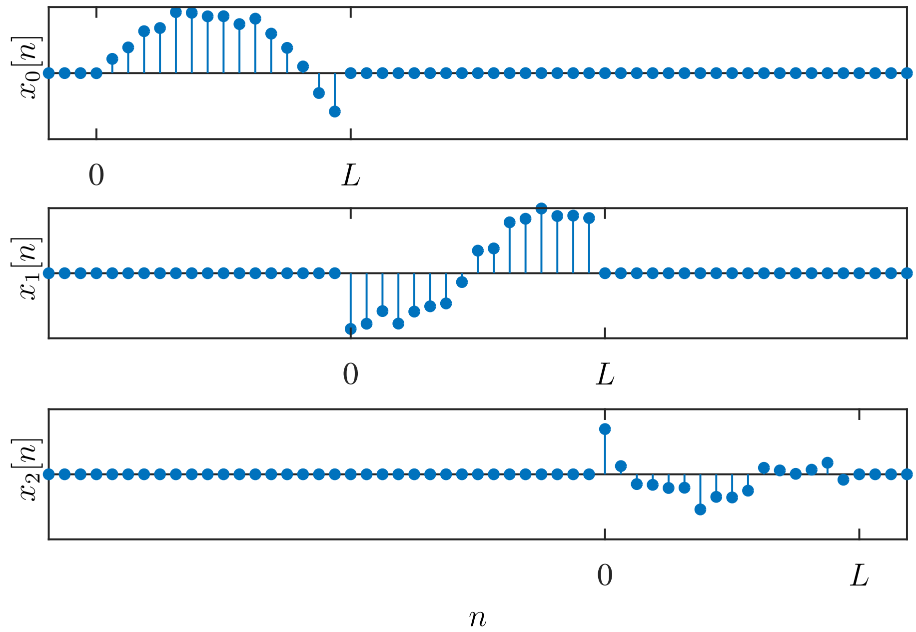


Assuming causality applies, the input signal $x[n]$ could be decomposed into blocks of length L (or portions) as follows:

$$x[n] = \sum_{r=0}^{\infty} x_r[n - rL],$$

for r an integer. The portions can thus be described by

$$x_r[n] = \begin{cases} x[n + rL] & 0 \leq n \leq L - 1 \\ 0 & \text{otherwise.} \end{cases}$$



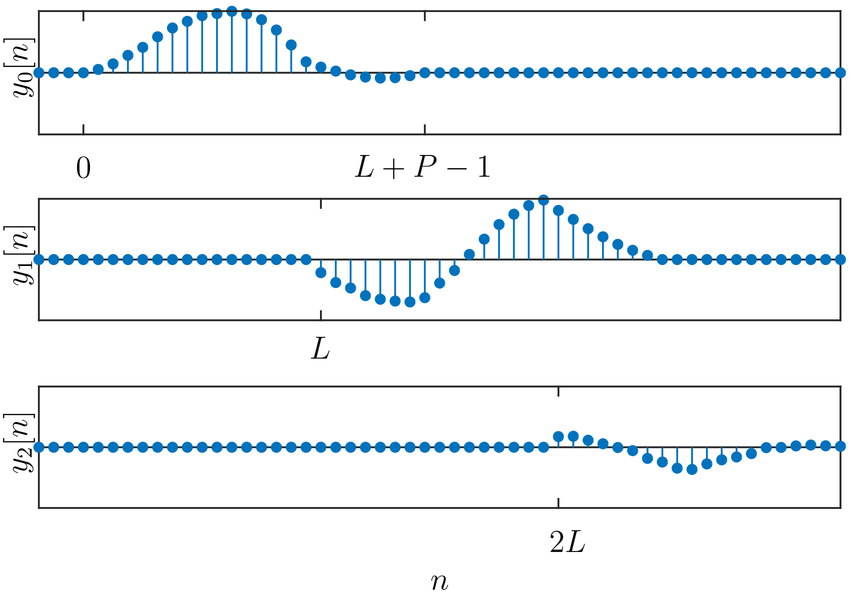
And the convolution product could therefore be written as the sum of the various output signal portions:

$$y[n] = x[n] * h[n] = \sum_{r=0}^{\infty} y_r[n - rL],$$

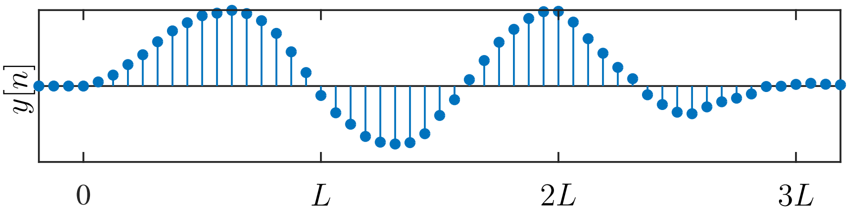
where $y_r[n]$ is the response of the system to each input portion, producing output portions in the form

$$y_r[n] = x_r[n] * h[n].$$

These sequences can then be plotted as below, with each portion starting at rL:



Finally, the output sequences can be added together:



Since the sequences $x_r[n]$ have only L non-zero points and $h[n]$ is of length P, each response term $y_r[n]$ has length $L + P - 1$. Thus linear convolution can be obtained using N-point DFTs with $N \geq L + P - 1$. Since the final result is obtained by summing the overlapping output regions, this is called the **overlap-add** method.

An alternative block convolution procedure, called the **overlap-save** method, corresponds to implementing an L-point circular convolution of a P-point impulse response $h[n]$ with an L-point segment $x_r[n]$. The portion of the output that corresponds to linear convolution is then identified (consisting of $L - (P - 1)$ points), and the resulting segments are patched together to form the output. This method effectively discards the output overlaps and then concatenates the non-overlapping regions together, resulting in a slightly more efficient approach since the discarded overlaps are accounted for in the next signal portion.

While block convolution is useful, it is important to keep in mind that the computations are always L samples behind, as we need to wait for the full L samples to arrive before we can compute the output for that signal portion. Logically, this introduces latency, which can be a problem for some applications where real-time processing is needed.

However, it is also worth noting that the DFT is more efficient for larger L (or N), so the size of L represents a trade-off in efficiency; if we increase L then the DFT operation itself becomes more efficient, but this also increases the latency of the system since we need to wait for more samples to arrive before being able to process the signal.

5.4 Spectrum Estimation Using the DFT

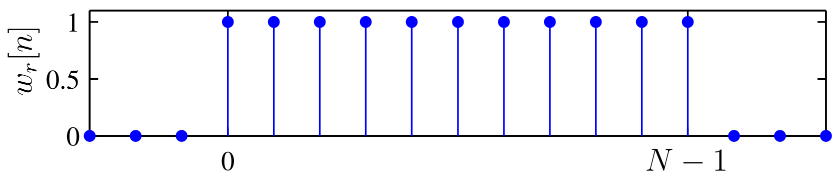
5.4.1 Relation to the DTFT

Spectrum estimation is the task of estimating the DTFT of a signal $x[n]$. The DTFT of a discrete-time signal $x[n]$ is

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}.$$

The signal $x[n]$ is generally of infinite duration, and $X(e^{j\omega})$ is a continuous function of ω that tells us how much of frequency ω is present in the signal. But because the DTFT is a continuous function, we cannot truly calculate it using a computer.

Consider now that we truncate $x[n]$ by multiplying with the rectangular window $w_r[n]$:



The windowed signal is then $x_w[n] = x[n]w_r[n]$. The DTFT of this (truncated) windowed signal is given by

$$X_w(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x_w[n]e^{-j\omega n} = \sum_{n=0}^{N-1} x_w[n]e^{-j\omega n}.$$

Noting that the DFT of $x_w[n]$ is

$$X_w[k] = \sum_{n=0}^{N-1} x_w[n]e^{-j\frac{2\pi kn}{N}},$$

it is evident that

$$X_w[k] = X_w(e^{j\omega})\big|_{\omega=2\pi k/N}.$$

The values of the DFT $X_w[k]$ of the signal $x_w[n]$ are therefore periodic samples of the DTFT $X_w(e^{j\omega})$, where the spacing between the samples is $2\pi/N$. Since the relationship between the discrete-time frequency variable and the continuous-time frequency variable is $\omega = \Omega T$, the DFT frequencies correspond to continuous-time frequencies

$$\Omega_k = \frac{2\pi k}{NT}.$$

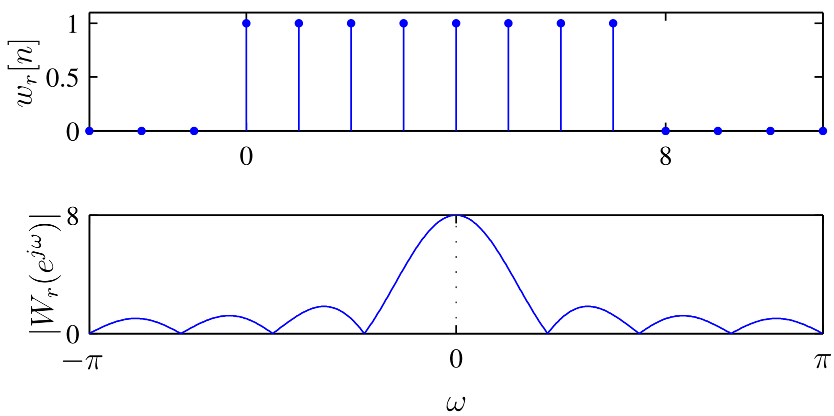
The DFT can therefore only find points on the DTFT of the *windowed* signal $x_w[n]$.

5.4.2 Windowing

The operation of windowing involves multiplication in the discrete time domain, which corresponds to continuous-time periodic convolution in the DTFT frequency domain. The DTFT of the windowed signal is therefore

$$X_w(e^{j\omega}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\theta})W(e^{j(\omega-\theta)})d\theta,$$

where $W(e^{j\omega})$ is the frequency response of the window function. For a simple rectangular window, the frequency response is as follows:



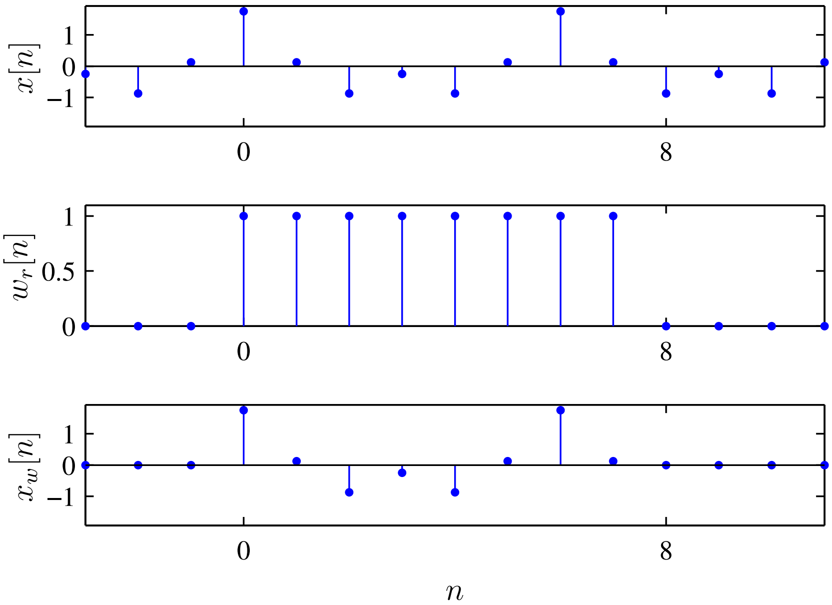
The DFT therefore effectively samples the DTFT of the signal convolved with the frequency response of the window.

Example: Spectrum analysis of sinusoidal signals

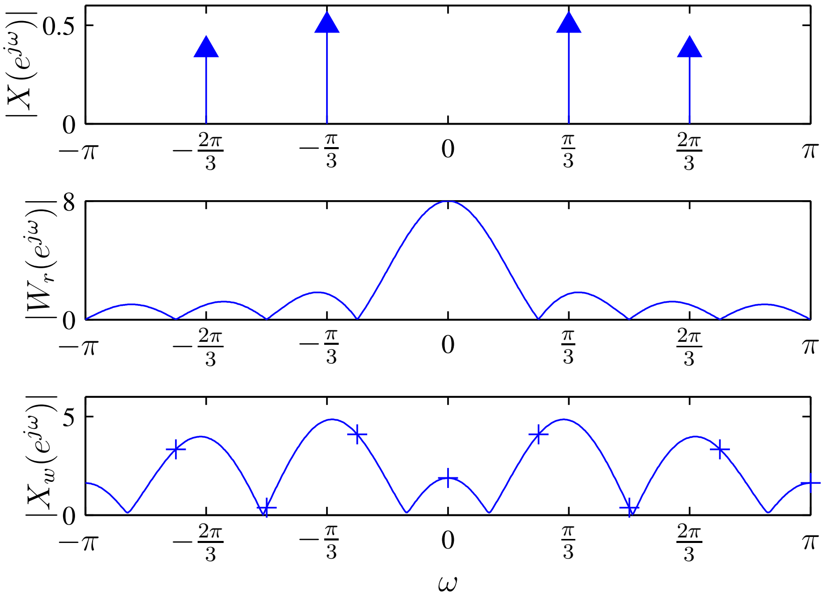
Suppose we have the sinusoidal signal combination

$$x[n] = \cos(\pi/3n) + 0.75\cos(2\pi/3n), \qquad -\infty < n < \infty.$$

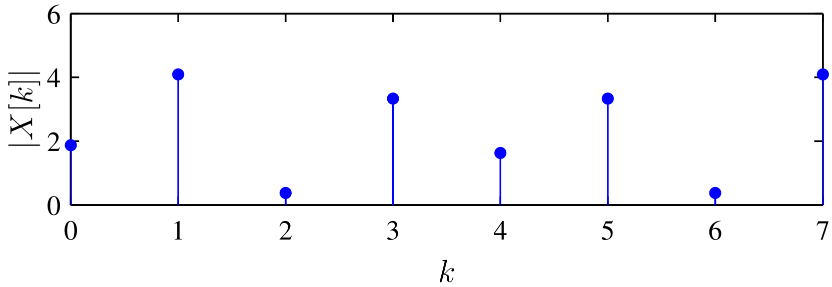
Since the signal is infinite in duration, the DFT cannot be computed numerically. We therefore window the signal (using a rectangular window, in this case) to make the duration finite:



The operation of windowing modifies the signal; in the time domain, this corresponds to a simple multiplication of the two signals, and hence in frequency we have a convolution occurring. This is reflected in the discrete-time Fourier transform domain by a *spreading* of the frequency components, as the ideal delta functions (i.e., the DTFT of the sum of cosines) effectively become sinc functions:



Windowing therefore limits the ability of the Fourier transform to resolve closely-spaced frequency components, e.g., if we wanted to resolve/separate the heartbeat frequencies of a mother and her fetus during pregnancy. Therefore, when the DFT is used for spectrum estimation, it effectively samples the spectrum of this modified signal at the locations of the crosses indicated, and thus the DFT for the above example would look like:



Note that since $k = 0$ corresponds to $\omega = 0$, there is a corresponding shift in the sampled values. Also note that we have used an 8-point DFT in the above example.

5.4.3 Improving the Estimation

In general, the elements of the N-point DFT of $x_w[n]$ contain N evenly-spaced samples from the DTFT $X_w(e^{j\omega})$. These samples span an entire period of the DTFT, and therefore correspond to frequencies at spacings of $2\pi/N$. We can obtain samples with a closer spacing by performing more computation. Suppose we form the zero-padded length M signal $x_M[n]$ as follows:

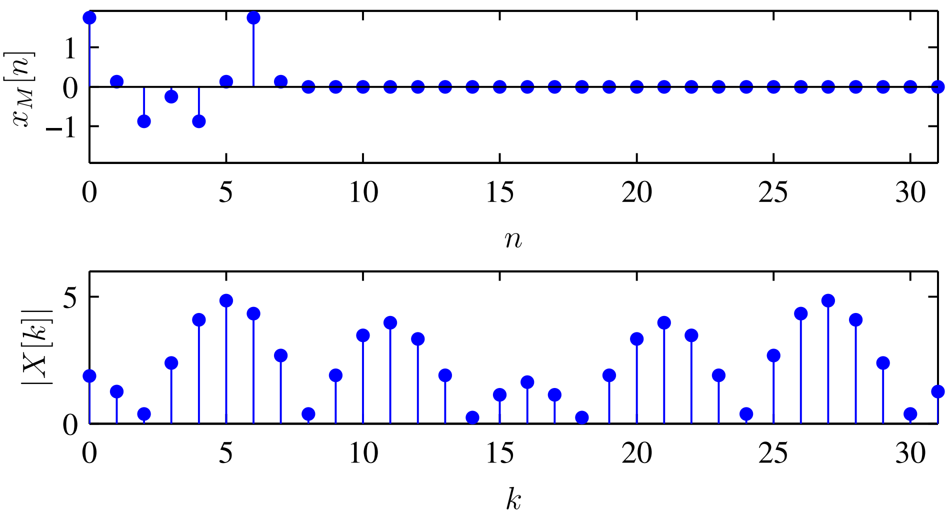
$$x_M[n] = \begin{cases} x[n] & 0 \le n \le N-1 \\ 0 & N \le n \le M-1. \end{cases}$$

The M-point DFT of this signal is

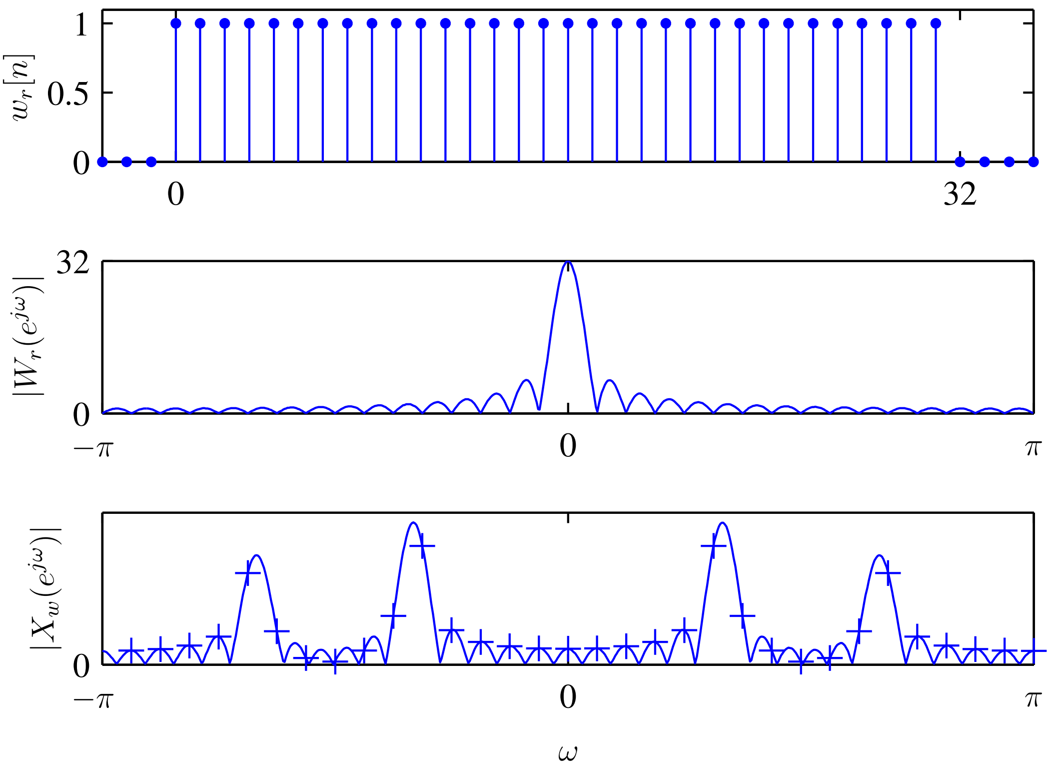
$$\begin{aligned} X_M[k] &= \sum_{n=0}^{M-1} x_M[n] e^{-j\frac{2\pi}{M}kn} = \sum_{n=0}^{N-1} x_w[n] e^{-j\frac{2\pi}{M}kn} \\ &= \sum_{n=-\infty}^{\infty} x_w[n] e^{-j\frac{2\pi}{M}kn}, \end{aligned}$$

where we write the final line in this form (with infinite limits) to denote that $X_p[k]$ corresponds to the DTFT of the windowed signal $x_w[n]$ at frequency $\omega_k = 2\pi k/M$. Since M is chosen to be larger than N, the transform values correspond to regular samples of $X_w(e^{j\omega})$ with a closer spacing of $2\pi/M$ (instead of $2\pi/N$).

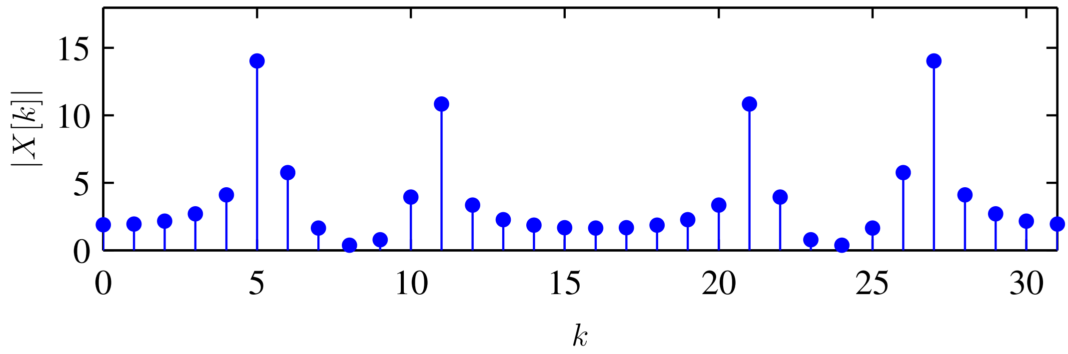
The following figure shows the magnitude of the DFT transform values for the 8-point signal shown previously, but zero-padded to use a 32-point DFT (i.e., we still have $N = 8$ but now $M = 32$):



Additionally, if $W(e^{j\omega})$ is sharply peaked, and approximates a Dirac delta function at the origin, then $X_w(e^{j\omega}) \approx X(e^{j\omega})$. The values of the DFT then correspond quite accurately to samples of the DTFT of $x[n]$. For a rectangular window, the approximation improves as N (the size of the window) increases:



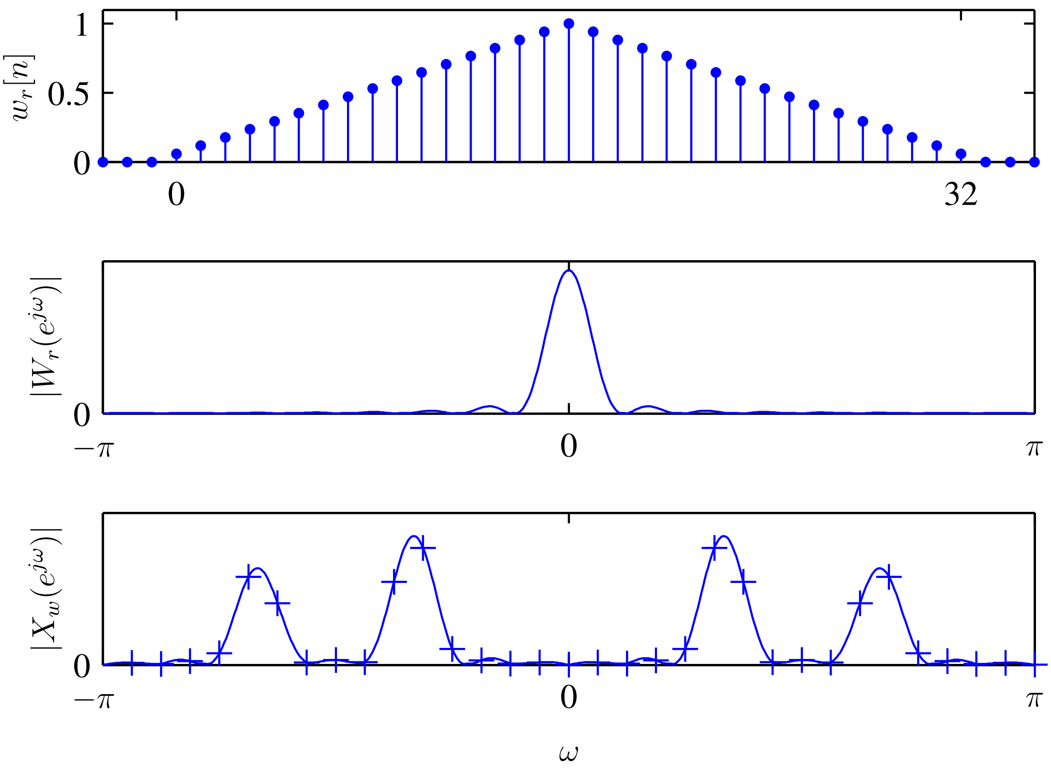
The magnitude of the DFT of the windowed signal is



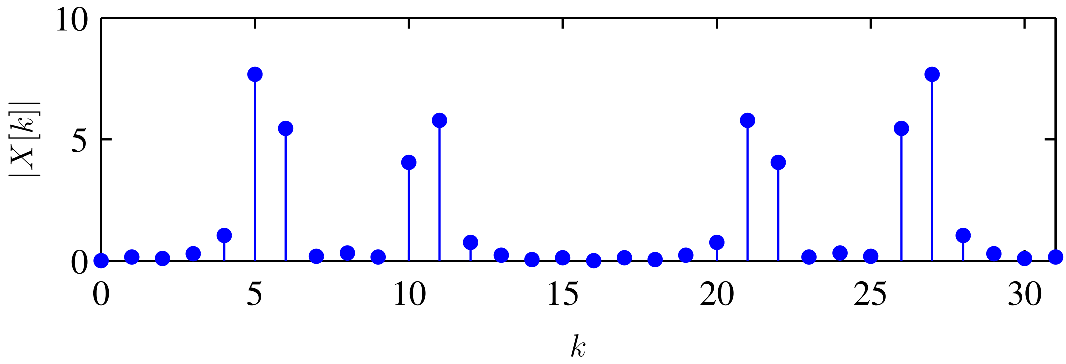
which is clearly easier to interpret than for the case of the shorter signal. As the window length tends to ∞, the relationship becomes exact and the sinc function approaches a delta function.

5.4.4 Other Windows

The rectangular window inherent in the DFT has the disadvantage that the peak sidelobe of $W_r(e^{j\omega})$ is high relative to the mainlobe. This limits the ability of the DFT to resolve frequencies. Alternative windows may be used which have preferred behaviour — the only requirement is that in the time domain the window function is of finite duration. For example, the triangular window



leads to DFT samples with the magnitudes below



This is a bit closer to the original $X(e^{j\omega})$, and the sidelobes have been reduced (less ringing) at the cost of diminished resolution — i.e., the mainlobe has become wider.

The method just described forms the basis for the *periodogram* spectrum estimate. It is often used in practice on account of its perceived simplicity. However, it has a poor statistical properties — *model-based* spectrum estimates generally have higher resolution and more predictable performance.

Finally, note that the discrete samples of the spectrum are only a complete representation if the Nyquist sampling criterion is met. The samples therefore have to be sufficiently closely spaced.

5.5 Fast Fourier Transforms

The widespread application of the DFT to convolution and spectrum analysis is due to the existence of fast algorithms for its implementation. This class of methods is referred to as **Fast Fourier Transforms** (FFTs).

Consider a direct implementation of an 8-point DFT:

$$X[k] = \sum_{n=0}^{7} x[n]W_8^{kn}, \qquad k = 0, \ldots, 7.$$

If the factors W_8^{kn} have been calculated in advance (and perhaps stored in a lookup table), then the calculation of $X[k]$ for each value of k requires 8 complex multiplications and 7 complex additions. The 8-point DFT therefore requires 8×8 multiplications and 8×7 additions. For an N-point DFT these become N^2 and $N(N-1)$ respectively. If $N = 1024$, then approximately one million complex multiplications and one million complex additions are required.

The key to reducing the computational complexity lies in the observation that the same values of $x[n]W_8^{kn}$ are effectively calculated many times as the computation proceeds — particularly if the transform is long.

The **conventional decomposition** (there are actually hundreds of different approaches!) involves **decimation-in-time**, where at each stage an N-point transform is decomposed into two $N/2$-point transforms. That is, $X[k]$ can be written as

$$\begin{aligned} X[k] &= \sum_{r=0}^{N/2-1} x[2r] W_N^{2rk} + \sum_{r=0}^{N/2-1} x[2r+1] W_N^{(2r+1)k} \\ &= \sum_{r=0}^{N/2-1} x[2r] (W_N^2)^{rk} + W_N^k \sum_{r=0}^{N/2-1} x[2r+1] (W_N^2)^{rk}. \end{aligned}$$

Noting that $W_N^2 = W_{N/2}$ this becomes

$$\begin{aligned} X[k] &= \sum_{r=0}^{N/2-1} x[2r] (W_{N/2})^{rk} + W_N^k \sum_{r=0}^{N/2-1} x[2r+1] (W_{N/2})^{rk} \\ &= G[k] + W_N^k H[k]. \end{aligned}$$

The original N-point DFT can therefore be expressed in terms of two $N/2$-point DFTs.

The $N/2$-point transforms can again be decomposed, and the process repeated until only 2-point transforms remain. In general this requires $\log_2 N$ stages of decomposition. Since each stage requires approximately N complex multiplications, the complexity of the resulting algorithm is of the order of $N \log_2 N$.

Generally the difference between N^2 and $N \log_2 N$ complex multiplications can become considerable for large values of N. For example, if $N = 2048$ then $N^2/(N \log_2 N) \approx 200$.

Note: This method requires N to be a power of 2, but modern FFTs do not have this requirement. Additionally, various textbooks will have different decomposition approaches with different algorithm complexities, so do not read too much into this specific derivation as there are numerous variations of FFT algorithms, and all exploit the basic redundancy in the computation of the DFT. In almost all cases, an off-the-shelf implementation of the FFT will be sufficient and there is seldom any reason to implement an FFT yourself.

Chapter 6

Transform Analysis of LTI Systems

(See Oppenheim and Schafer, Second edition pp. 240–339)

6.1 Introduction

For LTI systems we can write

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k].$$

Alternatively, this relationship can be expressed in the z-transform domain as

$$Y(z) = H(z)X(z),$$

where $H(z)$ is the **system function**, or the z-transform of the system impulse response. Note that this is not quite the frequency response of the system, but we know that there is a relationship between the z-transform and the DTFT.

Recall that a LTI system is completely characterised by its impulse response, or equivalently, its system function.

6.2 Frequency Response of LTI Systems

The frequency response $H(e^{j\omega})$ of a system is defined as the gain that the system applies to the complex exponential input $e^{j\omega n}$. The Fourier transforms of the system input and output are therefore related by

$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega}).$$

Keep in mind that the frequency response $H(e^{j\omega})$ is determined by evaluating the z-transform on the unit circle, and that this only exists for stable systems; if we have an unstable system, we need z-transforms to deal with it!

In terms of magnitude and phase,

$$|Y(e^{j\omega})| = |H(e^{j\omega})||X(e^{j\omega})|$$
$$\sphericalangle Y(e^{j\omega}) = \sphericalangle H(e^{j\omega}) + \sphericalangle X(e^{j\omega}).$$

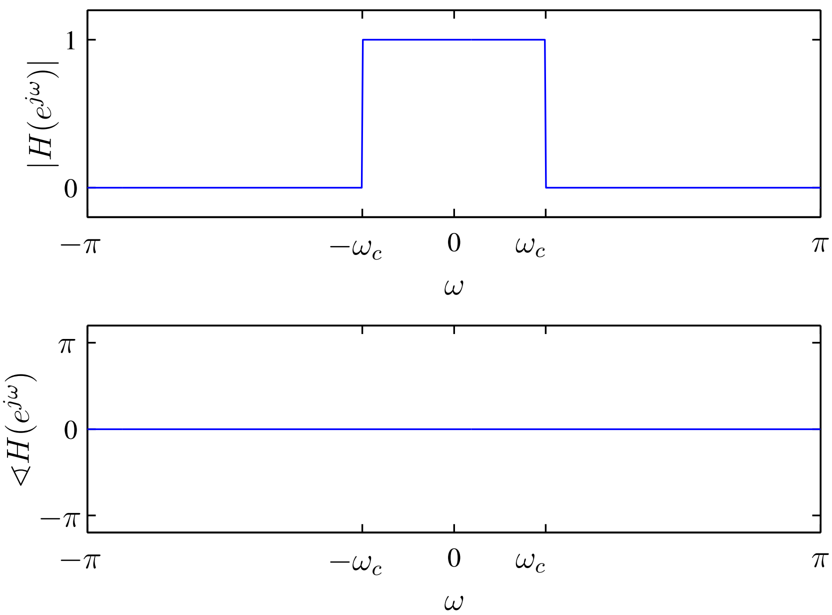
In this case $|H(e^{j\omega})|$ is referred to as the **magnitude response** or **gain** of the system (an *even* function if $y[n]$ is real), and $\sphericalangle H(e^{j\omega})$ is the **phase response** or **phase shift** (an *odd* function if $y[n]$ is real).

6.2.1 Ideal Frequency-selective Filters

Frequency components of the input are suppressed in the output if $|H(e^{j\omega})|$ is small at those frequencies. The **ideal lowpass filter** is defined as the LTI system with the frequency response shown below

$$H_{\mathrm{lp}}(e^{j\omega}) = \begin{cases} 1 & |\omega| \le \omega_c \\ 0 & \omega_c < |\omega| \le \pi. \end{cases}$$

Its magnitude and phase are

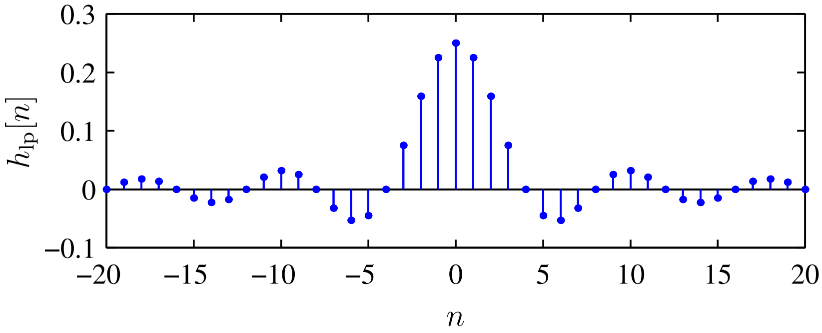


Note that the magnitude plot is essentially a "brick-wall" filter, and the phase plot demonstrates no phase distortion.

This response, as for all discrete-time signals, is periodic with period 2π. Its impulse response (for $-\infty < n < \infty$) is derived using the inverse DTFT equation as given below (and this is pretty much the only time we ever use it in this course):

$$\begin{aligned} h_{\text{lp}}[n] &= \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{j\omega n} d\omega = \frac{1}{2\pi}\left[\frac{1}{jn}e^{j\omega n}\right]_{-\omega_c}^{\omega_c} \\ &= \frac{1}{\pi n}\frac{1}{2j}(e^{j\omega_c n} - e^{-j\omega_c n}) = \frac{\sin(\omega_c n)}{\pi n} \end{aligned}$$

This is a sinc function, which for $\omega_c = \pi/4$ is visualised as



The ideal lowpass filter is non-causal, and its impulse response extends from $-\infty < n < \infty$. The system is therefore not computationally realisable. Also, the phase response of the ideal lowpass filter is specified to be zero — this is a problem in that causal ideal filters have nonzero phase responses.

The **ideal highpass filter** is

$$H_{\text{hp}}(e^{j\omega}) = \begin{cases} 0 & |\omega| \leq \omega_c \\ 1 & \omega_c < |\omega| \leq \pi. \end{cases}$$

Since $H_{\text{hp}}(e^{j\omega}) = 1 - H_{\text{lp}}(e^{j\omega})$, its impulse response is

$$h_{\text{hp}}[n] = \delta[n] - h_{\text{lp}}[n] = \delta[n] - \frac{\sin(\omega_c n)}{\pi n}.$$

6.2.2 Phase Distortion and Delay

Consider the ideal delay, with impulse response

$$h_{\text{id}}[n] = \delta[n - n_d]$$

and frequency response

$$H_{\text{id}}(e^{j\omega}) = e^{-j\omega n_d}.$$

The magnitude and phase of this response are

$$|H_{\text{id}}(e^{j\omega})| = 1,$$
$$\sphericalangle H_{\text{id}}(e^{j\omega}) = -\omega n_d, \qquad |\omega| < \pi.$$

The phase distortion of the ideal delay is therefore a linear function of ω. This is considered to be a rather mild (and therefore acceptable) form of phase distortion, since the only effect is to shift the sequence in time. In other words, a filter with linear phase response can be viewed as a cascade of a zero-phase filter, followed by a time shift or delay. This corresponds to a multiplication in frequency and thus a convolution in time.

In designing approximations to ideal filters, we are therefore frequently willing to accept linear phase distortion. The ideal lowpass filter with phase distortion would be defined as

$$H_{\text{lp}}(e^{j\omega}) = \begin{cases} e^{-j\omega n_d} & |\omega| \leq \omega_c \\ 0 & \omega_c < |\omega| \leq \pi, \end{cases}$$

with impulse response

$$h_{\text{lp}}[n] = \frac{\sin(\omega_c(n - n_d))}{\pi(n - n_d)}.$$

This corresponds to an impulse response that has been delayed in time in an attempt to make the filter causal, i.e., the filter should not respond before $n = 0$. However, this would also require that we *window* the signal to make it finite in time and thus computationally realisable (as discussed in the DFT chapter).

For instance, in the previous figure, we only plotted the impulse response for $n = -20$ to $n = 20$, so we could see this as a multiplication of the infinitely-long signal multiplied by a rectangular window over the same range in time. Then, to make the system causal, we delay the signal by 20 samples to that it spans from $n = 0$ to $n = 40$ and is thus a right-sided impulse response.

6.3 System Response For LCCD Systems

Ideal filters cannot be implemented with finite computation. Therefore we need approximations to ideal filters. Systems described by LCCD equations

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$$

are useful for providing one class of approximation.

The properties of this class of system are best developed in the z-transform domain. The z-transform of the equation is

$$\sum_{k=0}^{N} a_k z^{-k} Y(z) = \sum_{k=0}^{M} b_k z^{-k} X(z),$$

or equivalently

$$\left(\sum_{k=0}^{N} a_k z^{-k}\right) Y(z) = \left(\sum_{k=0}^{M} b_k z^{-k}\right) X(z).$$

The system function for a system that satisfies a difference equation of the required form is therefore

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}} = \left(\frac{b_0}{a_0}\right) \frac{\prod_{k=1}^{M}(1 - c_k z^{-1})}{\prod_{k=1}^{N}(1 - d_k z^{-1})}.$$

Each factor $(1 - c_k z^{-1})$ in the numerator contributes a zero at $z = c_k$ and a pole at $z = 0$. Each factor $(1 - d_k z^{-1})$ contributes a zero at $z = 0$ and a pole at $z = d_k$. Also note that we cannot compute b_0/a_0 from the pole-zero plot alone, and we require some kind of additional information about the system, such as an initial condition.

The difference equation and the algebraic expression for the system function are equivalent, as demonstrated by the next example.

Example: Second-order system

Given the system function

$$H(z) = \frac{(1 + z^{-1})^2}{(1 - \frac{1}{2}z^{-1})(1 + \frac{3}{4}z^{-1})},$$

we can find the corresponding difference equation by noting that

$$H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 + \frac{1}{4}z^{-1} - \frac{3}{8}z^{-2}} = \frac{Y(z)}{X(z)}.$$

Therefore

$$(1 + \frac{1}{4}z^{-1} - \frac{3}{8}z^{-2})Y(z) = (1 + 2z^{-1} + z^{-2})X(z),$$

and the difference equation is

$$y[n] + \frac{1}{4}y[n-1] - \frac{3}{8}y[n-2] = x[n] + 2x[n-1] + x[n-2].$$

Note that we do not require an ROC to be specified here because we are not solving for any of the signals $y[n]$, $h[n]$, or $x[n]$ – we are just representing the system's input-output relationship in the form of a difference equation, not inverting any one z-transform.

6.3.1 Stability and Causality

A difference equation does not uniquely specify the impulse response of a LTI system. For a given system function (expressed as a ratio of polynomials), each possible choice of ROC will lead to a different impulse response. However, they will all correspond to the same difference equation.

If a system is causal, it follows that the impulse response is a right-sided sequence, and the region of convergence of $H(z)$ must be outside of the outermost pole.

Alternatively, if we require that the LTI system be stable, then we must have

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty.$$

For $|z| = 1$ this is identical to the condition

$$\sum_{n=-\infty}^{\infty} |h[n]z^{-n}| < \infty,$$

so the condition for stability is equivalent to the condition that the ROC of $H(z)$ include the unit circle.

Example: Determining the ROC
The system function of the LTI system with difference equation

$$y[n] - \frac{5}{2}y[n-1] + y[n-2] = x[n]$$

is

$$H(z) = \frac{1}{1 - \frac{5}{2}z^{-1} + z^{-2}} = \frac{1}{(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})}.$$

There are thus three choices for the ROC:

- **Causal:** ROC outside of outermost pole $|z| > 2$ (but then not stable).

- **Stable:** ROC such that $\frac{1}{2} < |z| < 2$ (but then not causal).
- If $|z| < \frac{1}{2}$ then the system is neither causal nor stable.

For a causal and stable system the ROC must be outside the outermost pole and include the unit circle. This is only possible if all the poles are inside the unit circle.

6.3.2 Inverse Systems

The system $H_i(z)$ is the inverse system to $H(z)$ if

$$G(z) = H(z)H_i(z) = 1,$$

with the ROC being all z, which will (of course) contain the intersection of the ROCs of $H_i(z)$ and $H(z)$. This implies that

$$H(z) = \frac{1}{H_i(z)},$$

and the time-domain equivalent is

$$g[n] = h[n] * h_i[n] = \delta[n].$$

The question of which ROC to associate with $H_i(z)$ is answered by the convolution theorem — for the previous equation to hold, the ROCs of $H(z)$ and $H_i(z)$ must overlap.

Example: Inverse system for first-order system
Let $H(z)$ be

$$H(z) = \frac{1 - 0.5z^{-1}}{1 - 0.9z^{-1}}$$

with ROC $|z| > 0.9$. Then $H_i(z)$ is

$$H_i(z) = \frac{1 - 0.9z^{-1}}{1 - 0.5z^{-1}}.$$

Since there is only one pole, there are only two possible ROCs. The choice of ROC for $H_i(z)$ that overlaps with $|z| > 0.9$ is $|z| > 0.5$. Therefore, the impulse response of the inverse system is

$$h_i[n] = (0.5)^n u[n] - 0.9(0.5)^{n-1}u[n-1].$$

In this case the inverse is both causal and stable.

A LTI system is stable and causal with a stable and causal inverse if and only if both the poles and zeros of $H(z)$ are inside the unit circle — such systems are called **minimum phase** systems.

The frequency response of the inverse system, if it exists, is

$$H(e^{j\omega}) = \frac{1}{H_i(e^{j\omega})}.$$

Not every system has an inverse though; for example, there is no way to recover the frequency components (above the cutoff frequency) that were set to zero by the action of a lowpass filter.

6.3.3 Impulse Response For Rational System Functions

If a system has a rational transfer function – with at least one pole that is not at the origin and is not cancelled by a zero – then there will always be a term corresponding to an infinite length sequence in the impulse response. These are called **infinite impulse response (IIR)** systems.

On the other hand, if a system has no poles (except possibly at $z = 0$) such that $N = 0$ in the canonical LCCDE expression, then

$$H(z) = \sum_{k=0}^{M} b_k z^{-k}.$$

In this case the system is determined to within a constant multiplier by its zeros, so the impulse response has a finite length:

$$h[n] = \sum_{k=0}^{M} b_k \delta[n-k] = \begin{cases} b_n & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$$

The impulse response is finite in length, and the system is thus called a **finite impulse response (FIR)** system. Note that such a system has no poles, except possibly at $z = 0$.

Example: A first-order IIR system

Given a causal system satisfying the difference equation

$$y[n] - ay[n-1] = x[n],$$

the system function is

$$H(z) = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}, \qquad |z| > |a|.$$

The condition for stability is $|a| < 1$. The inverse z-transform is

$$h[n] = a^n u[n].$$

Example: A simple FIR system
Consider the truncated impulse response

$$h[n] = \begin{cases} a^n & 0 \le n \le M \\ 0 & \text{otherwise.} \end{cases}$$

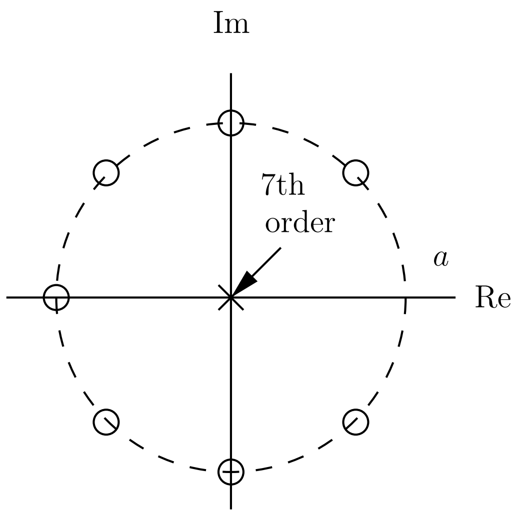
The system function is

$$H(z) = \sum_{n=0}^{M} a^n z^{-n} = \frac{1 - a^{M+1} z^{-M-1}}{1 - az^{-1}}.$$

By now, you should be able to compute that the zeros of the system function are at

$$z_k = ae^{j2\pi k/(M+1)}, \qquad k = 0, 1, \ldots, M.$$

With a assumed real and positive, the pole at $z = a$ is cancelled by a zero. The pole-zero plot for the case of $M = 7$ is therefore given by



The difference equation satisfied by the input and output of the LTI system is the convolution

$$y[n] = \sum_{k=0}^{M} a^k x[n-k].$$

The input and output also satisfy the difference equation

$$y[n] - ay[n-1] = x[n] - a^{M+1} x[n-M-1].$$

6.4 Rational System Functions

If a stable LTI system has a rational system function, its frequency response has the form

$$H(e^{j\omega}) = \frac{\sum_{k=0}^{M} b_k e^{-j\omega k}}{\sum_{k=0}^{N} a_k e^{-j\omega k}}.$$

We want to know the magnitude and phase associated with the frequency response. To this end, it is useful to express $H(e^{j\omega})$ in terms of the poles and zeros of $H(z)$:

$$H(e^{j\omega}) = \left(\frac{b_0}{a_0}\right) \frac{\prod_{k=1}^{M}(1 - c_k e^{-j\omega})}{\prod_{k=1}^{N}(1 - d_k e^{-j\omega})}.$$

It follows that

$$|H(e^{j\omega})| = \left|\frac{b_0}{a_0}\right| \frac{\prod_{k=1}^{M}|1 - c_k e^{-j\omega}|}{\prod_{k=1}^{N}|1 - d_k e^{-j\omega}|}.$$

Therefore $|H(e^{j\omega})|$ is the product of the magnitudes of all the **zero factors** of $H(z)$ evaluated on the unit circle, *divided by* the product of the magnitudes of all the **pole factors** evaluated on the unit circle.

The **gain** in dB of $H(e^{j\omega})$, also called the **log magnitude**, is given by

$$\text{Gain in dB} = 20 \log_{10} |H(e^{j\omega})|,$$

which for a rational system function takes the form

$$\begin{aligned} 20 \log_{10} |H(e^{j\omega})| = 20 \log_{10} \left|\frac{b_0}{a_0}\right| &+ \sum_{k=1}^{M} 20 \log_{10} |1 - c_k e^{-j\omega}| \\ &- \sum_{k=1}^{N} 20 \log_{10} |1 - d_k e^{-j\omega}|. \end{aligned}$$

Additionally, it is worth noting that

$$\text{Attenuation in dB} = -\text{Gain in dB}.$$

Thus a 60 dB attenuation at frequency ω corresponds to $|H(e^{j\omega})| = 0.001 = 10^{-3}$. Also,

$$20\log_{10}|Y(e^{j\omega})| = 20\log_{10}|H(e^{j\omega})| + 20\log_{10}|X(e^{j\omega})|.$$

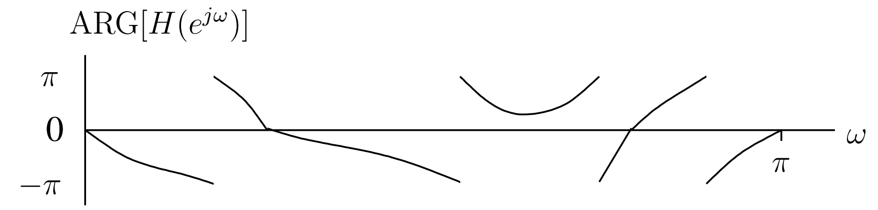
The phase response for a rational system function is

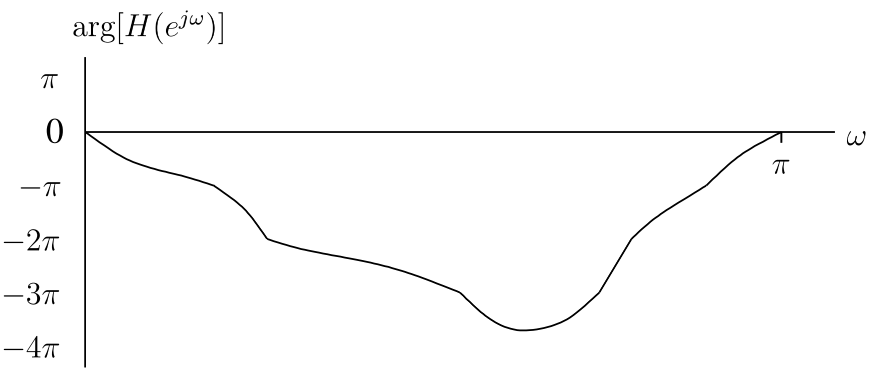
$$\sphericalangle H(e^{j\omega}) = \sphericalangle\left[\frac{b_0}{a_0}\right] + \sum_{k=1}^{M}\sphericalangle[1 - c_k e^{-j\omega}] - \sum_{k=1}^{N}\sphericalangle[1 - d_k e^{-j\omega}].$$

The zero factors contribute with a plus sign and the pole factors with a minus.

In the above equation, the phase of each term is ambiguous, since any integer multiple of 2π can be added at each value of ω without changing the value of the complex number.

As an aside: when calculating the phase with a computer, the angle returned will generally be the **principal** value $\text{ARG}[H(e^{j\omega})]$, which lies in the range $-\pi$ to π. This phase will generally be a discontinuous function, containing jumps of 2π radians whenever the phase wraps. Appropriate multiples of 2π can be added or subtracted, if required, to yield the continuous phase function $\arg[H(e^{j\omega})]$.





This process effectively removes discontinuities and ambiguity in the phase plot, and is sometimes referred to as "phase unwrapping" – which is non-trivial in some applications.

6.4.1 Frequency Response of a Single Pole or Zero

Consider a single zero factor of the form

$$(1 - c_k e^{-j\omega}) = (1 - re^{j\theta}e^{-j\omega}),$$

where $c_k = re^{j\theta}$ in the z-plane. The magnitude squared of this factor is

$$\begin{aligned}|1 - re^{j\theta}e^{-j\omega}|^2 &= (1 - re^{j\theta}e^{-j\omega})(1 - re^{-j\theta}e^{j\omega}) \\ &= 1 + r^2 - 2r\cos(\omega - \theta),\end{aligned}$$

so the log magnitude in dB is

$$20\log_{10}|1 - re^{j\theta}e^{-j\omega}| = 10\log_{10}[1 + r^2 - 2r\cos(\omega - \theta)].$$

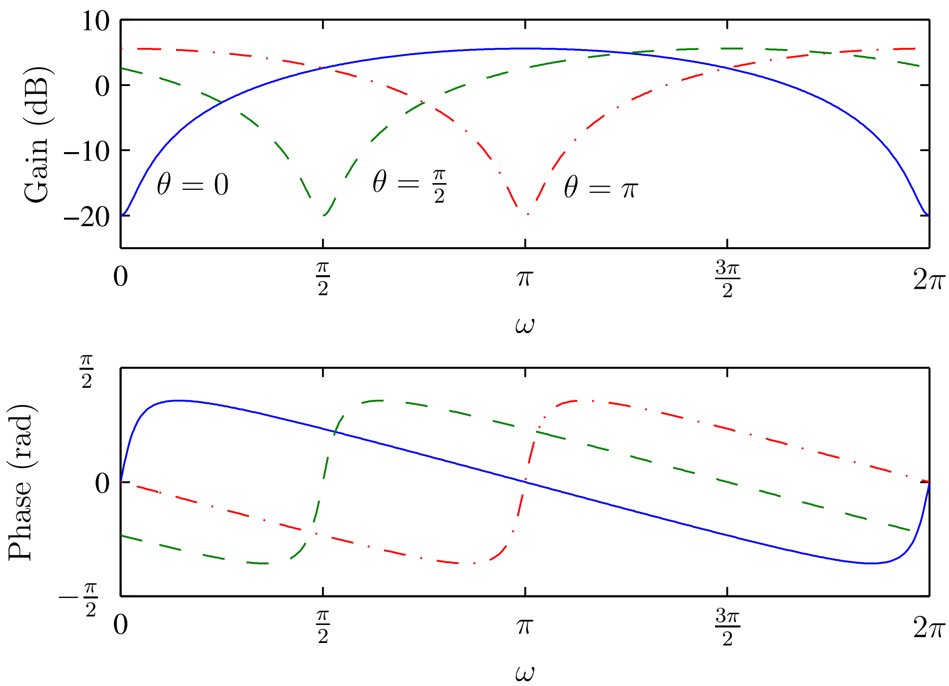
The principle value of the phase for the factor is

$$\mathrm{ARG}[1 - re^{j\theta}e^{-j\omega}] = \arctan\left[\frac{r\sin(\omega - \theta)}{1 - r\cos(\omega - \theta)}\right].$$

This can be computed (and ideally you should be able to do this yourself) by writing out the full expression for the zero and converting it from polar to rectangular form $(a+jb)$, computing the phase as $\arctan(b/a)$, and then exploiting trigonometric symmetry to get the equation into the above form.

These functions are periodic in ω with period 2π.

Plotted below are the frequency responses for $r = 0.9$ and three different values of θ:



Note that:

- The gain dips at $\omega = \theta$. As θ changes, the frequency at which the dip occurs changes.
- The gain is maximised for $\omega - \theta = \pi$ (since we want $\cos(\omega - \theta) = -1$), and for $r = 0.9$ the magnitude of the resulting gain is

$$10\log_{10}(1 + r^2 + 2r) = 20\log_{10}(1 + r) = 5.57\text{dB}.$$

- The gain is minimised for $\omega = \theta$ (since we want $\cos(\omega - \theta) = 1$), and for $r = 0.9$ the resulting gain is

$$10\log_{10}(1 + r^2 - 2r) = 20\log_{10}|1 - r| = -20\text{dB}.$$

- The phase is zero at $\omega = \theta$.

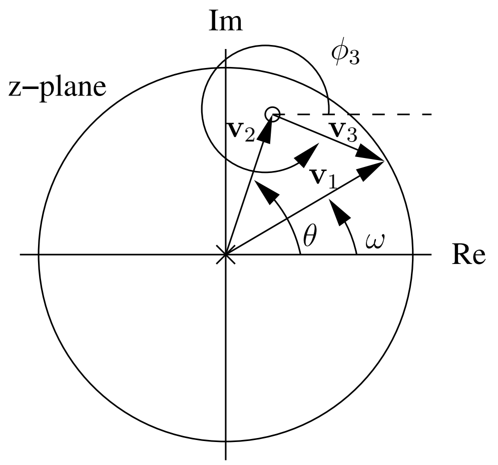
Note that if the factor $(1 - re^{j\theta}e^{j\omega})$ occurs in the denominator, thereby representing a pole factor, then the entire analysis holds with the exception that the sign of the log magnitude and the phase changes.

The frequency response can be sketched from the pole-zero plot using a simple geometric construction. Note firstly that the frequency response corresponds to an evaluation of $H(z)$ on the unit circle. Secondly, the complex value of each pole and zero can be represented by a vector in the z-plane from the pole or zero to a point on the unit circle.

Take for example the case of a single zero factor

$$H(z) = (1 - re^{j\theta}z^{-1}) = \frac{z - re^{j\theta}}{z} \qquad r < 1,$$

which corresponds to a pole at $z = 0$ and a zero at $z = re^{j\theta}$.



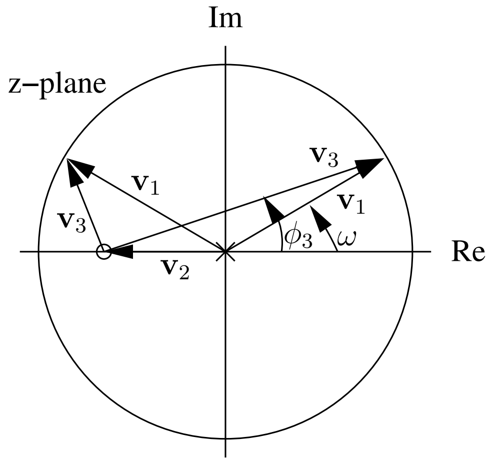
If the vectors $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3 = \mathbf{v}_1 - \mathbf{v}_2$ represent respectively the complex numbers $e^{j\omega}$, $re^{j\theta}$, and $e^{j\omega} - re^{j\theta}$, then

$$|H(e^{j\omega})| = |1 - re^{j\theta}e^{-j\omega}| = \left|\frac{e^{j\omega} - re^{j\theta}}{e^{j\omega}}\right| = \frac{|\mathbf{v}_3|}{|\mathbf{v}_1|} = |\mathbf{v}_3|$$

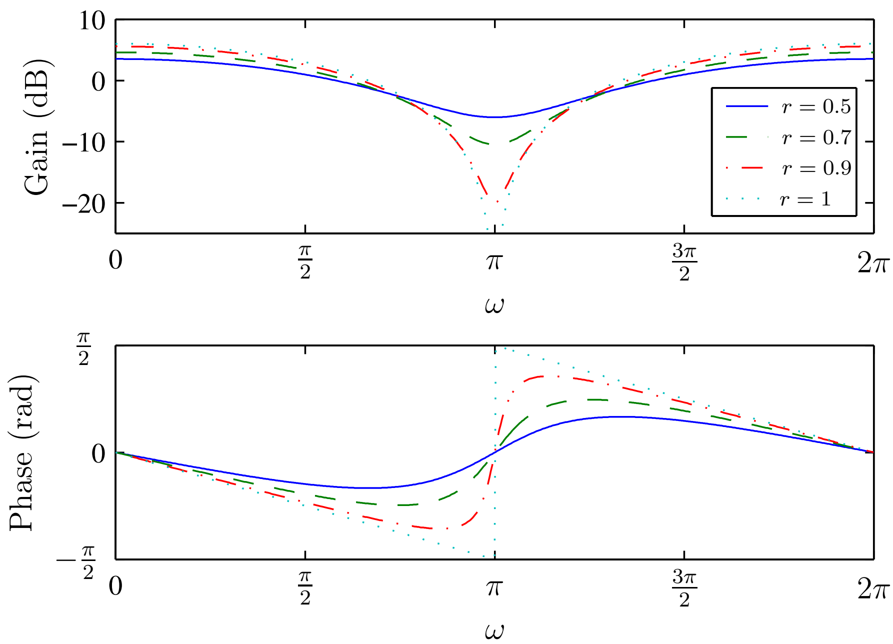
$$\begin{aligned} \sphericalangle H(e^{j\omega}) &= \sphericalangle(1 - re^{j\theta}e^{-j\omega}) = \sphericalangle(e^{j\omega} - re^{j\theta}) - \sphericalangle(e^{j\omega}) \\ &= \sphericalangle(\mathbf{v}_3) - \sphericalangle(\mathbf{v}_1) = \phi_3 - \phi_1 = \phi_3 - \omega. \end{aligned}$$

A vector such as $\mathbf{v}_3$ from a zero to the unit circle is referred to as a **zero vector**, and a vector from a pole to the unit circle is called a **pole vector**.

Consider now the pole zero system depicted below:



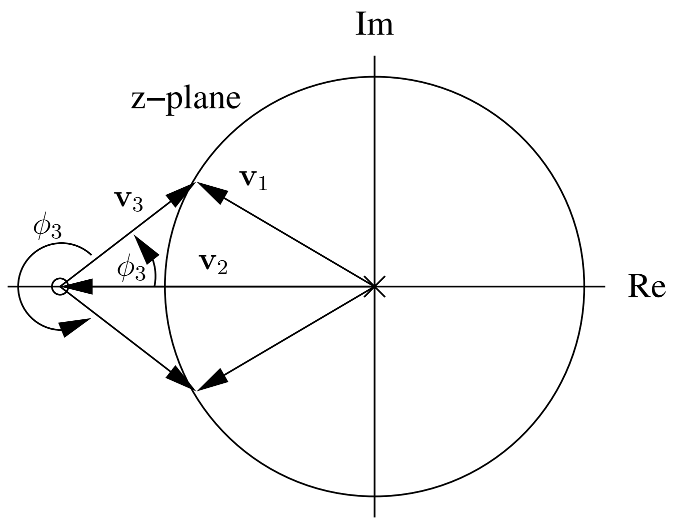
The frequency response for the single zero at different values of r and $\theta = \pi$ is



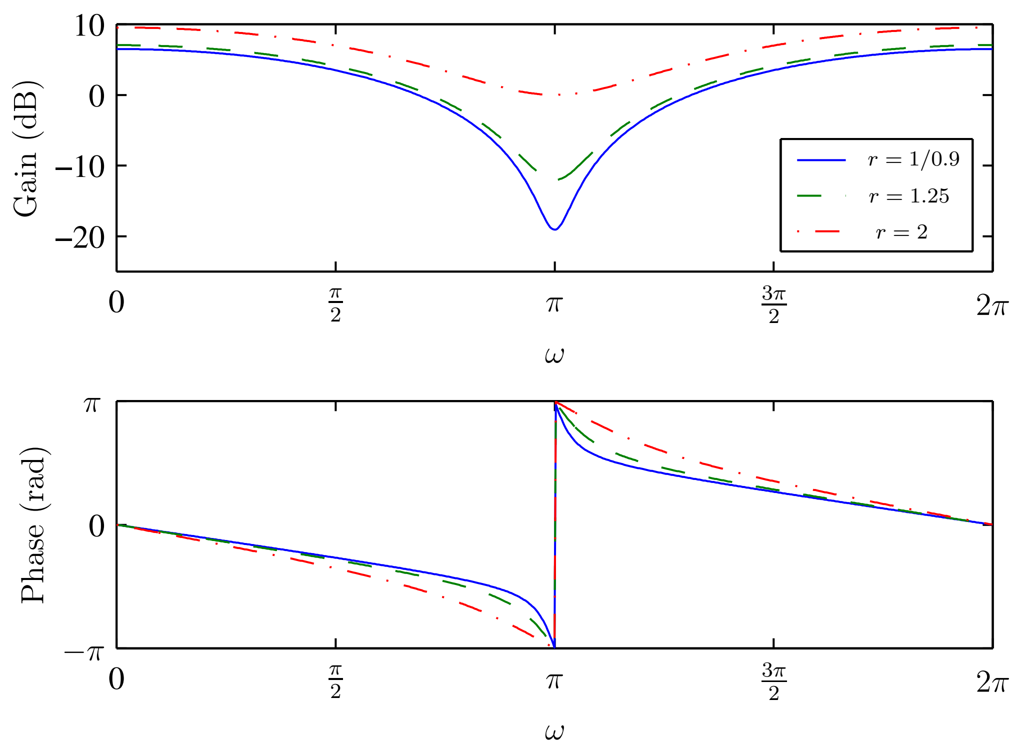
This is interesting because, as we vary the value of ω, we also vary the vector $\mathbf{v}_1$ and thus also the vector sums that are dependent on $\mathbf{v}_1$ – and this ultimately affects the gain and phase of the frequency response.

Note that the log magnitude dips more sharply as r approaches 1 (and at $\omega = \pi$ tends to $-\infty$ as r tends towards 1). The phase function has positive slope around $\omega = \theta$. This slope increases as r approaches 1, and becomes infinite for $r = 1$. In this case the phase function is discontinuous, with a jump of π radians at $\omega = \theta$.

If r increases still further, to lie outside of the unit circle,



then the frequency response becomes



6.4.2 Frequency Response With Multiple Poles and Zeros

In general, the z-transform of an LTI system can be factorised as

$$H(z) = \left(\frac{b_0}{a_0}\right) \frac{\prod_{k=1}^{M}(1 - c_k z^{-1})}{\prod_{k=1}^{N}(1 - d_k z^{-1})} = \left(\frac{b_0}{a_0}\right) z^{N-M} \frac{\prod_{k=1}^{M}(z - c_k)}{\prod_{k=1}^{N}(z - d_k)}.$$

Depending on whether N is greater than or less than M, the factor z^{N-M} represents either $N - M$ zeros at the origin, or $M - N$ poles at the origin. In either case, the z-transform can be written in the form

$$H(z) = K \frac{\prod_{k=0}^{M_0}(z - z_k)}{\prod_{k=0}^{N_0}(z - p_k)}$$

where $z_1, \ldots, z_{M_0}$ are the zeros, and $p_1, \ldots, p_{N_0}$ the poles of $H(z)$. This representation could also be obtained by merely factorising $H(z)$ in terms of z rather than z^{-1}.

The frequency response of this system is

$$H(e^{j\omega}) = K \frac{\prod_{k=0}^{M_0}(e^{j\omega} - z_k)}{\prod_{k=0}^{N_0}(e^{j\omega} - p_k)}.$$

The magnitude is therefore

$$|H(e^{j\omega})| = |K| \frac{\prod_{k=0}^{M_0}|e^{j\omega} - z_k|}{\prod_{k=0}^{N_0}|e^{j\omega} - p_k|},$$

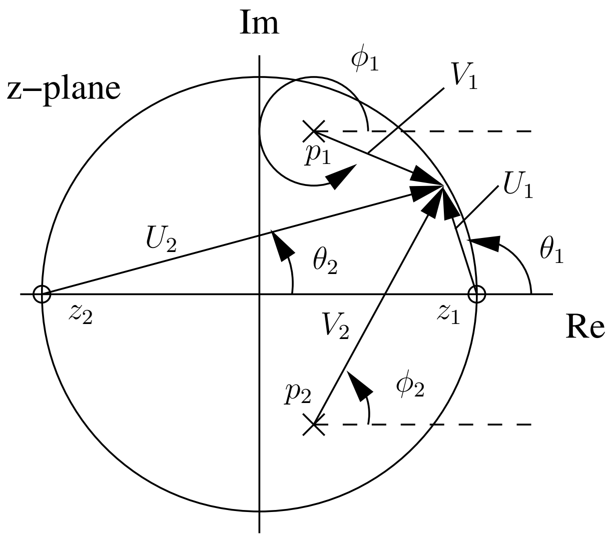
and the phase is

$$\sphericalangle H(e^{j\omega}) = \sum_{k=0}^{M_0} \sphericalangle(e^{j\omega} - z_k) - \sum_{k=0}^{N_0} \sphericalangle(e^{j\omega} - p_k).$$

In the z-plane, $(e^{j\omega} - z_k)$ is simply the vector from the zero z_k to the point on the unit circle. The term $|e^{j\omega} - z_k|$ is the length of this vector, and $\sphericalangle(e^{j\omega} - z_k)$ is the angle that it makes with the positive real axis. Similarly, the term $(e^{j\omega} - p_k)$ corresponds to the vector from the pole p_k to the point on the unit circle.

It follows then that the the magnitude response is the product of the lengths of the zero vectors, divided by the product of the lengths of the pole vectors. The phase response is the sum of the angles of the zero vectors, minus the sum of the angles of the pole vectors.

Thus, for the two-pole, two-zero system shown below



the frequency response is

$$|H(e^{j\omega})| = |K|\frac{U_1U_2}{V_1V_2}$$
$$\sphericalangle H(e^{j\omega}) = \theta_1 + \theta_2 - (\phi_1 + \phi_2),$$

where K is a constant multiplicative factor that cannot be determined from the pole-zero diagram alone, but only serves to scale the magnitude. Computing this value would thus require an additional piece of information, such as knowing an initial condition of the system.

Logically, at the zero locations, the frequency response will be zero, and the closer our observation point on the unit circle is to a zero, the smaller the frequency response will be. Conversely, the closer the observation point (on the unit circle) is to a pole, the larger the frequency response will get – and it will diverge to infinity if a pole lies on the unit circle.

Oftentimes, zeros (or poles) will also come in conjugate pairs when they are not on the real axis, meaning that the overall response will be real-valued.

These are important things to look out for, and together these foundational ideas allow us to get a rough idea of what a frequency response for a given system will look like just by quickly inspecting its pole-zero plot by eye.

6.5 Realisation Structures for Digital Filters

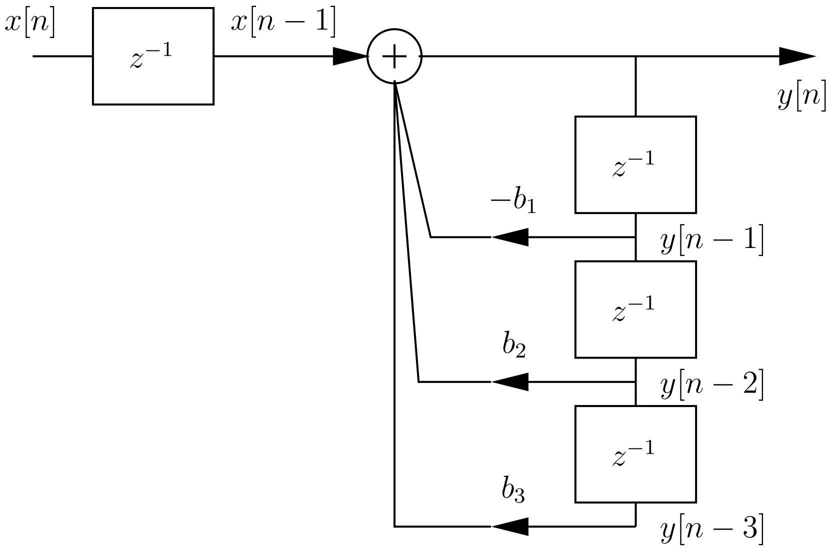
The difference equation, the impulse response, and the system function are all equivalent characterisations of the input-output relation for a LTI discrete-time system. For implementation purposes, systems described by LCCDEs can be implemented by structures consisting of an interconnection of the basic operations of addition, multiplication by a constant, and delay. The desired form for the interconnections depends on the technology to be used.

Discrete-time filters are often represented in the form of block or signal flow diagrams, which are convenient for representing the difference equations or transfer functions.

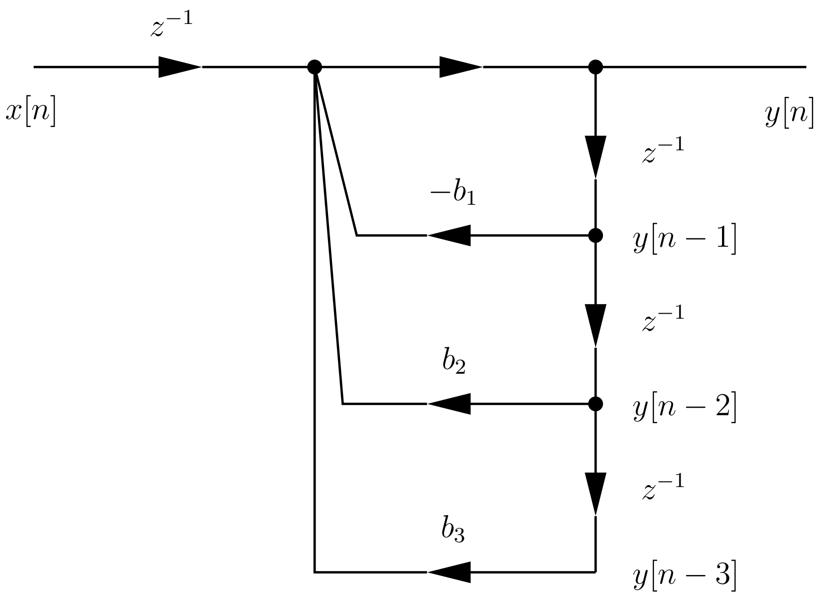
For example, the system with the difference equation

$$y[n] = x[n-1] - b_1 y[n-1] + b_2 y[n-2] + b_3 y[n-3]$$

can be represented in a block diagram form as



The symbol z^{-1} represents a delay of one unit of time, and the arrows represent multipliers (with the constant multiplication factors next to them). The equivalent signal flow diagram is



The relationship between the diagrams and the difference equation is clear.

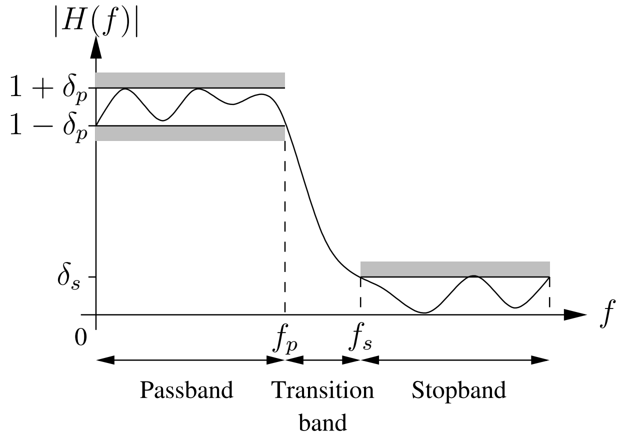
Many alternative filter structures can be developed, and they differ mainly with respect to their numerical stability and the effects of quantisation on their performance. A discussion of these effects can be found in many DSP texts, and a basic introduction to filter design is covered in the next chapter.

Chapter 7

Filter Design

Note: This chapter will not be examined in the course, but it is included for your perusal.

7.1 Introduction



The design of a digital filter involves five steps:

- **Specification:** The characteristics of the filter often have to be specified in the frequency domain. For example, for frequency selective filters (lowpass, highpass, bandpass, etc.) the specification usually involves tolerance limits as shown above.

- **Coefficient calculation:** Approximation methods have to be used to calculate the values $h[k]$ for a FIR implementation, or a_k, b_k for an IIR implementation. Equivalently, this involves finding a filter which has $H(z)$ satisfying the requirements.

- **Realisation:** This involves converting $H(z)$ into a suitable filter structure. Block or flow diagrams are often used to depict filter structures, and show the computational procedure for implementing the digital filter.

- **Analysis of finite wordlength effects:** In practice one should check that the quantisation used in the implementation does not degrade the performance of the filter to a point where it is unusable.

- **Implementation:** The filter is implemented in software or hardware. The criteria for selecting the implementation method involve issues such as real-time performance, complexity, processing requirements, and availability of equipment.

7.2 FIR Filter Design

A FIR filter is characterised by the equations

$$y[n] = \sum_{k=0}^{N-1} h[k]x[n-k]$$
$$H(z) = \sum_{k=0}^{N-1} h[k]z^{-k}.$$

The following are useful properties of FIR filters:

- They are always stable — the system function contains no poles. This is particularly useful for adaptive filters.

- They can have an exactly linear phase response. The result is no frequency dispersion, which is good for pulse and data transmission.

- Finite length register effects are simpler to analyse and of less consequence than for IIR filters.

- They are very simple to implement, and all DSP processors have architectures that are suited to FIR filtering.

- For large N (many filter taps), the FFT can be used to improve performance.

Of these, the linear phase property is probably the most important. A filter is said to have a generalised linear phase response if its frequency response can be expressed in the form

$$H(e^{j\omega}) = A(e^{j\omega})e^{-j\alpha\omega+j\beta}$$

where α and β are constants, and $A(e^{j\omega})$ is a real function of ω. If this is the case, then

- If A is positive, then the phase is

$$\sphericalangle H(e^{j\omega}) = \beta - \alpha\omega.$$

If A is negative, then

$$\sphericalangle H(e^{j\omega}) = \pi + \beta - \alpha\omega.$$

In either case, the phase is a linear function of ω.

It is common to restrict the filter to having a real-valued impulse response $h[n]$, since this greatly simplifies the computational complexity in the implementation of the filter.

A FIR system has linear phase if the impulse response satisfies either the even symmetric condition

$$h[n] = h[N - 1 - n],$$

or the odd symmetric condition

$$h[n] = -h[N - 1 - n].$$

The system has different characteristics depending on whether N is even or odd. Furthermore, it can be shown that all linear phase filters must satisfy one of these conditions. Thus there are exactly four types of linear phase filters.

Consider for example the case of an odd number of samples in $h[n]$, and even symmetry. The frequency response for $N = 7$ is

$$\begin{aligned}
H(e^{j\omega}) &= \sum_{n=0}^{6} h[n]e^{-j\omega n} \\
&= h[0] + h[1]e^{-j\omega} + h[2]e^{-j2\omega} + h[3]e^{-j3\omega} + h[4]e^{-j4\omega} \\
&\qquad + h[5]e^{-j5\omega} + h[6]e^{-j6\omega} \\
&= e^{-j3\omega}(h[0]e^{j3\omega} + h[1]e^{j2\omega} + h[2]e^{j\omega} + h[3] + h[4]e^{-j\omega} \\
&\qquad + h[5]e^{-j2\omega} + h[6]e^{-j3\omega}).
\end{aligned}$$

The specified symmetry property means that $h[0] = h[6]$, $h[1] = h[5]$, and $h[2] = h[4]$, so we have

$$
\begin{aligned}
H(e^{j\omega}) &= e^{-j3\omega}(h[0](e^{j3\omega} + e^{-j3\omega}) + h[1](e^{j2\omega} + e^{-j2\omega}) \\
&\qquad + h[2](e^{j\omega} + e^{-j\omega}) + h[3]) \\
&= e^{-j3\omega}(2h[0]\cos(3\omega) + 2h[1]\cos(2\omega) + 2h[2]\cos(\omega)) \\
&= e^{-j3\omega}\sum_{n=0}^{3} a[n]\cos(\omega n),
\end{aligned}
$$

where $a[0] = h[3]$, and $a[n] = 2h[3-n]$ for $n = 1, 2, 3$. The resulting filter clearly has a linear phase response for real $h[n]$. It is quite simple to show that in general for odd values of N the frequency response is

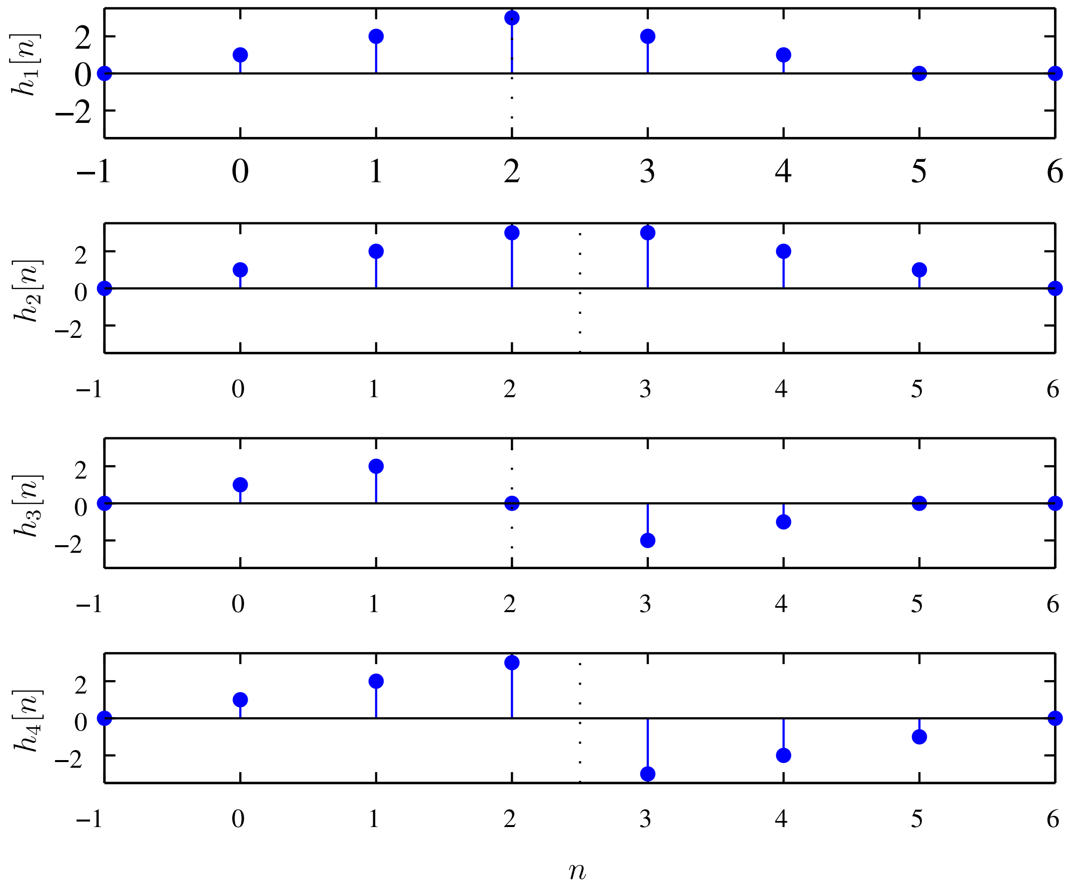
$$
H(e^{j\omega}) = e^{-j\omega(N-1)/2} \sum_{n=0}^{(N-1)/2} a[n]\cos(\omega n),
$$

for a set of real-valued coefficients $a[0], \ldots, a[(N-1)/2]$. As different values for $a[n]$ are selected, different linear-phase filters are obtained.

The cases of N odd and $h[n]$ antisymmetric are similar to that presented, and the frequency responses are summarised in the following table:

Symmetry	**N**	$H(e^{j\omega})$	**Type**
Even	Odd	$e^{-j\omega(N-1)/2} \sum_{n=0}^{(N-1)/2} a[n]\cos(\omega n)$	1
Even	Even	$e^{-j\omega(N-1)/2} \sum_{n=1}^{N/2} b[n]\cos(\omega(n-1/2))$	2
Odd	Odd	$e^{-j[\omega(N-1)/2-\pi/2]} \sum_{n=0}^{(N-1)/2} a[n]\sin(\omega n)$	3
Odd	Even	$e^{-j[\omega(N-1)/2-\pi/2]} \sum_{n=1}^{N/2} b[n]\sin(\omega(n-1/2))$	4

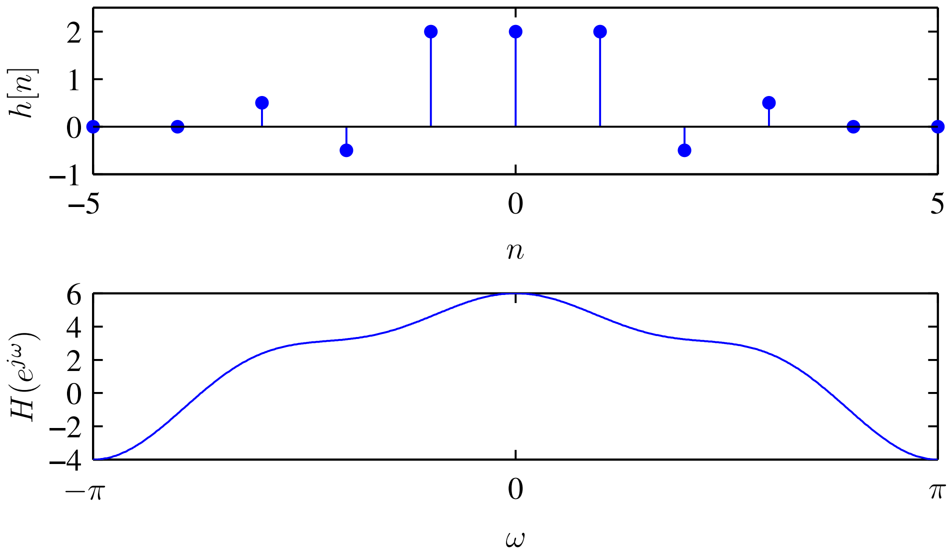
Recall that even symmetry implies $h[n] = h[N-1-n]$ and odd symmetry $h[n] = -h[N-1-n]$. Examples of filters satisfying each of these symmetry conditions are:



The center of symmetry is indicated by the dotted line.

The process of linear-phase filter design involves choosing the $a[n]$ values to obtain a filter with a desired frequency response. This is not always possible, however — the frequency response for a type II filter, for example, has the property that it is *always* zero for $\omega = \pi$, and is therefore not appropriate for a highpass filter. Similarly, filters of type 3 and 4 introduce a 90° phase shift, and have a frequency response that is always zero at $\omega = 0$ which makes them unsuitable for as lowpass filters. Additionally, the type 3 response is always zero at $\omega = \pi$, making it unsuitable as a highpass filter. The type I filter is the most versatile of the four.

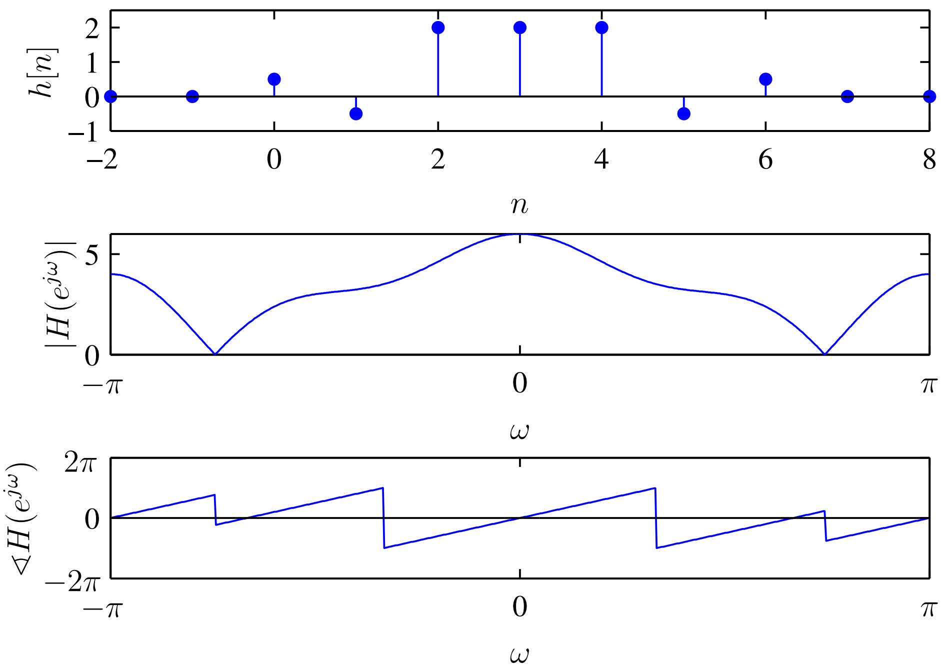
Linear phase filters can be thought of in a different way. Recall that a linear phase characteristic simply corresponds to a time shift or delay. Consider now a real FIR filter with an impulse response that satisfies the even symmetry condition $h[n] = h[-n]$:



Recall from the properties of the Fourier transform this filter has a real-valued frequency response $A(e^{j\omega})$. Delaying this impulse response by $(N-1)/2$ results in a causal filter with frequency response

$$H(e^{j\omega}) = A(e^{j\omega})e^{-j\omega(N-1)/2}.$$

This filter therefore has linear phase.



7.2.1 Group Delay

A convenient measure of the linearity of the phase is the **group delay**, which relates to the effect of the phase on a narrowband signal. (Note that this concept is not explicitly examinable in the course, but it is worth noting if you are interested in better understanding the concept of phase distortion.)

Consider the narrowband input $x[n] = s[n]\cos(\omega_0 n)$, where $s[n]$ is the envelope of the signal. Since $X(e^{j\omega})$ is non-zero only around $\omega = \omega_0$, the effect of the phase of the system can be approximated around $\omega = \omega_0$ by

$$\sphericalangle H(e^{j\omega}) \approx -\phi_0 - \omega n_d.$$

Thus the response of the system to $x[n] = s[n]\cos(\omega_0 n)$ is approximately $y[n] = s[n - n_d]\cos(\omega_0 n - \phi_0 - \omega_0 n_d)$.

The time delay of the envelope $s[n]$ of the narrowband signal $x[n]$ with Fourier transform centred at ω_0 is therefore given by the negative of the slope of the phase at ω_0. The group delay of a system is thus defined as

$$\tau(\omega) = \mathrm{grd}[H(e^{j\omega})] = -\frac{d}{d\omega}\left\{\arg[H(e^{j\omega})]\right\},$$

which essentially measures how the phase shifts with respect to frequency. The deviation of the group delay away from a constant thus indicates the degree of non-linearity of the phase.

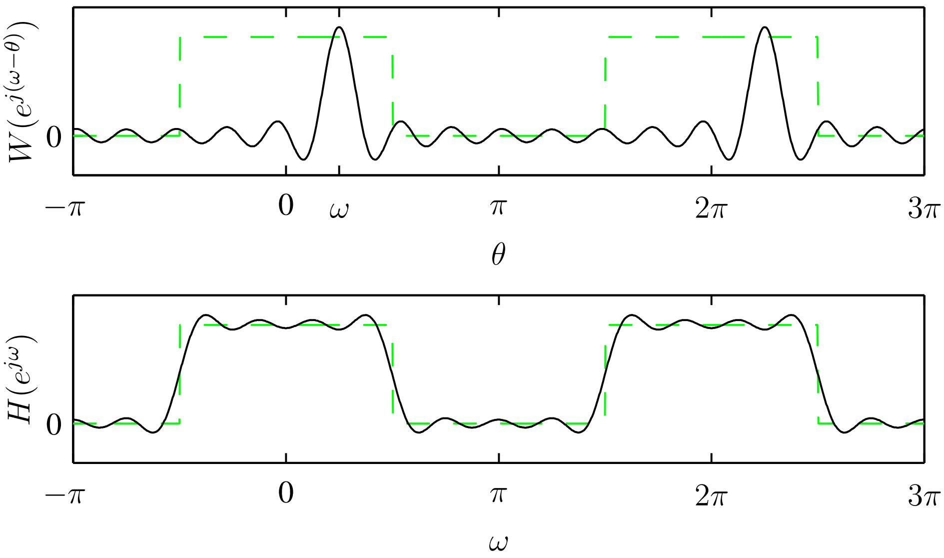
7.2.2 Window Method for FIR Filter Design

Assume that the desired filter response $H_d(e^{j\omega})$ is known. Using the inverse Fourier transform we can determine $h_d[n]$, the desired unit sample response. In the window method, a FIR filter is obtained by multiplying a window $w[n]$ with $h_d[n]$ to obtain a finite duration $h[n]$ of length N. This is required since $h_d[n]$ will in general be an infinite duration sequence, and the corresponding filter will therefore not be realisable. If $h_d[n]$ is even or odd symmetric and $w[n]$ is even symmetric, then $h_d[n]w[n]$ is a linear phase filter.

Two important design criteria are the *length* and *shape* of the window $w[n]$. To see how these factors influence the design, consider the multiplication operation in the frequency domain: since $h[n] = h_d[n]w[n]$,

$$H(e^{j\omega}) = H_d(e^{j\omega}) * W(e^{j\omega}).$$

The following plot demonstrates the convolution operation. In each case the dotted line indicates the desired response $H_d(e^{j\omega})$.



From this, note that

- The *mainlobe* width of $W(e^{j\omega})$ affects the *transition* width of $H(e^{j\omega})$. Increasing the length N of $h[n]$ reduces the mainlobe width and hence the transition width of the overall response.
- The *sidelobes* of $W(e^{j\omega})$ affect the passband and stopband tolerance of $H(e^{j\omega})$. This can be controlled by changing the shape of the window. Changing N does not affect the sidelobe behaviour.

Some commonly used windows for filter design are

- **Rectangular:**

$$w[n] = \begin{cases} 1 & 0 \leq n \leq N \\ 0 & \text{otherwise} \end{cases}$$

- **Bartlett (triangular):**

$$w[n] = \begin{cases} 2n/N & 0 \leq n \leq N/2 \\ 2 - 2n/N & N/2 < n \leq N \\ 0 & \text{otherwise} \end{cases}$$

- **Hanning:**

$$w[n] = \begin{cases} 0.5 - 0.5\cos(2\pi n/N) & 0 \le n \le N \\ 0 & \text{otherwise} \end{cases}$$

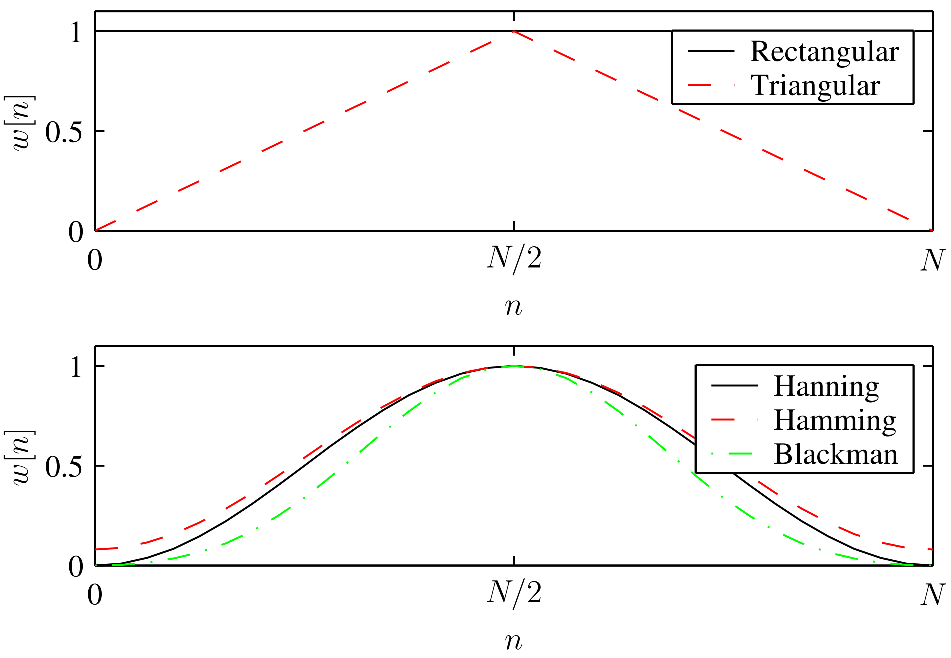
- **Hamming:**

$$w[n] = \begin{cases} 0.54 - 0.46\cos(2\pi n/N) & 0 \le n \le N \\ 0 & \text{otherwise} \end{cases}$$

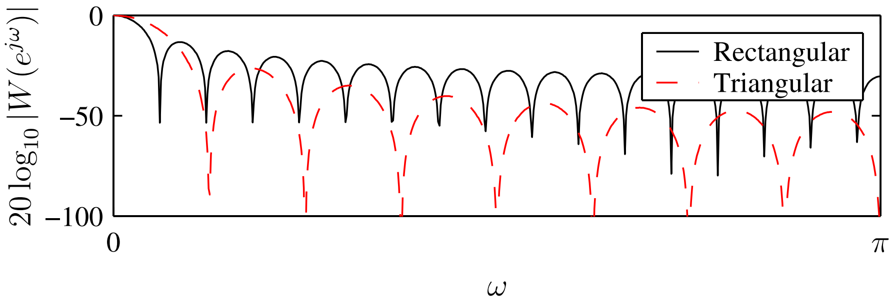
- **Kaiser:**

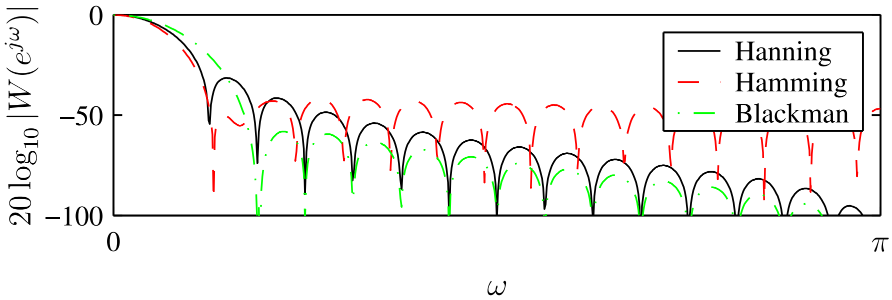
$$w[n] = \begin{cases} I_0[\beta(1 - [(n-\alpha)/\alpha]^2)^{1/2}] & 0 \le n \le N \\ 0 & \text{otherwise} \end{cases}$$

Examples of five of these windows are shown below:



All windows trade off a reduction in sidelobe level against an increase in mainlobe width. This is demonstrated below in a plot of the frequency response of each of the windows:





Some important window characteristics are compared in the following table:

Window	**Peak sidelobe amplitude (dB)**	**Mainlobe transition width**	**Peak approximation error (dB)**
Rectangular	-13	$4\pi/(N+1)$	-21
Bartlett	-25	$8\pi/N$	-25
Hanning	-31	$8\pi/N$	-44
Hamming	-41	$8\pi/N$	-53

The Kaiser window has a number of parameters that can be used to explicitly tune the characteristics.

In practice, the window shape is chosen first based on passband and stopband tolerance requirements. The window size is then determined based on transition width requirements. To determine $h_d[n]$ from $H_d(e^{j\omega})$ one can sample $H_d(e^{j\omega})$ closely and use a large inverse DFT.

7.2.3 Frequency Sampling Method for FIR Filter Design

In this design method, the desired frequency response $H_d(e^{j\omega})$ is sampled at equally-spaced points, and the result is inverse discrete Fourier transformed.

Specifically, letting

$$H[k] = H_d(e^{j\omega})\Big|_{\omega=\frac{2\pi k}{N}}, \qquad k = 0, \ldots, N-1,$$

the unit sample response of the filter is $h[n] = \mathrm{IDFT}(H[k])$, so

$$h[n] = \frac{1}{N} \sum_{k=0}^{N-1} H[k] e^{j2\pi nk/N}.$$

The resulting filter will have a frequency response that is exactly the same as the original response at the sampling instants. Note that it is also necessary to specify the *phase* of the desired response $H_d(e^{j\omega})$, and it is usually chosen to be a linear function of frequency to ensure a linear phase filter. Additionally, if a filter with real-valued coefficients is required, then additional constraints have to be enforced.

The *actual* frequency response $H(e^{j\omega})$ of the filter $h[n]$ still has to be determined. The z-transform of the impulse response is

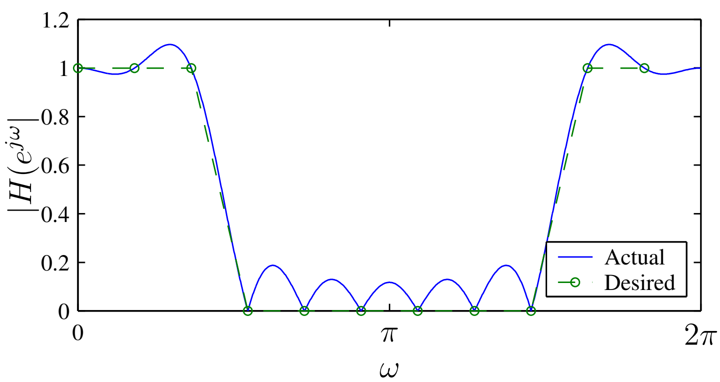
$$\begin{aligned} H(z) = \sum_{n=0}^{N-1} h[n] z^{-n} &= \sum_{n=0}^{N-1} \left[\frac{1}{N} \sum_{k=0}^{N-1} H[k] e^{j2\pi nk/N} \right] z^{-n} \\ &= \frac{1}{N} \sum_{k=0}^{N-1} H[k] \sum_{n=0}^{N-1} e^{j2\pi nk/N} z^{-n} \\ &= \frac{1}{N} \sum_{k=0}^{N-1} H[k] \left[\frac{1 - z^{-N}}{1 - e^{j2\pi k/N} z^{-1}} \right]. \end{aligned}$$

Evaluating on the unit circle $z = e^{j\omega}$ gives the frequency response

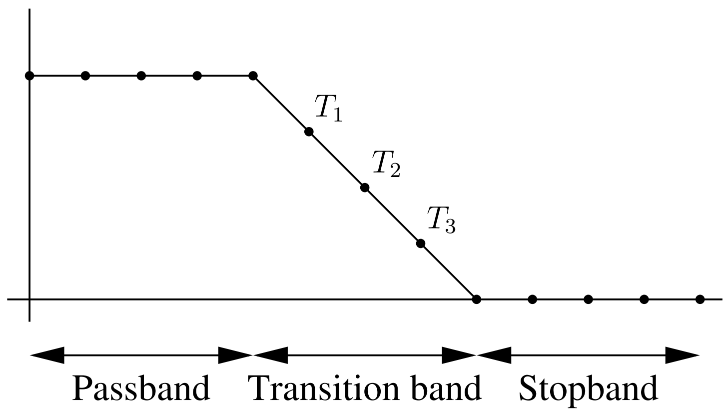
$$H(e^{j\omega}) = \frac{1 - e^{-j\omega N}}{N} \sum_{k=0}^{N-1} \frac{H[k]}{1 - e^{j2\pi k/N} e^{-j\omega}}.$$

This expression can be used to find the actual frequency response of the filter obtained, which can be compared with the desired response.

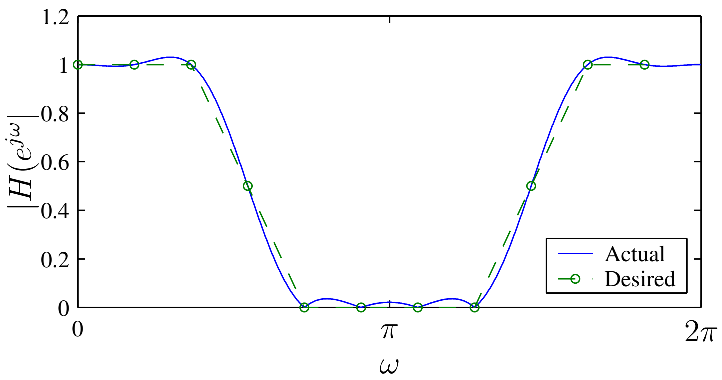
The method described only guarantees correct frequency response values at the points that were sampled. This sometimes leads to excessive ripple at intermediate points:



One way of addressing this problem is to allow **transition samples** in the region where discontinuities in $H_d(e^{j\omega})$ occur:



This effectively increases the transition width and can decrease the ripple, as observed below:



By leaving the value of the transition sample unconstrained, one can to some extent optimise the filter to minimise the ripple. Empirically, with three transition samples a stopband attenuation of 100dB is achievable. Recall however that for $h[n]$ real we require even or odd symmetry in the impulse response, so the values are not entirely unconstrained.

7.2.4 Optimum Approximations of FIR Filters

This method of filter design attempts to find the filter of length N that optimises a given design objective. In this case the objective is chosen to be the minimisation of

$$\max_{0\leq\omega\leq 2\pi} |E(e^{j\omega})|$$

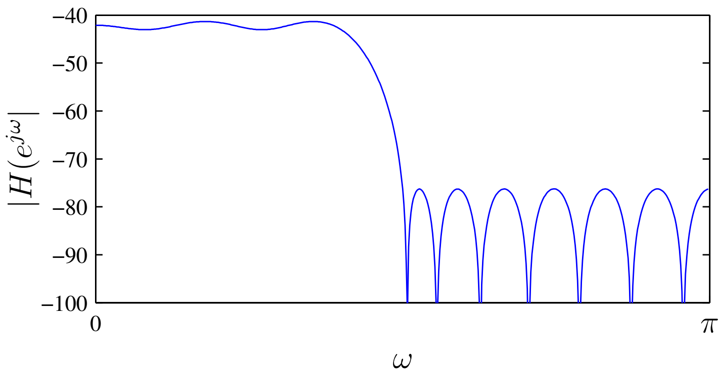
where $E(e^{j\omega})$ is a weighted error function

$$E(e^{j\omega}) = W(e^{j\omega})[H_d(e^{j\omega}) - H(e^{j\omega})].$$

The minimisation is performed over the filter coefficients $h[n]$.

In practice, the design problem can be specified as follows: given δ_p, δ_s, f_p, and f_s, determine $h[n]$ such that the design specification is satisfied with the smallest possible N. The optimal (or minimax) design method therefore yields the shortest filter that meets a required frequency response over the entire frequency range. It is widely used in practice.

Solutions to this optimisation problem have been explored in the literature, and many implementations of the method are available. It turns out that when $\max|E(e^{j\omega})|$ is minimised, the resulting filter response will have equiripple passband and stopband, with the ripple alternating in sign between two equal amplitude levels:



The maxima and minima are known as extrema. For linear phase lowpass filters, for example, there are either $r+1$ or $r+2$ extrema, where $r = (N+1)/2$ (for type 1 filters) or $r = N/2$ (for type 2 filters).

For a given set of filter specifications, the locations of the extremal frequencies, apart from those at band edges, are not known a priori. Thus the main problem in the optimal method is to find the locations of the extremal frequencies. Numerous algorithms exist to do this. Once the locations of the extremal frequencies are known, it is simple to specify the actual frequency response, and hence find the impulse response for the filter.

7.3 IIR Filter Design

An IIR filter has nonzero values of the impulse response for all values of n, even as $n \to \infty$. To implement such a filter using a FIR structure therefore requires an infinite number of calculations.

However, in many cases IIR filters can be realised using LCCDEs and computed recursively.

Example:

A filter with the infinite impulse response $h[n] = (1/2)^n u[n]$ has z-transform

$$H(z) = \frac{1}{1 - 1/2z^{-1}} = \frac{Y(z)}{X(z)}.$$

Therefore, $y[n] = 1/2y[n-1] + x[n]$, and $y[n]$ is easy to calculate.

IIR filter structures can therefore be far more computationally efficient than FIR filters, particularly for long impulse responses.

FIR filters are stable for $h[n]$ bounded, and can be made to have a linear phase response. IIR filters, on the other hand, are stable if the poles are inside the unit circle, and have a phase response that is difficult to specify. The general approach taken is to specify the magnitude response, and regard the phase as acceptable. This is a disadvantage of IIR filters.